Migrant Integration and Social Stability

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This Version: October, 2026
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Abstract

Societies around the world grapple with how to design migrant integration policies, amid concerns about these policies' impacts on social stability. This paper examines the causal link between migrant integration policy and social stability. We analyze the effect of China's 2014 reform of its restrictive internal migration system on migrant labor unrest (e.g., strikes and protests), exploiting a discontinuity in reform eligibility. We find that by offering migrants a pathway to obtaining full rights in host societies, the reform reduced labor unrest. We show that the reform strengthened migrants' intentions to settle in their destinations, thereby weakening their incentives to engage in labor unrest that could undermine their chances of staying due to employer or government retaliation. We develop a novel causal mediation analysis and show that settlement intentions can account for 26–63 percent of the reform's effect on labor unrest. We find no evidence that the reform altered migrant composition, improved migrants' well-being, or prompted governments to tighten social control. Our findings highlight the role of migrants' endogenous responses to migration regimes in shaping the social consequences of migration.

Keywords: Migration; Unrest; Integration; Household Registration; China.
JEL Codes: D74; D79; E24; O15.

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1 Introduction

Today, migration has become an important aspect of many societies. The foreign-born population now accounts for 14 percent of the population in the average EU country and 15 percent in the US (Frattini and Piovesan, 2026; US Census Bureau, 2026). In the developing world, over 1 billion people migrate within their own country (FAO, 2018). Therefore, societies around the world grapple with how to design policies governing migrants' integration, particularly the designation of legal status relevant to access to rights and services and opportunities for long-term settlement.[1]

A key policy consideration is maintaining social stability, namely, controlling clashes between migrants and host societies (European Commission, 2020; OECD, 2024). There are two competing views. One view favors restrictive integration policies to exclude migrants and deter migration, as it perceives migration as a source of social turmoil, such as crime, conflict, and cultural clashes, due to migrants' disadvantaged characteristics and differences from host societies (Marie and Pinotti, 2024; Hangartner et al., 2019; Campante et al., 2024, 2026).[2] In contrast, the other view favors more inclusive policies, at least for existing migrants. It contends that migrants' behavior is shaped by integration policies: settlement prospects of settlement can incentivize migrants to make the integration efforts expected by host societies, thereby promoting social cohesion (Gathmann and Keller, 2018; Adda et al., 2022). As such, it is important to empirically examine the causal link between integration policy and social stability and disentangle the mechanisms.

In this paper, we argue and provide evidence that pathways toward future integration can strengthen social stability, because they increase migrants' intentions to settle in host societies and hence reduce their incentives to engage in destabilizing activities that may hurt their chances of staying. In doing so, our paper highlights that the impacts of migration can be determined by migrants' endogenous responses to incentives set by the migration regime, not only driven by static native-migrant differences in individual characteristics. It also highlights the central role that prospects play in shaping social stability (Cantoni et al., 2024).

Our analysis builds on China's restrictive internal migration regime, known as the household registration (*hukou*) system, which governs domestic population flows in ways similar to an international migration regime (Chan, 2019; Vortherms, 2023). The system functions like an internal passport system. Although people are free to move and are entitled to work in their migration destinations, they face substantial barriers to integration: they are generally denied access to local public services (e.g., public schools, healthcare, and social security), unless they meet demanding

[1] Internal migrants also face policy restrictions. In 2013, among 185 countries with available data, 80 percent of them had policies to reduce rural-to-urban migration (United Nations, 2013, p. 8; cited in Imbert et al., 2025). Governments may also exclude rural migrants from public services in cities (Feler and Henderson, 2011).

[2] The perceived link between migration and social turmoil is well documented. Marie and Pinotti (2024) find that in most OECD countries, more people worry that immigrants increase crime than that "immigrants take jobs away from natives." Hangartner et al. (2019) show that mere exposure to refugee arrivals induces lasting concerns about social turmoil (e.g., crime and terrorism) and support for exclusionary policies. Campante et al. (2024, 2026) show that despite Ebola's negligible impact on American society in 2014, the Ebola scare increased voters' anti-immigration attitudes, and Republican politicians linked Ebola to immigration in their messaging.

criteria and officially transfer their residential registration (i.e., *hukou*) to the destination.[3] Therefore, Chinese migrants have limited hope to settle permanently in the destination and tend to stay for a short period (Meng, 2012).

Against this backdrop, in the first part of the paper, we establish the causal relationship between migrant integration and social stability. We exploit a natural experiment generated by the nationwide reform of the *hukou* system in 2014. The reform made it easier for migrants to transfer their *hukou* and settle permanently in cities with urban populations under 3 million, provided that they met minimum eligibility requirements, most notably having a job. Exploiting the discontinuity, we estimate the causal effect of the *hukou* reform on social stability. The outcome of interest is labor unrest, which largely represents *migrant workers'* collective action to protest employer exploitation (e.g., wage arrears and job displacement). In China, migrant workers face disproportionately more labor rights violations and make up the majority of participants in labor unrest (e.g., Lee, 2007; Chan, 2010; Friedman, 2014; Rho, 2023; among others). Meanwhile, the Chinese state is vigilant in controlling labor unrest (Friedman, 2014).[4] Therefore, in this setting, the impact of the *hukou* reform on labor unrest can reflect how migrants trade off the benefits of participating in labor unrest against the risk that doing so may reduce their chances of settlement due to employer or government retaliation. We expect that, because eventual settlement is valuable, migrants may disengage from labor unrest to safeguard their chances of settlement. Although we focus on labor unrest, the underlying logic extends to other activities that threaten social stability and jeopardize migrants' prospects for permanent settlement.[5]

We implement a difference-in-discontinuity (DiDC) design, which combines difference-in-differences and regression discontinuity designs. Intuitively, the causal effect of the *hukou* reform is identified by comparing changes in labor unrest between reform cities just below 3 million people and non-reform cities just above 3 million people. We validate that this research design yields clean comparisons by showing: (i) there is no evidence of selection into a specific reform status; (ii) baseline characteristics are balanced across the 3-million cutoff; (iii) no other potentially confounding policies have provisions correlated with population and may affect labor unrest; and (iv) there is no evidence of anticipation of the reform.

We find that the *hukou* reform significantly *reduced* labor unrest. Annual labor unrest fell by 1.419 incidents per million working-age population (ages 25–54) after the reform, and there were no differential trends leading up to the reform, favoring a causal interpretation. This effect amounts to

[3] Section 2.1 provides more details about the *hukou* system.

[4] We provide a more detailed discussion of labor unrest in China in Section 2.2.

[5] Our analysis should be interpreted as positive rather than normative. We study labor unrest because it is an available measure of migrants' behavior. We note that labor unrest involves workers' actions to protect their legal rights. The mechanism we argue can be applied to migrants' other activities that may undermine social stability and threaten their chances of settlement, such as crime. However, to the best of our knowledge, existing crime data in China do not separately report crimes committed by local residents and migrants. Because aggregate crime data would primarily reflect local residents' behavior, they are not suited for studying migrants' criminal activity.

42 percent of the mean level in non-reform regions. It is also comparable to the impact of a large economic windfall—a $3,576 increase in exports per worker (Campante et al., 2023).

A battery of evidence indicates that the decrease in labor unrest is not due to changes in the overall sociopolitical environment or statistical artifacts. First, a placebo test shows the decrease in labor unrest occurs *only* at the true cutoff of 3 million, not at other cutoffs, implying that our results are not driven by factors correlated with population size. Second, the reform has no discernible effect on population growth during the period we study, pointing to the importance of understanding existing migrants' responses. Third, our results do not simply reflect a natural reduction in discontent with the *hukou* system itself. In fact, the share of protests potentially related to the *hukou* system is too small to account for the detected effect. Fourth, we find no evidence of differential reporting of labor unrest events. Lastly, our results are robust to controlling for the recent decline in non-governmental organizations (NGOs) in China. In addition, the results are robust to a variety of checks, including alternative inference procedures, alternative specifications and estimators, excluding potential outliers, and covariate balancing.

The second part of the paper demonstrates how integration opportunities shape social stability via strengthening migrants' intentions to settle in host societies. Our analysis proceeds in stages. First, we estimate the effect of the *hukou* reform on migrants' intentions to settle in a destination, which yields an empirical measure of migrants' settlement intentions based on their decisions to outmigrate from a destination. In the second stage, building on this measure, we develop a novel, simple causal mediation analysis to quantify how the reform reduces labor unrest by changing migrants' settlement intentions (the mediator). We use an instrumental variable (IV) approach to address the endogeneity of the mediator, which would arise if migrants' settlement intentions (measured by outmigration decisions) are affected by the destination's economic and political factors conducive to labor unrest. Our main methodological contribution is a simple exercise to bound two sources of bias that are often underappreciated in causal mediation analysis: (i) the first arises when the causal effect of settlement intentions (the mediator) identified by IV for a particular subpopulation differs from the average effect for the main population; and (ii) the second is due to *sorting*, that is, places where settlement intentions (the mediator) respond most strongly to the reform (the treatment) are also those where labor unrest (the outcome) responds most strongly to settlement intentions (the mediator) (Baron and Kenny, 1986; Imai et al., 2011; Glynn, 2012; Imbens and Angrist, 1994).[6] Our method is applicable to broad settings where a researcher wants to quantitatively assess the importance of a mechanism in explaining the treatment effect. Finally, we test a variety of alternative explanations.

Using an individual-level panel dataset with information on migration history, we find that migrants in a city eligible for the *hukou* reform were 7 percentage points less likely to leave that city after the reform, implying that integration opportunities brought by the reform made migrants more inclined to stay. We use changes in outmigration rates as a measure of changes in migrants'

[6]This bias is specific to using IV to identify a causal effect of the mediator. Please see Section 5.2.1 for discussion.

settlement intentions. Then, we formally quantify the extent to which settlement intentions explain the reform's effect on labor unrest. We instrument for migrants' settlement intentions using quasi-exogenous trade shocks to migrants' *origins*, which alter their outside options and incentives to return. The IV is strongly associated with outmigration from the destination. We find that migrants' stronger settlement intentions (equivalently, decreased outmigration rates) can explain 26–63 percent of the *hukou* reform's total effect on labor unrest under moderate assumptions.

Additional evidence confirms the importance of migrants' stronger settlement intentions. First, we show that the *hukou* reform led to larger reductions in labor unrest in places that were more likely to attract migrants to stay when the reform made it possible, such as destinations where migrants were geographically closer to their origins or more culturally similar to natives. Second, we rule out several alternative mechanisms through which the *hukou* reform might reduce labor unrest. We find no changes in migrants' characteristics. We also rule out the possibility that the reform simply appeased migrants by delivering material benefits: we fail to detect immediate improvements in migrants' well-being (e.g., labor market outcomes). Furthermore, we find no evidence that the *hukou* reform triggered tighter social control.

In sum, the results show that relaxing integration restrictions—in our case, driven by a *hukou* reform that provides easier access to local *hukou*—can enhance social stability by raising migrants' intentions to settle and incentivizing desired behavior. Although the *hukou* system is specific to internal migration in China, the mechanism we identify is relevant to a broader class of institutions that govern migrants' long-term integration into host societies, such as citizenship and legal status. Indeed, the citizenship literature finds that access to citizenship can catalyze migrants' assimilation efforts, such as labor force participation, language training, and saving (Gathmann and Keller, 2018; Zillessen, 2026). We note two points when generalizing our findings. First, the impact of loosening integration restrictions should depend on how binding the barrier to full integration is, or in other words, how beneficial full integration would be. In our case, obtaining a local *hukou* is life-changing: it allows a migrant to fully access local public services in the host society. This is consistent with findings that naturalization policy tends to induce more assimilation efforts in Europe (than in the US), as noncitizens in Europe face high barriers to full engagement with host societies (Fouka, 2023). Second, the behavior change we detect here is relevant to migrants who do want to stay in host societies. As we show, reducing barriers to integration has a more pronounced effect when migrants are more inclined to stay because of cultural similarity and geographic proximity to home. Taken together, we argue that inclusive integration policies can strengthen social stability when migrants face binding integration constraints and do want to settle in host societies.

Our paper engages with several strands of literature. First, it speaks to the causal link between migration and social stability. The theoretical prediction is ambiguous, as it depends on both migrants' characteristics and their endogenous responses to migration policies. A large body of empirical literature in this domain studies the effect of international migration on crime (Spenkuch, 2014; Bianchi et al., 2012; Bell et al., 2013; Nunziata, 2015; Ajzenman et al., 2023), with mixed

findings. When significant effects are detected, they are typically explained by immigrants' disadvantaged characteristics and poor labor market opportunities. Our paper departs from migrants' characteristics and highlights a distinct mechanism: migrants' behavior endogenously responds to their expected stay in host societies, which is influenced by integration rules. In this regard, we add to a growing literature on how stay plans shape migrants' choices (Dustmann and Görlach, 2016; Adda et al., 2022, 2025; Zaiour, 2023).

Second, our paper contributes to the literature on China's *hukou* system, a core institution of Chinese society. Several papers have documented the *economic* consequences of the *hukou* system, highlighting efficiency losses due to labor mobility frictions (Au and Henderson, 2006; Ngai et al., 2019; Adamopoulos et al., 2024; Gai et al., 2025). To the best of our knowledge, we are the first to examine the *political* impact of the *hukou* system. While there is a perception shared by Chinese officials that loosening the *hukou* system may induce instability (Vortherms, 2023),[7] our results suggest that ironically, migrants' inability to integrate into host societies can contribute to instability. In this regard, our paper relates to the political economy literature on state-citizen relations, in particular, how rights and benefits extended by the government shape citizens' political support and social cohesion (Manacorda et al., 2011; Marcucci et al., 2023; Wen, 2025; Bai and Jia, 2016; Pan, 2020).

Last but not least, our paper adds to the econometric literature on causal mediation analysis, which aims to formally estimate the treatment effect mediated by a mechanism (Baron and Kenny, 1986; Imai et al., 2011; Glynn, 2012; Pearl, 2009). As in Imai et al. (2011) and Bai and Kung (2015), we use an IV to address the endogeneity of the mediator. Our contribution is to allow effect heterogeneity and develop a simple exercise to bound potential bias in the estimated mediation effect stemming from heterogeneous effects of the mediator. This complements approaches that accommodate heterogeneous effects using a richer structural and nonparametric framework (Frölich and Huber, 2017).

The remainder of this paper is organized as follows. Section 2 introduces the institutional context. Section 3 describes our data and research design. Section 4 reports the effect of the *hukou* reform on labor unrest. Section 5 explores underlying mechanisms. Section 6 concludes. Additional results and discussions can be found in the Online Appendices.

2 Institutional Context

In Section 2.1, we first introduce China's *hukou* (household registration) system, and then we provide key information on the 2014 *hukou* reform that we study in this paper.[8] In Section 2.2, we discuss labor unrest in China. Finally, in Section 2.3, we outline how the *hukou* reform may affect labor unrest to help fix ideas for our empirical investigation.

[7] Vortherms (2023) provides interview evidence that Chinese officials view migrants as a source of crime and instability, and they take it into account when designing local *hukou* policies.

[8] We refer interested readers to Song (2014) and Chan (2019) for a more comprehensive account of the *hukou* system.

2.1 China's *Hukou* System and Internal Migration

The *Hukou* System. China's *hukou* (household registration) system functions like an "internal passport" system (Au and Henderson, 2006). Each Chinese citizen is assigned a *hukou* at birth, which includes an *official* location of residence and an "agricultural" or "non-agricultural" category. People typically inherit their parents' *hukou* locations and categories. An individual is only eligible for local public services in their *hukou* registration location, regardless of where they actually reside. The geographic scope of eligibility varies by service, with the prefecture level—the administrative level between the county and the province—being the most consequential.[9] In today's China, people can move freely within the country, but it is generally difficult to transfer one's *hukou* registration across regions. Local governments typically restrict transfers to migrants with substantial financial means, such as investors, home buyers, or highly educated professionals. Some local governments have established point-based systems to evaluate migrants' qualifications and determine eligibility for *hukou* transfers. Consequently, migrants have limited prospects of fully integrating into their destinations and gaining full access to local public services.

The 2014 *Hukou* Reform. On July 30, 2014, the Chinese central government launched a nationwide reform of the *hukou* system (State Council, 2014). The reform was *centrally mandated* and required local governments to adjust their *hukou* transfer policies according to their urban population. Importantly, it substantially relaxed transfer requirements in cities with fewer than 3 million people, while requiring cities above 3 million to maintain strict control.

Cities (with fewer than 3 million people) are set to accept a broader base of migrants: instead of the high bars based on financial capacity, obtaining a local *hukou* now generally requires only a stable job, stable residence, and a minimum period of enrollment in the local social security system, with requirements more lenient in lower population tiers (1–3 million, 0.5–1 million, and < 0.5 million).[10] By contrast, larger cities (those with more than 3 million people) are required to maintain tight control over *hukou* transfers, and they were urged to establish points-based systems that only incorporate select migrants.

Therefore, the 2014 reform generated a sharp change in access to a local *hukou* around the 3-million population cutoff. We focus on this discontinuity in our empirical analysis. The 3-million cutoff is arguably the most relevant one, as most very small cities already had relaxed *hukou* transfer restrictions before 2014 (Song, 2014). Moreover, the central government reiterated in 2016 and 2024 that cities with an urban population below 3 million must abolish all barriers to *hukou* transfers (State Council, 2016, 2024). In Section 4, we further show empirically that only the 3-million cutoff, rather than the other population cutoffs, is the relevant discontinuity.

[9] In 2010, an average prefecture had 4.2 million residents. Primary education is administered at the district or township level, while secondary education falls under county-level management. Social security programs, such as pensions, healthcare, and housing funds, are generally managed at the prefecture level.

[10] Having a stable job means having an employment contract or being a business owner (with minimum requirements on tax payments and/or registered capital). Having a stable residence means either having a rental contract registered with the government or owning an apartment.

2.2 Labor Unrest in China

Despite the autocratic regime, labor unrest is not uncommon in China. Migrant workers are central to this phenomenon. The literature on China's labor politics underscores the institutional discrimination faced by migrant workers under the *hukou* system (e.g., Lee, 2007; Chan, 2010; Friedman, 2014; Rho, 2023; Elfstrom, 2021; among others). Restricted social and economic mobility for migrants due to limited rights, combined with employers' rampant violations of basic statutory protections, fuels grievances among migrants and contributes to the occurrence of labor unrest, including strikes, protests, and riots. Although there are no official data on migrant workers' participation in labor unrest, ample anecdotes and fieldwork suggest that migrant workers indeed make up the majority of participants in labor unrest, especially in confrontational actions that demand better treatment beyond minimum rights (Friedman, 2014; Elfstrom, 2021; Rho, 2023; Goebel, 2019).[11] Consistent with this view, Figure A1 shows a strong positive relationship between labor unrest and the share of migrants in a region.

Participation in labor unrest is risky and costly for workers. The Chinese state has been vigilant about labor unrest, especially in the last decade (Franceschini and Nesossi, 2018; Rho, 2023). While the central government strategically tolerates localized labor unrest to extract information about discontent and monitor local officials (Lorentzen et al., 2013), protests that cross implicit boundaries, such as involving broad mobilization or threatening social stability, are subject to repression (Rho, 2023; Chen and Gallagher, 2018). Local officials are incentivized to contain labor unrest under the cadre evaluation system that stresses social stability (Edin, 2003; Campante et al., 2023; You et al., 2022; Mueller, 2025). Workers also face the risk of employer retaliation. Anecdotes suggest that employers share blacklists of worker activists, raising the cost of protesting (Leung, 2015). Moreover, unions in China are government-affiliated and do not independently mobilize protests (Friedman, 2014). In sum, participation in labor unrest entails nontrivial risks and costs, making individual workers' willingness to participate particularly important.

2.3 The *Hukou* Reform and Labor Unrest

The *hukou* reform offers migrants a pathway toward permanent settlement in their destinations (i.e., obtaining local *hukou*). Migrants who want to pursue this path may become unwilling to participate in labor unrest. The *hukou* reform does not immediately entitle migrants to local *hukou*; rather, migrants still need to meet minimum requirements, notably having a job, and obtain government approval. Participation in labor unrest may make it harder to maintain the stable employment and housing required for local *hukou* because of employer retaliation, landlord discrimination, or

[11] Friedman (2014, pp. 14) writes that "anecdotal evidence suggests that they [migrants] are the primary actors in contemporary insurgency." Elfstrom (2021, pp. 42) concludes that "nonlocals are now both the most likely to mobilize against their bosses and the most likely to choose extra-institutional channels to do so. The Beijing municipal government reported in 2010 that migrant workers' labor disputes comprised nearly 70 percent of all labor disputes in Beijing in 2010 (Rho, 2023, pp. 8). Goebel (2019) analyzes social unrest on social media and finds that migrant workers have engaged in the largest number of protests." Reassuringly, he finds that local migrant density is positively associated with strike incidents but negatively associated with formal disputes through legal channels.

government sanctions. Also, since migrants now depend on the government for future settlement, they may be psychologically reluctant to engage in labor unrest that the government dislikes.[12] Put plainly, the pathway toward settlement gives migrants more to lose and raises their opportunity cost of participating in labor unrest. In Appendix C, we present a simple model to formalize this argument (Proposition C1).

We acknowledge that the *hukou* reform may affect labor unrest through other channels. For instance, the reform may improve migrant workers' well-being, reducing their grievances and, in turn, their engagement in labor unrest. Reform cities may attract more population and experience changes in population composition, resulting in differential dynamics of labor unrest. When implementing the reform, local governments may tighten social control in preparation for *perceived* risks of social turmoil. Throughout the paper, we carefully examine these possibilities.

3 Data and Research Design

3.1 Data

Our unit of analysis is the prefecture (often referred to as a prefectural city or simply a city), which is an administrative level between provinces and counties. There are 337 prefectures in total.[13] For our analysis, we exclude prefectures in Tibet and Xinjiang due to their distinct political environments. Our final sample consists of 287 prefectures with available urban population data, allowing us to define each prefecture's reform status.[14] According to the 2010 population census, these prefectures account for 94.4 percent of China's total population and 95.8 percent of its urban population.

Reform Status. As discussed in Section 2, prefectures with urban populations below 3 million should reduce barriers to *hukou* transfers from 2014 onward. We collect urban population data from the Urban Construction Statistical Yearbook (UCSY), compiled by the Ministry of Housing and Urban-Rural Development based on local governments' reports. Using the 2014 urban population, denoted by $P_{i,2014}$, we create a binary variable, $Reform_i = \mathbb{1}\{P_{i,2014} \leq 3 \text{ million}\}$, which equals one if prefecture i's 2014 urban population is below 3 million, and equals zero otherwise. Under this definition, 250 prefectures are classified as reform prefectures and 37 as non-reform prefectures.

We show that the population-based reform status captures actual policy changes (see Appendix D for details). We conduct a thorough review of *hukou* policy documents issued by local governments and manually code each prefecture's actual reform status. Then, we compare it to the population-defined status. We find that by 2015, most prefectures already had guidelines for implementing the reform, and no amendments were made in subsequent years. Moreover, the population-based

[12] Indeed, previous studies find that migrant workers are concerned about their actions irritating the government (Chen and Gallagher, 2018).

[13] This is based on the 2010 delineation; there are no significant changes to the delineation of prefectures over our sample period. We include 4 provincial-level municipalities: Beijing, Tianjin, Shanghai, and Chongqing.

[14] We exclude 34 prefectures because urban population data are unavailable and an additional 16 prefectures in Tibet and Xinjiang. Appendix G.1 shows that our results are not driven by these sample restrictions.

definition proves accurate: among 287 prefectures in the sample, there are only 17 discrepancies (6 percent) between the manually-coded and population-based reform statuses.[15] In subsequent analysis, we rely on the population-based reform status. Our results are robust to excluding the 17 prefectures with discrepancies (see Section 4.3).

Labor Unrest. Our data on labor unrest are from the China Labor Bulletin (CLB), a non-profit organization based in Hong Kong that has monitored workers' collective actions across China since 2011.[16] Our sample consists of events from 2011 to 2019. Due to the lack of administrative data on labor unrest in China, this dataset has been frequently cited by news outlets outside China (e.g., Hernández, 2016) and used in research on social unrest in China (e.g., Campante et al., 2023; Qin et al., 2024; Chen, 2026).

CLB coders actively collect and verify information on unrest events from China's domestic social media platforms. Between 2011 and 2019, the dataset reports 11,733 labor unrest events.[17] Given the collection method, the dataset primarily captures *severe* labor conflicts, where workers take to the streets to demand public attention. In 83 percent of events, workers demonstrate in public spaces through protests, marches, sit-ins, traffic obstruction, and even suicide threats. CLB often records multiple causes of an event. Wage arrears are the most common cause, cited in 72 percent of all protest events. Other common causes include job displacement (11 percent) and demands for welfare benefits (7 percent).

CLB captures similar patterns as in other data sources. We compare it with the Global Database of Events, Language, and Tone (GDELT), a widely used dataset on social unrest at the global level (Cantoni et al., 2024). Both datasets display quite similar national trends in labor unrest (see Figure A2).[18] Other authors also show that CLB data are correlated with labor disputes handled by government agencies and courts (Campante et al., 2023; Chen, 2026). The consistency across data collection efforts suggests that CLB data capture underlying patterns of labor conflict in China.

Auxiliary Data. We use a range of auxiliary data to validate the research design and explore mechanisms, including prefecture-level covariates, population censuses, migrant surveys, trade data, and biographical data on local officials, among others. Appendix E describes these data sources, and we introduce them when they become pertinent to the analysis.

3.2 Research Design

To estimate the causal effect of the *hukou* reform, a natural starting point is a difference-in-differences (DiD) design comparing changes in labor unrest between reform and non-reform prefectures. It

[15] In Appendix D, we discuss possible reasons for these discrepancies.

[16] CLB website: `https://clb.org.hk/en`. CLB was shut down on June 12, 2025 (Leung, 2025).

[17] The vast majority (11,451 events) occurred in the 287 prefectures in our main sample.

[18] However, GDELT is not suitable for regression analysis at the subnational level. As CLB focuses specifically on labor unrest and uses human coders for data collection and verification, it records more labor unrest events than GDELT: CLB identifies 11,733 events, whereas GDELT identifies only 4,681. Also, GDELT geocodes only 1,112 labor unrest events at the prefecture level.

can be implemented by a two-way fixed effects (TWFE) model:

$$\frac{Unrest_{it}}{L_{i,2010}} = \beta\left(Reform_i \times Post_t\right) + \lambda_i + \mu_t + \varepsilon_{it}. \tag{1}$$

The dependent variable, $\frac{Unrest_{it}}{L_{i,2010}}$, is the labor unrest rate, measured as the number of labor unrest events per million prime-age population (aged 25–54). $Reform_i$ is an indicator for prefecture i's reform status, which equals one if prefecture i's urban population in 2014 is below 3 million. $Post_t$ is an indicator that equals one for years from 2014 onward. λ_i and μ_t are the prefecture and year fixed effects, respectively. Standard errors are clustered at the prefecture level.

The coefficient β identifies a causal effect of the *hukou* reform under the standard parallel trends assumption: absent the *hukou* reform, labor unrest would have evolved similarly between reform and non-reform prefectures. However, this assumption is questionable because the urban population itself, which determines the reform status, may affect labor unrest. For instance, unrest activities may be easier to organize in more populous areas, and large populations may also make it difficult for governments to counter unrest threats (Wallace, 2014). It is also likely that populous areas have more resources for social control.

To address this concern, we exploit sharp variation in reform status around the 3-million population cutoff, and compare changes in labor unrest between reform and non-reform prefectures with close population sizes. Thus, we combine difference-in-differences (DiD) and regression discontinuity (RD) designs, creating a *difference-in-discontinuity* (DiDC) design:[19]

$$\frac{Unrest_{it}}{L_{i,2010}} = \beta\left(Reform_i \times Post_t\right) + \lambda_i + \mu_t + f\left(\tilde{p}_i; \zeta_{Reform,t}\right) + \varepsilon_{it}. \tag{2}$$

The running variable, $\tilde{p}_i = \log(3) - \log\left(P_{i,2014}\right)$, is the centered log urban population, which captures the deviation of log urban population from log(3), the cutoff deciding reform status.[20] f is a polynomial function of $\tilde{p}_i$, and $\zeta_{Reform,t}$ is a vector of coefficients on $\tilde{p}_i$ in the polynomial function. Importantly, as the subscripts indicate, coefficients in $\zeta_{Reform,t}$ are allowed to vary by reform status and over time. Following Gelman and Imbens (2019), we let f be a first-order polynomial function. In most of our analysis, we use the full sample of prefectures. Because there are more reform prefectures than non-reform prefectures (250 versus 37), restricting the sample to a narrow bandwidth around $\tilde{p}_i = 0$ would exclude many non-reform prefectures and reduce statistical power. Prefectures are weighted equally (i.e., we use a uniform kernel). The results are stable to alternative choices of polynomial functions, bandwidths, and kernels (see Section 4).

Akin to RD designs, β identifies the average causal effect of the *hukou* reform on labor unrest at $\tilde{p}_i = 0$. The causal interpretation hinges on the comparability between reform and non-reform prefectures near $\tilde{p}_i = 0$. Formally, a continuity assumption is needed: in the absence of the *hukou*

[19]The results are similar if we implement an RD design for each period (see Figure A3).

[20]We consider the deviation in logs because urban population $P_{i,2014}$ has a skewed distribution (see Figure A5). The results are similar when using the deviation in levels, $3 - P_{i,2014}$, as the running variable (see Table B1).

reform, the *trends* in labor unrest would have been similar around $\tilde{p}_i = 0$. Put differently, this is a *local* parallel trends assumption, weaker than the global version required by Equation 1. In Section 4.1, we provide various evidence to validate this assumption.

We also estimate an event-study specification:

$$\frac{Unrest_{it}}{L_{i,2010}} = \sum_{\tau \neq 2013} \beta_\tau \left(Reform_i \times \mathbb{1}\{t = \tau\} \right) + \lambda_i + \mu_t + f\left(\tilde{p}_i; \zeta_{Reform,t}\right) + \varepsilon_{it}. \tag{3}$$

2013 is the reference year. The coefficients β_τ ($\tau \geq 2014$) capture the dynamic reform effects. The pre-reform coefficients β_{2011} and β_{2012} constitute a pretrend test: we would expect them to be small if the continuity assumption holds.

4 Main Results

This section presents the main results of this paper. We start by showing the validity of our research design in Section 4.1. Section 4.2 reports the estimated effects of the *hukou* reform on labor unrest using the DiDC design. Section 4.3 presents a battery of robustness checks.

4.1 Validity of the Research Design

Our research design leverages variation in labor unrest trends around the cutoff of 3 million urban population. Therefore, the main threat to identification is other factors that may drive differential trends in labor unrest *precisely* around the cutoff. We discuss and address several concerns below.

Concern 1: Selection into a Specific Reform Status. The identification assumption is violated if some prefectural governments *precisely* manipulate urban population statistics to sort into a preferred reform status (Lee and Lemieux, 2010). However, there is little evidence for such manipulation.

First, if there were systematic manipulation of urban population to select a preferred reform status for potential benefits, we would expect significant bunching near the 3-million cutoff (or equivalently, $\tilde{p}_i = 0$). However, Figure 1, which displays the density of the running variable $\tilde{p}_i$, shows no such bunching. The McCrary (2008) test confirms that the density function is continuous at $\tilde{p}_i = 0$. Figure A6 similarly shows no discontinuity in the density of the 2015 urban population. Therefore, there is no evidence that prefectural governments manipulated urban population statistics either *ex ante* or *ex post* to justify their reform choices.

Second, substantial manipulation of urban population statistics would be difficult, as the new urban population cannot deviate too much from the historical level. In fact, from 2013 to 2014, only one prefecture's urban population grew from below 3 million to above 3 million, while none dropped from above 3 million to below 3 million. This suggests that manipulating population statistics to select a preferred reform status was rare.

Lastly, for manipulation to occur in practice, local officials must benefit from doing so. In the Chinese context, such benefits are predominantly bureaucratic promotion (e.g., Wang et al., 2020; He et al., 2020; Jia, 2024). For instance, Zeng and Zhou (2024) find that promotion-motivated local officials manipulate GDP statistics to improve observed performance. However, unlike GDP, urban population is not a performance metric used by upper-level governments to evaluate local officials. Moreover, we find no association between population growth from 2013–2014 and a local official's promotion prospects (see Table B2).[21] This suggests that local officials have little incentive to manipulate urban population statistics.

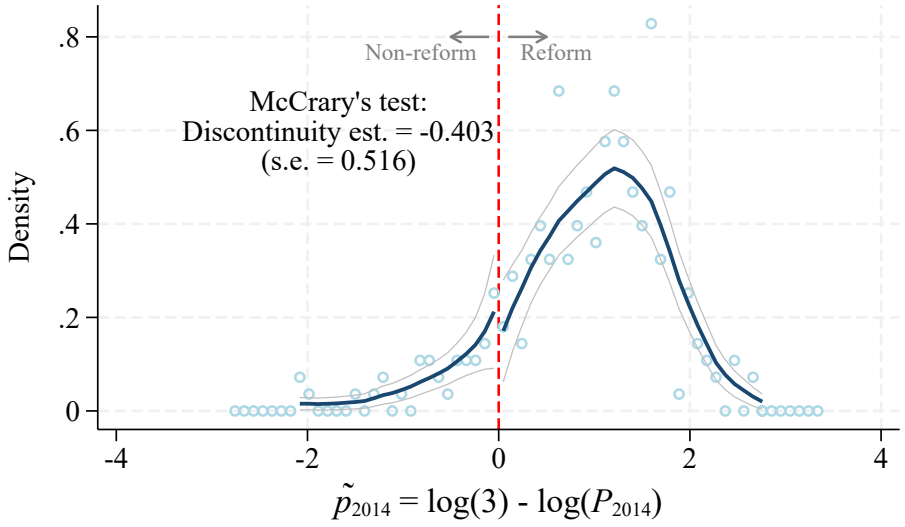


Figure 1. Density of Centered Log Urban Population of 2014

Note: This figure depicts the density of centered log urban population of 2014, $\tilde{p}_{i,2014} = \log(3) - \log(P_{i,2014})$. We report the McCrary test of density discontinuity at $\tilde{p}_{i,2014} = 0$.

Concern 2: Comparability Between Prefectures. Our identification assumption requires that, in the absence of the *hukou* reform, prefectures near the 3-million cutoff would have had similar *trends* in labor unrest. Although the assumption is inherently untestable because the counterfactual is not observed, we can examine the *pretrends* and show that prefectures are comparable near the 3-million cutoff. Specifically, we run the following RD regression:

$$\Delta W_{it} = \alpha_0 + \alpha_1 Reform_i + f\left(\tilde{p}_i; \zeta_{Reform,t}\right) + \mu_t + \nu_{it}, \quad t \leq 2013. \tag{4}$$

Here, $\Delta W_{it} = W_{it} - W_{i,t-1}$ is the change in a covariate. Our sample includes two pre-reform episodes, 2011–2012 and 2012–2013. Equation 4 stacks these two episodes and includes year fixed effects μ_t.

Panel A of Table 1 reports the results. We first estimate Equation 4 without the polynomial function $f\left(\tilde{p}_i; \zeta_{Reform,t}\right)$, so α_1 captures the average difference in pretrends between all reform and all non-reform prefectures. Columns (1) and (2) report the estimated α_1 and the standard error.

[21] Following Wang et al. (2020), we estimate an index for *ex-ante* promotion prospects (see Appendix E for estimation details).

Before the reform, reform prefectures experienced lower growth in unrest rates and population, but interestingly, higher GDP growth. There are no discernible differential trends in fiscal expenditures as well as "carrot and stick," measured by expenditures on social security and public security (i.e., police). These patterns suggest that a simple DiD design may not reliably estimate the causal effect of the *hukou* reform on labor unrest.

We next estimate Equation 4 with polynomials $f\left(\tilde{p}_i; \zeta_{Reform,t}\right)$. α_1 now captures local differences between reform and non-reform prefectures at $\tilde{p}_i = 0$. Columns (3) and (4) in Panel A of Table 1 show that when comparing prefectures barely eligible for the reform with those barely ineligible, the differential pretrends found in Columns (1) and (2) substantially shrink and all three statistically significant differences are eliminated. Importantly, there are no differential trends in labor unrest. The lack of differential pretrends is also evident in the RD plots displayed in Figure A7A.

The pretrend tests support our research design. We also examine pre-reform characteristics. Specifically, we estimate a cross-sectional variant of Equation 4, where the dependent variable is a characteristic measured in 2010. We consider factors potentially conducive to social unrest, including the share of males, the share of migrants, the shares of workers in industries with high frequencies of labor unrest (construction, manufacturing, and transportation), and the share of internet users. Panel B of Table 1 presents the results. As above, Columns (1) and (2) report estimates without a polynomial function, and Columns (3) and (4) report estimates with a polynomial function. Clearly, including polynomials substantially shrinks differences in predetermined characteristics, although the share of migrants differs slightly at the 10 percent significance level. Figure A7B visualizes the RD regressions, confirming the balance in predetermined characteristics. In Section 4.3, we also show that our results are robust to a variety of strategies to control for potential covariate-related differential trends.

Concern 3: Other Population-Correlated Policies. One may worry that the urban population determines policies other than the *hukou* reform, *and* those policies can also influence labor unrest. If so, our estimates would be contaminated. We show that this is not of primary concern. We conduct a comprehensive review of policies related to urban population (see Appendix F for details). We use the *PKULaw* database—a large database of Chinese laws maintained by Peking University and frequently used in research on policy-making in China (Wang and Yang, 2025; Tian, 2024)—to identify policies that mention "urban population" or other similar terms. Most policies merely reference "urban population" as part of a description, rather than specifying provisions that vary by urban population. For example, a prefecture's urban planning may include a projection of urban population. A small number of policies do include provisions based on population tiers, but they focus on domains unrelated to labor unrest, such as prefabricated construction, public transit, and domestic services. Additionally, in Section 4.2, we conduct placebo tests estimating "effects" at cutoffs other than 3 million. We find null effects, indicating that our estimates are not confounded by other policies correlated with the urban population.

Table 1. Continuity in Covariates

Dependent	(1) Coef. on *Reform*	(2) SE	(3) Coef. on *Reform*	(4) SE
Panel A: Pretrends (2011-2012, 2012-2013)				
ΔUnrest/L	-0.352**	(0.161)	0.070	(0.158)
ΔLog population	-0.019***	(0.003)	-0.005	(0.005)
ΔLog GDP	0.009*	(0.005)	0.001	(0.008)
ΔLog expenditure	0.009	(0.007)	0.001	(0.009)
ΔLog expenditure on social security	0.018	(0.013)	0.014	(0.021)
ΔLog expenditure on public security	0.012	(0.007)	-0.005	(0.011)
Year FE	Yes		Yes	
Polynomials			Yes	
Panel B: Predetermined characteristics (2010)				
Share of males	-0.001	(0.002)	-0.003	(0.003)
Share of migrants	-0.166***	(0.031)	-0.067*	(0.036)
Share of construction workers	-0.006	(0.004)	0.008	(0.006)
Share of manufacturing workers	-0.140***	(0.029)	-0.054	(0.042)
Share of transportation workers	-0.016***	(0.003)	-0.008	(0.005)
Share of internet users	-0.165***	(0.061)	0.044	(0.086)
Polynomials			Yes	

Note: This table examines continuity in covariates. Panel A examines pretrends over 2011–2012 and 2012–2013, and Panel B examines predetermined characteristics measured in 2010. Columns (1) and (2) report results from regressions of the dependent variable on the reform indicator $Reform_i$ (controlling for year fixed effects for Panel A). Columns (3) and (4) include a flexible linear polynomial function of $\tilde{p}_{i,2014}$. Standard errors are clustered at the prefecture level in Panel A and are heteroskedasticity-robust in Panel B.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

Concern 4: Anticipation of the Reform. Another concern is that people may have anticipated the reform and altered their behavior preemptively, or that local governments, with insider information, may have adjusted policies in preparation for the upcoming change in the migration regime. This is not very likely, given that Table 1 shows no pretrends in labor unrest, population, GDP, and local fiscal expenditures. Furthermore, we use online searches to test anticipation. Figure A8 presents searches for *hukou*- and *hukou*-reform related terms on Baidu, Google's Chinese counterpart. We see a transitory spike in searches for these terms after July 30, 2014, the date when the *hukou* reform was publicly announced. There were no increases in searches leading up to the announcement, and the spike quickly dissipated. Taken together, these results rule out anticipation of the reform.

4.2 The Effect of the *Hukou* Reform on Labor Unrest

4.2.1 Main Findings

We begin by examining the dynamic effects of the *hukou* reform on labor unrest. Figure 2 presents the event-study estimates based on Equation 3. Comparing prefectures around the reform cutoff, as in Table 1, we observe no differential trends in labor unrest leading up to the reform. After the reform went into effect, however, reform prefectures experienced a decline in unrest rates relative to

non-reform prefectures. The decrease in labor unrest trends stabilized after 2015.[22] Several checks confirm the significance of this trend break. First, in Figure A9, we find similar patterns when we control for province-specific linear trends. Second, using additional data on labor unrest from Elfstrom (2017), we extend the pre-reform period back to 2003. Reassuringly, Figure A10 shows a lack of pretrends. Lastly, to further evaluate the significance of the trend break, we implement a sensitivity test developed by Rambachan and Roth (2023), with results reported in Appendix G.2. The test extrapolates the differential trends implied by estimated pretrends to the post-reform period and examines, net of the extrapolated trends, whether the post-reform effects are still statistically significant. The results imply that one can conclude with high confidence that the *hukou* reform significantly reduced labor unrest, unless there exist extremely nonlinear differential trends.

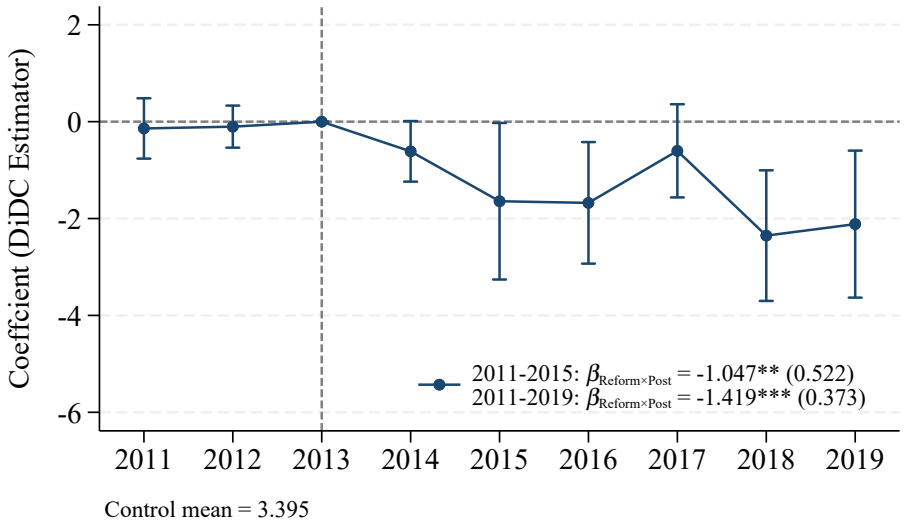


Figure 2. Dynamic Effects of the *Hukou* Reform on Labor Unrest

Note: This figure presents the dynamic effects of the *hukou* reform on labor unrest. We estimate Equation 3. The solid dots are point estimates, and the spikes are 95 percent confidence intervals. Standard errors clustered at the prefecture level are used for constructing the confidence intervals.

Table 2 reports the estimates of Equation 2 to summarize the overall effects. The results show that from 2011–2015, the *hukou* reform reduced annual labor unrest by 1.047 events per million prime-age population, and from 2011–2019, reform prefectures experienced 1.419 fewer unrest events per million prime-age population annually. Testing based on both clustered standard errors and permutation tests confirms that the estimated effects are statistically distinguishable from zero. In Appendix G.3, we show that our results are robust to alternative choices of bandwidths, kernels, and polynomial orders. Across specifications, we consistently find a strong negative effect of the *hukou* reform on labor unrest rates.

The effects we find are remarkable. Consider the estimated effect for 2011–2019. The magnitude amounts to about 42 percent (= 1.419/3.395) of the mean of non-reform prefectures. The estimated

[22] For completeness, Figure A4 compares the DiD and DiDC estimates. The DiD estimates show significant downward pretrends, as suggested by Table 1, but the post-trends are similar to those displayed by DiDC estimates.

effect of the *hukou* reform is also comparable to that of a large economic windfall. Following Campante et al. (2023), we estimate that a per $1,000 decrease in per-worker exports causes 0.397 more labor unrest events per million workers.[23] Therefore, the estimated effect of the *hukou* reform amounts to a $3,576 increase in per-worker exports; put this number in perspective, it is about one standard deviation of per-worker exports across Chinese prefectures during 2013–2015. CLB also reports coarse information on the scale of each unrest event: small (1–100 participants), middle (101–1,000 participants), or large (1,001–10,000 participants). Exploiting this information, we find that the *hukou* reform mainly reduced small- and middle-scale labor unrest events, but not large-scale unrest events (see Table B3). Using those estimates for a back-of-the-envelope calculation (and taking the midpoint of scale bins), the *hukou* reform reduced the share of protesting workers in the working-age population by 0.022 percentage points. This is a sizeable effect given that it is estimated that on average 0.108 percent of the working-age population protested annually in non-reform prefectures.

Table 2. The Effect of the *Hukou* Reform on Labor Unrest

	(1) Unrest/L 2011–2015	(2) Unrest/L 2011–2019
Reform × Post	-1.047**	-1.419***
	(0.522)	(0.373)
Control mean	2.501	3.395
Permutation test *p*-value	0.000	0.000
Prefecture FE	Yes	Yes
Year FE	Yes	Yes
Polynomials	Yes	Yes
Observations	1,435	2,583
R^2	0.700	0.601

Note: This table presents the results for the average effect of the *hukou* reform on labor unrest rates. Column (1) examines the period 2011–2015, while Column (2) examines the period 2011–2019. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

4.2.2 Ruling Out Alternative Interpretations

We now address whether the effects found above merely reflect changes in the sociopolitical environment. We present evidence to rule out interpretations that overlook the role of the *hukou* reform *per se*, including unobserved determinants of labor unrest, population growth, discontent with the *hukou* system, differential reporting of local events, and declining NGO activities in China.

[23] Campante et al. (2023) use a Bartik IV strategy to estimate the impact of China's export slowdown on labor unrest. In their main specification, they use an unbalanced panel of 278 prefectures from 2013–2015 and estimate that a $1,000 decrease in exports per worker causes 0.173 (SE = 0.075, N = 822) events per million workers in labor unrest. We use only the reform prefectures to hold constant the impacts of the *hukou* reform. Restricting to 243 reform prefectures, we estimate an effect of 0.397 (SE = 0.195, N = 718).

Unobserved Determinants of Labor Unrest. One interpretation is that our results reflect changes in unobserved determinants of labor unrest that are associated with urban population *and* have time-varying impacts. This is not very likely given the lack of urban population-correlated policies and the balanced pretrends. We perform a placebo test to further rule out unobserved confounders. Specifically, we use our research design to estimate the "effect" of being below a population cutoff other than the true 3-million cutoff.[24] Figure 3 reports the results. We detect a significant negative effect only at the true 3-million cutoff but a null effect elsewhere. Therefore, our results are unlikely to be driven by confounders correlated with urban population.

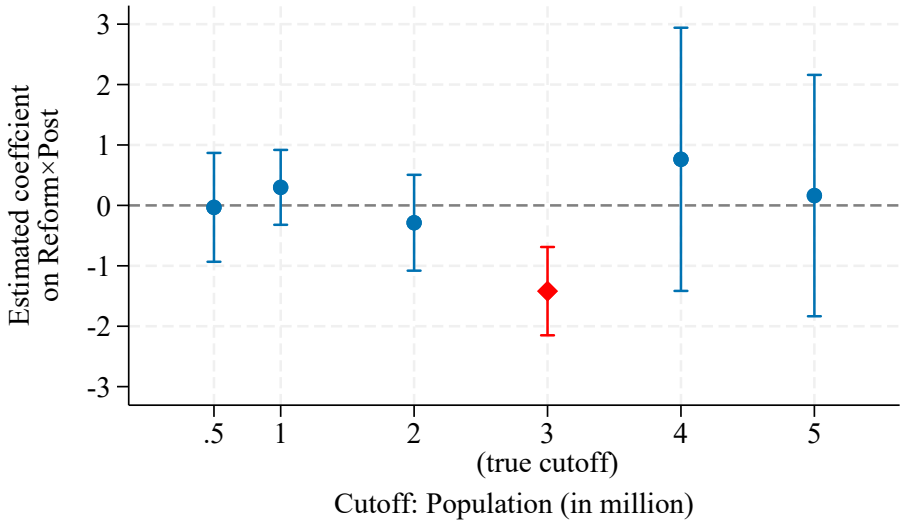


Figure 3. Placebo Test

Note: This figure presents the effects at falsified cutoffs: 0.5 million, 1 million, 2 million, 4 million, and 5 million. To avoid contamination from the true treatment effect at the 3-million cutoff, following Cattaneo and Titiunik (2022), we use only prefectures below 3 million for 0.5-million, 1-million, and 2-million cutoffs, and only prefectures above 3 million for 4-million and 5-million cutoffs. The solid dots are point estimates, and the spikes are 95 percent confidence intervals. Confidence intervals are constructed using standard errors clustered at the prefecture level.

Population Growth. Our findings can hardly be explained by population growth. A conventional concern is that relaxing migration restrictions may increase population. However, this would work against finding a decrease in the labor unrest rate: note that our dependent variable scales labor unrest events by *fixed* population (see Equation 2). Moreover, in Appendix G.5, we find that the reform had no discernible effects on total and urban population growth; if anything, the gaps in population and urbanization levels between reform and non-reform prefectures kept enlarging over time. These findings echo recent evidence that Chinese migrants prefer to go to large cities because of the high wages there and that the 2014 *hukou* reform failed to

[24]For a given cutoff c, we redefine $Reform_i$ as $\mathbb{1}\{P_{i,2014} \leq c\}$ and $\tilde{p}_i$ as $\log(c) - \log(P_{i,2014})$ in Equation 2. To avoid contamination from the treatment effect at the true 3-million cutoff, we follow Cattaneo and Titiunik (2022) and use only prefectures with urban populations below (above) 3 million when estimating the effect at a cutoff below (above) 3 million.

promote urbanization of small cities (Chen and Fu, 2023; Chan, 2021).[25] Furthermore, in Appendix G.6, we show that the results hold when accounting for the time-varying population size. These results point to the importance of understanding existing migrants' responses, which we concentrate on in Section 5.

Discontent with the *Hukou* System. The decrease in labor unrest is not about changes in discontent with the *hukou* system. Protesting against the *hukou* policy is not the primary cause of labor unrest. In CLB data, only a small share of unrest events may be effectively about protesting the *hukou* system, where workers demand social security and housing subsidies from employers. In a robustness check, we show that the *hukou* reform reduced labor unrest unrelated to *hukou*, but had a null effect on *hukou*-related unrest (see Table B4).

Differential Reporting of Local Events. Since our labor unrest data from the CLB are based on online posts, one concern is that our findings are driven by differential reporting of local events. However, Table 1 shows no significant difference in internet penetration between prefectures around the 3-million cutoff. In Appendix G.4, we further address this concern in two ways. First, we show that the *hukou* reform does not affect the coverage of local events, measured by the number of general events or protests recorded in GDELT, a database that focuses on a wide range of topics including but not limited to labor unrest (see Columns (1) and (2) of Table G2). Second, our results barely change when controlling for regional variation in internet censorship intensities (see Column (3) of Table G2).

The Role of NGOs. One may wonder if growing restrictions over NGOs in China can explain our results. Most NGOs in China assist labor by providing legal aid. However, some labor NGOs are movement-oriented. These movement-oriented NGOs are particularly active in Guangdong province due to the province's proximity to Hong Kong (Li, 2021; Fu, 2018). We control for the influence of NGOs in several ways: (i) controlling for interactions between distance to Hong Kong and year fixed effects; (ii) dropping prefectures within 500km or 1,000km of Hong Kong; and (iii) dropping Guangdong province. Table B5 shows a stable negative effect of the *hukou* reform on labor unrest. Thus, our results cannot be explained by variation in NGO activity in China.

4.3 Robustness Checks

Alternative Standard Errors. In Appendix H.1, we show that our results are robust to alternative standard errors, including clustering by province and Conley standard errors (Conley, 1999).

Alternative Specifications and Estimators. In Appendix H.2, we adopt alternative specifications and estimators: (i) different forms of the dependent variable (log and inverse hyperbolic sine transformations); (ii) a Poisson regression in light of the non-negative nature of the dependent variable (Silva and Tenreyro, 2006); and (iii) a spatial autoregressive (SAR) model to take into

[25] Chen and Fu (2023) find that big cities' policies that restrict migrant children's access to public schools increase the likelihood that migrants work in the cities alone while leaving their children at home. This suggests that, despite lower migration costs for relocating to small cities, migrants are still inclined to move to big cities.

account spatial spillovers. Across specifications and estimators, we consistently find that the *hukou* reform lowered labor unrest rates (see Table H2 and Figure H1).

Excluding Potential Outliers. In Appendix H.3, we show that our results are not driven by outlier observations. First, we leave out one province at a time and re-estimate Equation 2; the estimated effect remains similar when excluding any single province (see Figure H2). Second, inspired by the "donut-hole RD" exercise, we exclude prefectures immediately around the 3-million cutoff to address potential selection into reform status or measurement error (Column (1) of Table H3). Third, we exclude 17 prefectures with discrepancies between population-defined and actual reform status (Column (2) of Table H3). Lastly, following Hansen (2022, pp. 84–86), we identify and exclude observations most influential to estimation (Column (3) of Table H3). Across these exercises, our results remain robust.

Covariate Balancing. Despite the balances documented in Table 1, in Table H4 in Appendix H.4, we show that our results are robust to different strategies for controlling for baseline covariates: (i) directly controlling for interactions between the covariates and year dummies in regressions; (ii) balancing the propensity score (i.e., the probability of being under the reform) predicted by the covariates; and (iii) balancing the *distributions* of covariates using the coarsened exact matching (CEM) proposed by Iacus et al. (2012).

Summary. In this section, we provide robust evidence that the *hukou* reform led to a significant decrease in labor unrest. This decrease is not due to population-correlated confounders, population growth, changes in discontent with the *hukou* system, differential coverage of local events, or declining NGO activities. In the next section, we investigate mechanisms through which the *hukou* reform reduced labor unrest.

5 Unpacking Mechanisms

As discussed in Section 2.3, a primary mechanism we argue is that the *hukou* reform enables migrants to pursue permanent settlement in their destinations, which raises the opportunity cost of participating in labor unrest. In this section, we shed light on this argument by establishing that (i) the *hukou* reform does make migrants more willing to settle in the destination (Section 5.1), and (ii) this increase in settlement intentions does play an important role in explaining the decrease in labor unrest (Section 5.2). In Section 5.3, we discuss other potential mechanisms.

5.1 The Effect of the *Hukou* Reform on Migrants' Settlement Intentions

5.1.1 Data and Empirical Strategy

For our investigation, we use microdata from the 2015 mini-census, which report an individual's places of residence by the end of 2010, 2014, and 2015. This allows us to construct an individual's residential history at three time points, and we can infer whether an individual migrated between

these points. For example, consider the following residential history: A, 2010 → B, 2014 → B, 2015. We can infer that the individual must have migrated between 2010–2014 (from A to B) but not between 2014–2015.[26]

Using the data, we estimate how the *hukou* reform affects a migrant's decision to *outmigrate from their prefecture of residence in 2010*. We restrict the sample to individuals who were migrants in 2010. An individual is defined as a migrant if their 2010 prefecture of residence differs from their 2015 *hukou* registration prefecture.[27] Our analysis resembles a survival analysis, where the event of interest is outmigration from the 2010 prefecture of residence. Specifically, we estimate the following linear probability model (LPM):

$$\Pr\left(Outmigration_{jit} \mid \mathbf{W}_{jit}\right) = \rho\left(Reform_i \times Post15_t\right) + \lambda_i + \mu_t + f\left(\tilde{p}_i; \zeta_{Reform,t}\right),\ t \in \{2014, 2015\}. \quad (5)$$

In Equation 5, the dependent variable, $Outmigration_{jit}$, is a dummy variable that equals one if individual j has left their 2010 prefecture of residence i by year t. Given the structure of our data, we observe this outcome for 2014 and 2015, and we estimate Equation 5 using these two periods. $\mathbf{W}_{jit}$ collects explanatory variables on the right-hand side. $Reform_i$ is the reform status of prefecture i. $Post15_t$ is a dummy variable that equals one for 2015 and zero for 2014. We therefore treat 2015 as the post-reform period and 2014 as the pre-reform period.[28] We also include prefecture and year fixed effects as well as the polynomial function. In the spirit of survival analysis, once an individual leaves their 2010 prefecture of residence, we drop the individual's subsequent observations to focus on initial outmigration.

The coefficient of interest is ρ. It is estimated by comparing trends in outmigration rates among not-yet-migrated individuals between reform and non-reform prefectures. Thus, it captures the impact of the *hukou* reform in the 2010 prefecture of residence on a migrant's outmigration decision. We expect $\rho < 0$, as the *hukou* reform should make it easier for migrants to remain in their initial migration destinations.

5.1.2 Results

Table 3 reports estimates of Equation 5. Columns (1)–(3) include all migrants in the estimation sample. Column (1) uses a specification that includes only prefecture and year fixed effects and polynomials of urban population. Column (2) further includes fixed effects for migrants' origins,

[26]The data do not allow us to identify circular migration. For instance, an individual recorded as living in A in both 2010 and 2014 may have moved temporarily to B in 2012. This movement would not be observed. However, because circular migration should reflect continued attachment to the original location, migration captured in the data likely represents a more *permanent* departure. This focus aligns with our primary interest in investigating how the *hukou* reform influences migrants' pursuit of permanent settlement.

[27]We use this definition because while we observe an individual's residential history, we do not observe the *hukou* registration history. The definition could introduce potential measurement error if an individual changed *hukou* registration between 2010 and 2015. We discuss and address potential measurement error in Section 5.1.2.

[28]Because the census only reports the residential prefecture by the end of 2014, we can only define outmigration during 2010–2014 but not the exact timing of outmigration. However, for individuals who moved during 2010–2014, most should have moved before the *hukou* reform took effect in July 2014.

allowing us to compare migrants exposed to the same origin shocks but different reform status in their destinations. In Column (3), we additionally control for interactions between individual covariates and year indicators to account for differential integration dynamics associated with migrant characteristics. The covariates include birth cohort, gender, educational attainment, and employment status. Across specifications, consistent with our hypothesis, all estimates of ρ are similar and negative. For example, Column (3) implies that the *hukou* reform reduces the probability of outmigration from the initial destination by 5.7 percentage points, which amounts to 42.9 percent of the average outmigration rate in non-reform prefectures.

Columns (4) and (5) examine the effect separately for rural and urban migrants, defined based on the *hukou* category in 2015. The effect is more pronounced for rural migrants: they were 7.1 percentage points less likely to leave the destination after the reform. One explanation is that rural migrants face greater barriers to integration due to limited wealth, making them more responsive to the new opportunity brought by the *hukou* reform. Indeed, in our sample, rural migrants are twice as likely as urban migrants to leave their initial destinations.

Table 3. The Effect of the *Hukou* Reform on Outmigration

	Outmigration from 2010 residing prefecture				
	(1)	(2)	(3)	(4)	(5)
Reform × Post [ρ]	-0.057**	-0.056**	-0.057**	-0.071**	-0.019
	(0.026)	(0.026)	(0.027)	(0.034)	(0.016)
Sample	All migrants	All migrants	All migrants	Rural migrants	Urban migrants
Control mean	0.133	0.133	0.133	0.165	0.089
Prefecture FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes	Yes	Yes
Origin FE		Yes	Yes	Yes	Yes
Individual covariates × Year FE			Yes	Yes	Yes
Observations	100,875	100,875	100,875	58,647	42,228
R^2	0.048	0.087	0.095	0.146	0.066

Note: This table reports the effect of *hukou* reform on outmigration. We estimate Equation 5. Individual covariates include birth cohorts, gender, educational attainment, and employment status. Columns (1)–(3) include all migrants in the estimation sample, while Columns (3) and (4) include only rural and urban migrants, respectively. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

We perform a battery of robustness checks for these results. First, the results are similar when using a Cox proportional hazards model, a common method in survival analysis (see Appendix I.1). Second, we address potential measurement error in our migrant definition. Recall that we define migrants as individuals whose 2010 prefecture of residence differs from their *hukou* registration prefecture reported in 2015. This may misclassify some individuals as migrants: natives who later migrated *and* transferred their *hukou* would appear to be migrants in 2010. These misclassified individuals would mechanically exhibit higher outmigration rates from their 2010 prefectures of residence. However, our results are not driven by measurement error. Measurement error should be minimal among rural migrants, because urban-to-rural *hukou* transfers are very rare and even prohibited (Song, 2014). Table 3 shows a strong reform effect among rural migrants, suggesting that

measurement error is not a primary driver of our results. In addition, results are similar when we exclude potentially misclassified migrants (see Appendix I.2 for details). The estimated effects are in fact larger than those in Table 3, albeit with larger standard errors due to smaller sample sizes. The effects are again driven by rural migrants, indicating that the heterogeneous effects between rural and urban migrants are not due to differential measurement error between the two groups.

In conclusion, we find that the *hukou* reform significantly reduced outmigration rates among migrants, even after conditioning on origin fixed effects and individual characteristics. The effects are particularly pronounced among rural migrants who otherwise have a low chance of integration. Because rural migrants have not yet made *hukou* transfers, we interpret their decisions not to outmigrate as increases in settlement intentions. Next, we use rural migrants' outmigration rates to measure settlement intentions and investigate how they help explain the *hukou* reform's impact on labor unrest.

5.2 Linking Settlement Intentions to Labor Unrest

Thus far, we have shown that the *hukou* reform increases settlement intentions. One natural question is: how much of the *hukou* reform's effect on labor unrest can be attributed to increased settlement intentions? To answer this question, we need to conduct a *causal mediation analysis*, which decomposes the causal effect of the *hukou* reform on labor unrest into two parts: (i) the effect that goes through enhancing settlement intentions, i.e., the *indirect effect*, and (ii) the effect that goes through other mechanisms, that is, the *direct effect*. Part (i) tells us the explanatory power of the settlement intentions mechanism.

However, this task is challenging. In Section 5.2.1, we first briefly discuss issues with the conventional approach and then propose improvements. We focus on intuitions, leaving full technical details in Appendix J. Then, in Section 5.2.2, we illustrate how we apply our method to estimate the extent to which settlement intentions mediate the *hukou* reform's impact on labor unrest. Results are reported in Sections 5.2.3 and 5.2.4.

5.2.1 Methodology

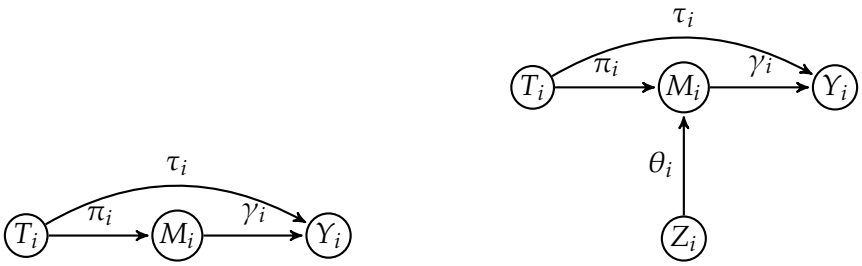
Setup and Conventional Approach. We consider a more general setup. Let i denote the unit of observation. Let T_i denote the quasi-exogenous treatment (e.g., the *hukou* reform), M_i denote the mechanism of interest (e.g., the decrease in outmigration rates), and Y_i denote the outcome of interest (e.g., the decrease in labor unrest). Figure 4A presents a directed acyclic graph (DAG) for the relationship among these variables.[29] A unit increase in treatment T_i affects outcome Y_i in two ways:

1. the *indirect effect* $\gamma_i \pi_i$, where π_i is the change in M_i when T_i increases by one unit, and γ_i is the marginal effect of M_i on Y_i; and

[29] We use the DAG here for the ease of presentation. In Appendix J, we provide technical details using the potential outcome framework, which is equivalent to the DAG framework (Imbens, 2020).

2. the *direct effect* τ_i.

Hence, $\beta_i \equiv \tau_i + \gamma_i \pi_i$ is the *total effect*. Note that parameters $(\tau_i, \pi_i, \gamma_i)$, as the subscripts indicate, may vary across i, which extends the homogeneous effect framework in Imai et al. (2011).



A. Conventional Approach **B.** IV-Augmented Approach

Figure 4. Directed Acyclic Graphs for Causal Mediation Analysis

A researcher typically estimates the average total effect (ATotE): $ATotE = E(\beta_i) = E(\tau_i) + E(\gamma_i \pi_i)$. $E(\tau_i)$ is the average direct effect (ADE), and $E(\gamma_i \pi_i)$ is the average indirect effect (AIE). A researcher may want to estimate AIE or $\frac{\text{AIE}}{\text{ATotE}}$, which reveals how important a particular mechanism is in explaining the total effect. Conventionally, this is done by estimating the following two regression models:

$$Y_i = \alpha_1 + \beta T_i + e_{i1}, \tag{6}$$

$$Y_i = \alpha_2 + \tau T_i + \gamma M_i + e_{i2} \tag{7}$$

For illustration, also consider a third regression model:

$$M_i = \alpha_3 + \pi T_i + e_{i3}. \tag{8}$$

Let $\hat{\kappa}$ denote the OLS estimand of κ. Since we assume that T_i is exogenous, $\hat{\beta}$ from Equation 6 captures the average total effect (ATotE). In Equation 7, intuitively, as M_i is controlled for, $\hat{\tau}$ should capture the average direct effect (ADE), i.e., the effect *not* mediated by M_i. Then, the change in coefficients on T_i,

$$\widehat{AIE} \equiv \hat{\beta} - \hat{\tau} = \hat{\gamma}\hat{\pi}, \tag{9}$$

is interpreted as an estimand for the average indirect effect (AIE), i.e., the effect mediated by M_i. The second equality follows from properties of least squares.

Despite the simplicity, $\widehat{AIE}$ can be biased for AIE (see Appendix J.1 for a full proof). To illustrate, note that $\hat{\gamma}\hat{\pi}$ is the product of (i) $\hat{\pi}$, the estimated effect of T_i on M_i; and (ii) $\hat{\gamma}$, the estimated effect of M_i on Y_i. With exogeneity of T_i, $\hat{\pi}$ identifies a causal effect of T_i on M_i. However, $\hat{\gamma}$ typically does not identify a causal effect of M_i on Y_i because M_i is endogenous. For instance, in our setting, migrants' settlement intentions (as measured by outmigration rates) may be associated with local economic and political conditions, which are also conducive to labor unrest. In addition,

even if $\hat{\gamma}$ is causal, there can be bias resulting from effect heterogeneity. To see this, note that in general, $AIE = E(\gamma_i \pi_i) \neq E(\gamma_i)E(\pi_i)$, unless γ_i is homogeneous. Intuitively, homogeneity in γ_i avoids an ecological bias: the average relationship ($E(\gamma_i)E(\pi_i)$) may not mimic individual relationships ($E(\gamma_i \pi_i)$) (Glynn, 2012).[30] Taken together, for the conventional approach to identify the AIE, two strong assumptions are necessary: (i) exogeneity of the mechanism variable M_i and (ii) homogeneity of the effects of M_i on Y_i.

Proposed Approach. The literature has mainly tackled assumption (i), while keeping assumption (ii) (Dippel et al., 2022; Dix-Carneiro et al., 2018; Bai and Kung, 2015). To address assumption (i), we follow Imai et al. (2011) and Bai and Kung (2015) to use an instrumental variable (IV) for M_i, denoted by Z_i. Therefore, we refer to this approach as an "IV-augmented approach." Figure 4B visualizes the relationship between variables in our proposed approach. Note that instrument Z_i affects outcome Y_i only through mechanism M_i, reflecting the exclusion restriction. In Appendix J.2, we show that under the general setup and in an extension to RD designs, if assumption (ii) holds, one can estimate Equation 6 by OLS and Equation 7 by two-stage least squares using Z_i as an IV for M_i, and

$$\widehat{AIE} = \hat{\beta} - \hat{\tau} = \hat{\gamma}_{\text{IV}} \hat{\pi} \tag{10}$$

identifies the average indirect effect (AIE). In other words, the mediation analysis can still be done by simply running two regressions, *if* assumption (ii) holds.

Bounding Exercise. Our main contribution here is to further relax the homogeneity assumption (ii) (see full technical details in Appendix J.3). In a nutshell, we offer a simple exercise to bound AIE. Using the bounding exercise, a researcher can also obtain an estimate of AIE for a given degree of effect heterogeneity they are comfortable with.

Let us provide the main intuition of this exercise. For simplicity, we allow γ_i to be heterogeneous, and instead assume that γ_i is independent of π_i. In Appendix J.3, we present a general framework without this independence assumption, and the same intuition discussed below holds. When γ_i is heterogeneous, it is well known that the IV estimand $\hat{\gamma}_{\text{IV}}$ only identifies a *local* average effect of M_i on Y_i within compliers (Imbens and Angrist, 1994), which may differ from the population average effect $E(\gamma_i)$.[31] This introduces a bias when using $\widehat{AIE}$ to estimate AIE:

$$\widehat{AIE} = \hat{\gamma}_{\text{IV}} \hat{\pi} = \underbrace{E(\gamma_i \pi_i)}_{\text{AIE}} + \underbrace{\left[\hat{\gamma}_{\text{IV}} - E(\gamma_i)\right] E(\pi_i)}_{\text{IV bias}}, \tag{11}$$

[30] This happens if γ_i is systematically correlated with π_i. For instance, prefectures where settlement intentions increase more after the reform also have larger effects of settlement intentions on labor unrest. Formally, $AIE = E(\gamma_i \pi_i) = E(\gamma_i)E(\pi_i) + Cov(\gamma_i, \pi_i)$, where the covariance represents the bias.

[31] This explains why assumption (ii) is useful. If γ_i is constant, then $\hat{\gamma}_{\text{IV}} = E(\gamma_i)$. A researcher can extrapolate the local average effect identified in the complier group to the broader population and then evaluate the treatment effect mediated by a mechanism.

where the second term represents the bias.[32] We refer to it as the "IV bias." The bias is particularly salient in the presence of *sorting*, i.e., when compliers have systematically different γ_i. For instance, this can happen in our case if migrants' settlement intentions have much larger effects on labor unrest in prefectures more responsive to the IV we use.

In Appendix J.3, we show that under moderate distributional assumptions, *AIE* can be estimated by

$$\widehat{\widehat{AIE}} = \hat{\hat{\gamma}}\left(\hat{\gamma}_{\text{IV}}, \rho_{\phi\gamma}\right)\hat{\pi}. \tag{12}$$

In this expression, $\hat{\hat{\gamma}}\left(\hat{\gamma}_{\text{IV}}, \rho_{\phi\gamma}\right)$ is an estimand for the population average $E(\gamma_i)$, which is a function of the IV estimand $\hat{\gamma}_{\text{IV}}$ and a measure for compliers' *sorting* that leads to IV bias, $\rho_{\phi\gamma}$. A larger $\rho_{\phi\gamma}$ means compliers tend to have higher γ_i. $\rho_{\phi\gamma}$ is equivalent to the homogeneity assumption. $\hat{\hat{\gamma}}\left(\hat{\gamma}_{\text{IV}}, \rho_{\phi\gamma}\right)$ is monotonic in $\rho_{\phi\gamma}$. Importantly, $\rho_{\phi\gamma}$ is a correlation coefficient and thus bounded. Therefore, a researcher can

1. vary $\rho_{\phi\gamma}$ arbitrarily to obtain the lowest $\widehat{\widehat{AIE}}$ as the lower bound of *AIE*; or
2. pick an acceptable $\rho_{\phi\gamma}$ and calculate $\widehat{\widehat{AIE}}$ to estimate *AIE*.

5.2.2 Empirical Strategy

Estimating Equation. To implement our approach, we convert the panel regression model in the previous analysis to its cross-sectional equivalent. We estimate the following equations in parallel to Equations 6 and 7:

$$\frac{\Delta Unrest_i}{L_{i,2010}} = \alpha_1 + \beta Reform_i + f\left(\tilde{p}_i; \zeta_{Reform}\right) + e_{i1}, \tag{13}$$

$$\frac{\Delta Unrest_i}{L_{i,2010}} = \alpha_2 + \tau Reform_i + \gamma \Delta Outmigration_i + f\left(\tilde{p}_i; \zeta_{Reform}\right) + e_{i2}. \tag{14}$$

In both equations, the dependent variable, $\frac{\Delta Unrest_i}{L_{i,2010}} = \left(\Sigma_{t=2014}^{2015} \frac{Unrest_{it}}{L_{i,2010}}\right)/2 - \left(\Sigma_{t=2011}^{2013} \frac{Unrest_{it}}{L_{i,2010}}\right)/3$, is the change in the average annual unrest rate between the pre-reform period 2011–2013 and the post-reform period 2014–2015. We focus on 2011–2015 because it aligns with the period for which we can measure outmigration rates using census data. The estimated β from Equation 13 would be numerically identical to the baseline estimate reported in Column (1) of Table 2. The mechanism variable in Equation 14, $\Delta Outmigration_i$, measures the change in the annual outmigration rate between the pre- and post-reform periods:

$$\Delta Outmigration_i = Outmigration_{i,2014\text{–}2015} - Outmigration_{i,2010\text{–}2014}, \tag{15}$$

[32] If we do not assume that γ_i and π_i are independent, then $\widehat{AIE} = \hat{\gamma}_{\text{IV}}\hat{\pi} = E(\gamma_i\pi_i) + [\hat{\gamma}_{\text{IV}} - E(\gamma_i)]\,E(\pi_i) - Cov(\gamma_i, \pi_i)$, where the third term is ecological bias as in the conventional approach if homogeneity is not imposed (see footnote 30). In Appendix J.3, we consider and bound both IV and ecological biases.

where $Outmigration_{ip}$ denotes the annualized outmigration rate in prefecture i during period p.[33] We calculate these rates among rural migrants, who are more affected by the *hukou* reform (cf. Table 3).

Instrumental Variable. We instrument for $\Delta Outmigration_i$ using plausibly exogenous trade shocks in migrants' *origins*. Trade shocks in origins are not related to economic and political factors in destinations, and as a pull factor, positive (negative) shocks can attract (discourage) migrants to return. In our migrant sample, 88 percent of migrants who eventually left their 2010 prefectures of residence returned to their home prefectures, suggesting that the origin represents an important outside option when migrants decide whether to remain in their destinations.

We construct the IV in two steps. First, we measure the trade shock to a prefecture h in a shift-share (Bartik) fashion, following Campante et al. (2023):

$$TradeShock_{ht} = \sum_k \underbrace{\frac{X^{CN}_{hk,2010}}{L_{h,2000}}}_{\text{share}} \times \underbrace{\frac{\Delta X^{ROW}_{kt}}{X^{CN}_{k,2010}}}_{\text{shift}}. \tag{16}$$

$X^{CN}_{hk,2010}$ is prefecture h's exports in industry k (6-digit Harmonized System level); $X^{CN}_{k,2010} = \sum_h X^{CN}_{hk,2010}$ is the national aggregate exports in industry k; $\Delta X^{ROW}_{kt} = X^{ROW}_{kt} - X^{ROW}_{k,t-1}$ denotes the annual change in exports *within* the rest of the world (less China; ROW in short) in year t; and $L_{h,2000}$ denotes working-age population (ages 25–54) in 2000.[34] The "shift" component, $\frac{\Delta X^{ROW}_{kt}}{X^{CN}_{k,2010}}$, is the change in global demand for product k in year t, scaled by the size of industry k in 2010. These industry-level shocks are apportioned across prefectures according to the "share" term that captures industrial specialization, $\frac{X^{CN}_{hk,2010}}{L_{h,2000}}$, the per-worker export value of industry k in prefecture h. Thus, $TradeShock_{ht}$ can be interpreted as an average worker's gains or losses due to global trade fluctuations.

To be an exogenous pull factor, $TradeShock_{ht}$ needs to be (i) orthogonal to other determinants of return migration and (ii) predictive of return migration. Regarding condition (i), Borusyak et al. (2022) show that the exogeneity of a shift-share variable (i.e., $TradeShock_{ht}$) can be established by the exogeneity of shifts $\frac{\Delta X^{ROW}_{kt}}{X^{CN}_{k,2010}}$. Because the shifts leverage trade shocks in the ROW, they are not likely to be associated with local factors within China. Reassuringly, following the methodology developed by Borusyak et al. (2022), we show that the distribution of industry-level trade shocks is not correlated with local characteristics (see Appendix K.1). Regarding condition (ii), we show that

[33] Recall that in the 2015 census, we observe a migrant's residing prefectures by the end of 2010, 2014, and 2015. $Outmigration_{i,2014–2015}$ is the share of migrants who resided in prefecture i in 2014 but had moved out by 2015. For $Outmigration_{i,2010–2014}$, we first calculate the outmigration rate during the period 2010–2014 and then convert it to an annual term, assuming a constant annual hazard rate.

[34] As in Campante et al. (2023), we consider only manufacturing industries. $X^{CN}_{hk,2010}$ and $X^{CN}_{k,2010}$ are measured using Chinese customs data from 2010. ΔX^{ROW}_{kt} is measured using the UN Comtrade database. All trade variables are in 1,000 US dollars. The working-age population is measured in 2000 to avoid simultaneity bias.

individuals who face favorable trade shocks at home are more likely to migrate out of the destination (see Appendix K.2).

In the second step, we construct Z_i as the average origin trade shock faced by migrants living in prefecture i. To do so, we first measure initial migration patterns in 2010: $s_{h\to i}$ denotes the share of migrants residing in prefecture i who are originally from prefecture h. Then, we interact migration patterns with trade shocks in 2015:

$$Z_i = \sum_h s_{h\to i} \times TradeShock_{h,2015} = \sum_k \underbrace{\left(\sum_h s_{h\to i} s_{hk}\right)}_{\text{share}} \underbrace{\frac{\Delta X^{ROW}_{k,2015}}{X^{CN}_{k,2000}}}_{\text{shift}}. \tag{17}$$

As the second equality shows, Z_i can be written as a shift-share variable: shares are industrial specialization adjusted by migration patterns, and shifts are global demand fluctuations. That said, like $TradeShock_{ht}$, Z_i exploits differential exposure to exogenous trade shocks, and Z_i has a higher value if migrants in prefecture i tend to come from places that specialize in industries with high global demand fluctuations.

For Z_i to be a valid IV, it must satisfy an exclusion restriction: it affects labor unrest only through the outmigration rate. The exclusion restriction is violated if industries with high global demand fluctuations tend to send migrants to prefectures with high labor activism. Following Borusyak et al. (2022), we address this concern by showing that trade shocks in the origins are not associated with destination characteristics (see Appendix K.3). In Section 5.2.3, we perform several robustness checks, which support the IV strategy's validity. Z_i strongly predicts aggregate outmigration rates. Figure 5 shows the first stage: there is a strong positive relationship between Z_i and $\Delta Outmigration_i$ (effective F statistic = 16.126), consistent with the individual-level results (cf. Appendix K.2).

5.2.3 Results: Migrants' Settlement Intentions as a Determinant of Labor Unrest

Main Findings. Table 4 reports the causal mediation analysis, with results visualized in Figure 6. In Column (1), we estimate Equation 13, showing that the reform reduces the annual labor unrest rate by 1.047 events per million working-age population. This estimate captures the total effect.

We now examine how much of this total effect is explained by changes in migrants' settlement intentions, measured by changes in outmigration rates $\Delta Outmigration_i$. We start with the conventional approach by directly adding $\Delta Outmigration_i$ to the regression in Column (1). As Column (2) shows, a higher outmigration rate (i.e., stronger settlement intentions) is associated with lower labor unrest. Comparing Columns (1) and (2), controlling for $\Delta Outmigration_i$ attenuates the coefficient on $Reform_i$ by 9.1 percent, suggesting that migrants' stronger settlement intentions can explain 9.1 percent of the *hukou* reform's total effect on labor unrest. However, as discussed in Section 5.2.1, this interpretation is only valid if (i) $\Delta Outmigration_i$ is exogenous and (ii) it has homogeneous effects on labor unrest.

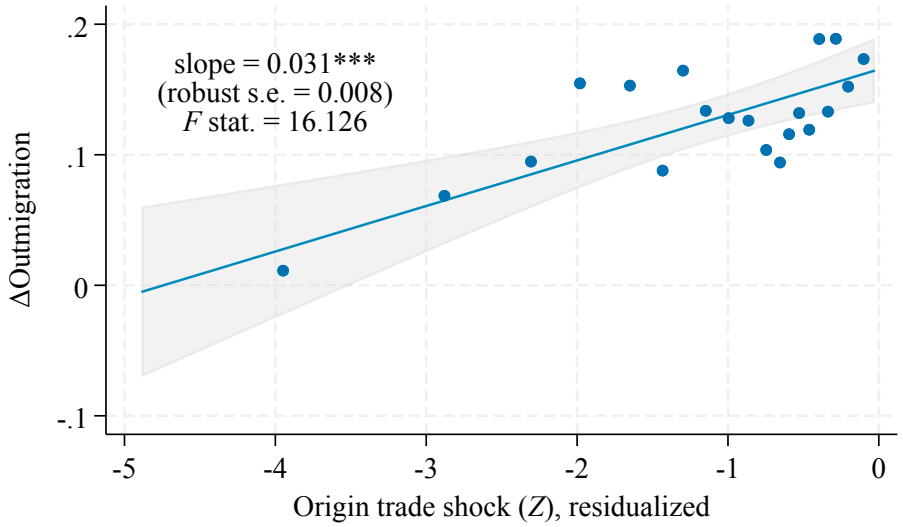


Figure 5. Origin Trade Shocks and Outmigration Rates

Note: This figure plots the change in the outmigration rate ($\Delta Outmigration_i$) against the average trade shock at origins of migrants in a destination (Z_i, partialing out variation attributed to reform status and polynomials of urban population). The binscatter plot uses the data-driven method to estimate conditional means proposed by Cattaneo et al. (2024). The fitted line is created using all observations (rather than just binscatter points). The effective F statistic is computed following Olea and Pflueger (2013).

Table 4. The *Hukou* Reform, Outmigration, and Labor Unrest

	DV: ΔUnrest/L, 2011–2015		
	(1)	(2)	(3)
	Baseline	Mediation-OLS	Mediation-IV
Reform [β or τ]	-1.047**	-0.951*	-0.387
	(0.518)	(0.535)	(0.740)
ΔOutmigration [γ]		1.689**	11.651***
		(0.680)	(4.077)
% Total effect explained		0.091	0.630
Effective F stat.			16.126
Observations	287	287	287

Note: This table reports causal mediation analysis that quantifies the importance of the settlement intentions mechanism, measured by the outmigration rate, during 2011–2015. Column (1) reports the baseline estimate. Column (2) represents the conventional approach. Column (3) uses the IV-augmented approach. The coefficient on *Reform* in Column (1) is denoted by β, while the coefficients on *Reform* in Columns (2) and (3) are denoted by τ. The share of the total effect explained is calculated as $\frac{\beta-\tau}{\beta}$. The effective F statistic is calculated following Olea and Pflueger (2013). Robust standard errors are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

We now implement our proposed approach. As a first pass, we instrument outmigration using migrants' origin trade shocks to relax the exogeneity assumption (i), while maintaining the homogeneity assumption (ii). In Column (3), we find that a low outmigration rate significantly

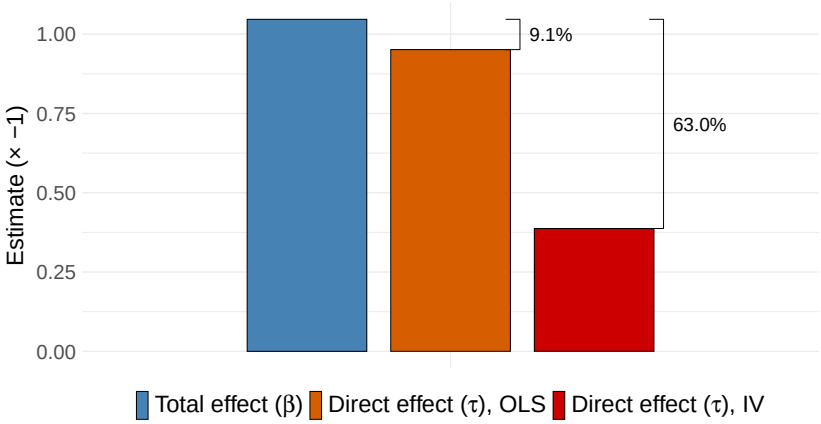


Figure 6. Mediation Visualized

Note: This figure visualizes causal mediation analysis in Table 4. The first bar corresponds to the estimated total effect (β) in Column (1); the second bar corresponds to the estimated direct effect (τ) in Column (2); and the third bar corresponds to the estimated direct effect (τ) in Column (3). All estimates are multiplied by −1 to improve visualization. The percent changes are calculated as $\frac{\beta-\tau}{\beta}$.

reduces labor unrest.[35] Comparing Columns (1) and (3), we can conclude that 63 percent of the total effect of the *hukou* reform on unrest can be attributed to the increase in settlement intentions, assuming that settlement intentions have homogeneous effects on labor unrest.[36]

Homogeneity is nonetheless a strong assumption. We use the bounding exercise proposed in Section 5.2.1 to relax it. For each degree of effect heterogeneity imposed, denoted by $\rho_{\phi\gamma}$, $\widehat{\widehat{AIE}}$ identifies AIE. Figure 7 plots $\frac{\widehat{\widehat{AIE}}}{\hat{\beta}}$, the estimated share of the total effect explained, against $\rho_{\phi\gamma}$. We focus on values of $\rho_{\phi\gamma}$ such that the results in Table 4 *overestimate* the share of the total effect explained by settlement intentions. When $\rho_{\phi\gamma} = 0$, i.e., there is no heterogeneity-induced bias, 63 percent of the total effect can be attributed to migrants' settlement intentions, as reported in Table 4. As $\rho_{\phi\gamma}$ increases and heterogeneity results in larger bias, the share of the total effect attributable to migrants' settlement intentions declines. Nevertheless, in the most extreme case, $\rho_{\phi\gamma} = 1$, the settlement intentions mechanism can still explain 16 percent of the *hukou* reform's effect on labor unrest. We regard this as a nontrivial fraction. This lower bound is likely overly conservative by assuming maximal IV bias. In reality, $\rho_{\phi\gamma}$ may not be that large. If we assume a moderate degree of

[35]Two remarks on the IV estimate are in order. First, the IV estimate remains statistically significant using the robust inference proposed by Lee et al. (2023), which yields a 95 percent confidence interval [4.405, 21.355]. Second, the IV estimate is substantially larger than the OLS estimate. To understand this IV-OLS gap, we apply the decomposition method developed by Ishimaru (2024), which separates the IV-OLS gap into: (i) endogeneity bias purged by IV and (ii) the different weighting schemes used by IV and OLS to aggregate empirical nonlinear effects. The decomposition shows that nearly all of the IV-OLS gap can be attributed to the purged endogeneity bias. This further highlights the importance of addressing endogeneity in causal mediation analysis.

[36]For completeness, we also conduct a similar mediation analysis for the change in labor unrest during 2011–2019. In Table B6, the IV-augmented approach shows that heightened settlement intentions, measured from 2010–2015, can explain 32.2 percent of the decrease in labor unrest from 2011–2019 resulting from the *hukou* reform. This magnitude is notably smaller than that for the 2011–2015 period. The attenuation may be because $\Delta Outmigration_i$ does not fully capture migrants' settlement intentions over the longer term, or $\Delta Outmigration_i$ has zero effect in the longer term as migrants have successfully attained local *hukou*.

heterogeneity, $\rho_{\phi\gamma} = 0.5$, then 26 percent of the total effect can be attributed to migrants' settlement intentions.[37]

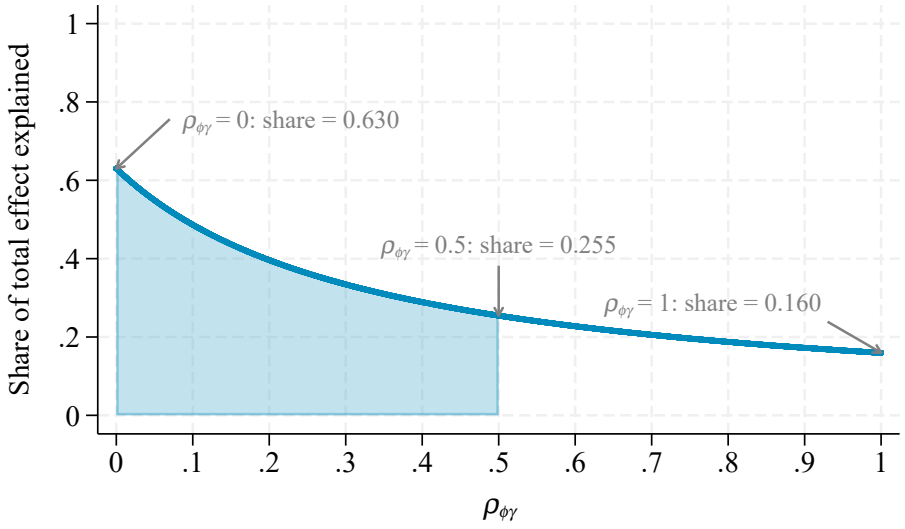


Figure 7. Bounding Exercise for Causal Mediation Analysis

Note: This figure presents the bounding exercise for causal mediation analysis. $\rho_{\phi\gamma}$ is the correlation coefficient measuring the degree of complier sorting. A larger $\rho_{\phi\gamma}$ leads to a larger IV bias, i.e., the larger gap between the IV estimand and the population average. We consider $\rho_{\phi\gamma} \in [0, 1]$ such that the baseline mediation results overestimate the importance of outmigration by imposing $\rho_{\phi\gamma} = 0$.

Robustness Checks. Our analysis rests on the validity of the IV, i.e., origin trade shocks affect labor unrest *only* through their effects on outmigration. We conduct several robustness checks to alleviate concerns about IV validity. We summarize them here and leave details in Appendix K.4. First, one may worry that both origins and destinations are exposed to correlated trade shocks, and origin trade shocks may induce in-migration to the destinations. However, Table K4 and Figure K1 show that the results are robust to controlling for trade shocks and population growth in the destination; if anything, the outmigration channel becomes more pronounced. Second, we address the concern that trade shocks at some origins may have spatial spillover impacts on labor unrest at the destination. We construct the IV Z_i in two alternative ways to mitigate spillovers: (i) we exclude origins within 100km of a destination; (ii) we exclude origins that are regional economic centers, e.g., provincial capitals and four provincial-level municipalities; and (iii) following Borusyak and Hull (2023), we recenter Z_i by subtracting the expected trade shock to tease out potential spillovers and other forms of omitted variables due to endogenous migration networks and industrial specialization. Table K5 and Figure K2 present the results. Reassuringly, they show that settlement intentions are a quantitatively important mechanism. Third, using the methodology developed by Conley et al. (2012), we show that outmigration still has a significant impact on labor unrest even if allowing for violations of the exclusion restriction (see Figure K3). In

[37] In Appendix K.5, we show that when we further relax the independence between γ_i and π_i (see footnote 32), under moderate assumptions, we can still conclude that at least 25 percent of the total effect is explained by migrants' settlement intentions.

sum, the results support the validity of our IV strategy and the conclusion that outmigration is an important mechanism through which the *hukou* reform influences labor unrest.

5.2.4 Additional Evidence: Heterogeneous Effects by Destination Attractiveness

Although the *hukou* reform reduces barriers to migrant integration, migrants' decisions to stay also depend on other destination characteristics, such as similarities to their origins and connections to local communities. Thus, the reform should induce stronger settlement intentions among migrants in destinations that are more attractive along these other dimensions. If settlement intentions are really an important mechanism, we would expect the reform to reduce labor unrest more in destinations where migrants have stronger incentives to stay. In Appendix C, we demonstrate this heterogeneous effect in a simple model (see Proposition C2).

To test this heterogeneity, we use two measures of destination attractiveness. The first measure is migrants' *geographic distance to home*. Migrants may prefer living closer to their origins because of proximity to families, similar cultures, similar climate, etc. Migrants in destinations closer to their origins should exhibit stronger settlement intentions after the reform. Therefore, we measure the average geographic distance to origins of migrants in prefecture i:

$$D_i^M = \sum_h s_{h \to i} D_{hi}, \tag{18}$$

where $s_{h \to i}$ is the share of migrants from prefecture h to prefecture i (measured in 2010), as in Equation 17, and D_{hi} is the distance between prefectures h and i.

Our second measure is migrants' *cultural distance to natives*. Migrants may be more willing to settle in destinations where they face fewer cultural barriers to integration, such as differences in languages, lifestyles, and social norms. We follow Guarnieri (2025) to measure cultural distance using linguistic distance. China has 106 languages and dialects, and even the official language, Mandarin, has 44 variants (Lavely and Berman, 2012). As in Guarnieri (2025), we can calculate the linguistic distance between residents in two prefectures h and i, denoted by LD_{hi} (see Appendix E for construction details). Then, we measure the average linguistic distance between migrants and natives in prefecture i:

$$LD_i^M = \sum_h s_{h \to i} LD_{hi}. \tag{19}$$

For each measure, we divide prefectures into two groups based on whether their geographic or linguistic distance is above or below the sample median, and estimate the effect of the *hukou* reform separately for each group. Figure 8A reports heterogeneous effects by distance to migrants' origins. The reform leads to a larger reduction in labor unrest in prefectures where migrants are closer to their origins. Similarly, Figure 8B shows that the reduction in labor unrest is driven by prefectures where migrants are linguistically similar to natives. Taken together, these findings are consistent with the settlement intentions mechanism: the *hukou* reform has a larger effect on labor unrest in

destinations where migrants have stronger incentives to settle because of geographic proximity to their origins and cultural similarity to local communities.

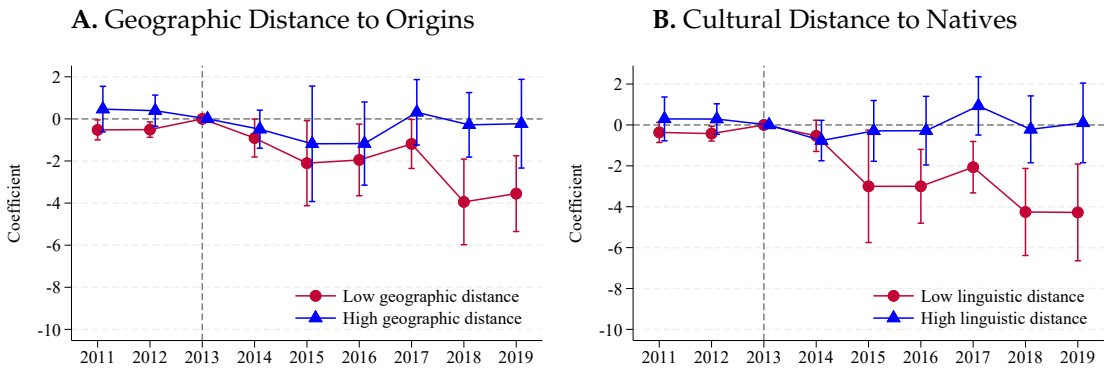


Figure 8. Heterogeneous Effects by Destination Attractiveness

Note: This figure presents the heterogeneous effects of the *hukou* reform on labor unrest by migrants' geographic distance to home or cultural distance to natives. In Panel A, the "low geographic distance migrants" group includes prefectures with a D_i^M (migrants' average distance to origins) below the median, and the "high geographic distance migrants" group includes prefectures with a D_i^M above the median. In Panel B, the "low linguistic distance migrants" group includes prefectures with an LD_i^M (migrants' linguistic distance to natives) below the median, and the "high linguistic distance migrants" group includes prefectures with an LD_i^M above the median. The solid dots are point estimates, and the spikes are 95 percent confidence intervals.

5.3 Other Mechanisms

Compositional Changes. An alternative explanation for our findings is that the *hukou* reform changed the composition of migrants toward those with low propensities to participate in labor unrest, thereby reducing unrest rates. To examine this possibility, we use the China Migrants Dynamic Survey (CMDS), a large, nationally representative survey of migrants. We estimate Equation 2 at the individual level, using migrants' characteristics as dependent variables. Because the CMDS is conducted annually in May, we treat years from 2015 onward as post-reform for these regressions. Table 5 reports the results. We find no statistically significant changes in the characteristics of the overall migrant population or new arrivals (defined as migrants who arrived over the past year), including gender, ethnicity, age, marital status, educational attainment, migration distance, and family unity. These findings suggest that changes in migrant composition cannot explain the observed decrease in labor unrest.

Migrants' Well-Being. If the *hukou* reform expanded migrants' access to benefits or improved labor practices, it could reduce workers' discontent and, consequently, labor unrest. We examine these possibilities in Table 6. We find no discernible effects on migrants' workforce participation,

Table 5. The *Hukou* Reform and Migrant Characteristics

	(1) Female	(2) Han ethnic	(3) Age below 35	(4) Married	(5) High school completion	(6) Cross-province migrant	(7) Co-residence w. spouse	(8) Co-residence w. child
Panel A: All migrants								
Reform × Post	0.006	-0.003	0.011	0.005	-0.005	-0.000	-0.020*	-0.006
	(0.013)	(0.004)	(0.010)	(0.010)	(0.014)	(0.011)	(0.010)	(0.015)
Control mean	0.474	0.953	0.536	0.876	0.389	0.571	0.885	0.654
Sample period	2011–2018	2011–2018	2011–2018	2011–2018	2011–2018	2011–2018	2011–2018	2011–2018
No. prefectures	255	255	255	255	255	255	255	255
Observations	990,912	990,912	990,912	990,912	990,912	990,912	867,392	661,846
Panel B: New arrivals								
Reform × Post	-0.001	0.005	0.007	0.025	-0.016	0.004	-0.010	-0.008
	(0.016)	(0.005)	(0.015)	(0.016)	(0.016)	(0.019)	(0.015)	(0.019)
Control mean	0.459	0.947	0.639	0.809	0.395	0.524	0.833	0.537
Sample period	2011–2018	2011–2018	2011–2018	2011–2018	2011–2018	2011–2018	2011–2018	2011–2018
No. prefectures	255	255	255	255	255	255	255	255
Observations	238,161	238,161	238,161	238,161	238,161	238,161	192,728	146,772

Note: This table presents the effects of *hukou* reform on migrants' characteristics. All regressions control for prefecture and year fixed effects and a polynomial function of centered log urban population. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

wages, and access to social security.[38] Taken together, these results suggest that the decrease in labor unrest is unlikely to be driven by improvements in migrants' well-being.

Table 6. The *Hukou* Reform and Migrants' Well-Being

	(1) Working	(2) Log wage	(3) Access to social security
Reform × Post	-0.003	0.003	0.006
	(0.007)	(0.019)	(0.029)
Control mean	0.883	8.153	0.522
Sample period	2011–18	2011–18	2011, 13, 16
No. prefectures	255	255	255
Observations	990,912	810,696	162,239

Note: This table presents the effects of the *hukou* reform on migrants' well-being. The dependent variables are an indicator for working, log wages, and an indicator of having access to social security. All regressions control for prefecture and year fixed effects and a polynomial function of centered log urban population. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

Governments' Social Control. The *hukou* reform may reduce labor unrest if it induces local officials to tighten social control out of the *perceived* risk of social instability associated with the reform (e.g., population inflows). In Table 7, we examine this possibility using several measures of local governments' social control. Columns (1) and (2) show no statistically significant effects of the

[38] Access to social security is only available from the CMDS in 2011, 2013, and 2016. A recent paper by An et al. (2024) find significant negative effects of the *hukou* reform on wages and access to social security. However, their definition of reform status and empirical strategy (DiD) are different from ours. In Appendix L, we replicate their results and compare them under a series of different empirical decisions. We find that their results are due to the use of a DiD strategy that simply compares more populous prefectures with less populous prefectures. In Section 4.1, we show why such a comparison may not be reliable.

hukou reform on local governments' expenditures on public security (police) and social security, suggesting that the reform did not induce more financial investments in social control.

Alternatively, local governments may prioritize stability maintenance by reallocating bureaucratic resources rather than increasing expenditures. To examine this possibility, we measure the share of stability-related keywords in the following year's government work report (GWR). GWRs are comprehensive policy documents presented annually by local government heads to local People's Congresses, summarizing accomplishments from the previous year and outlining policy priorities for the coming year.[39] However, Column (3) shows no discernible shift in policy priorities toward stability maintenance following the reform.

Finally, if the decrease in labor unrest were driven by tighter social control, we would expect to observe a higher share of labor unrest events repressed by the government, even as the overall number of unrest events declined. Using information on government responses in the CLB data, we calculate the share of unrest events known to have been repressed by the government, through worker detention or police intervention.[40] Column (4) shows no statistically significant effect of the *hukou* reform on this share.

In sum, these results collectively suggest that tighter social control is not a primary mechanism underlying the decrease in labor unrest.

Table 7. The *Hukou* Reform and Social Control

	(1) Log expenditure on public security	(2) Log expenditure on social security	(3) Share of stability related keywords	(4) Share of unrest events known to be repressed
Reform × Post	-0.000	0.058	0.011	0.088
	(0.035)	(0.050)	(0.009)	(0.077)
Control mean	12.908	13.316	0.086	0.228
Sample period	2007–2017	2007–2017	2010–2015	2011–2019
No. prefectures	287	287	287	285
Observations	3,064	3,091	1,676	1,806

Note: This table presents the effects of *hukou* reform on social control. Dependent variables are: log expenditure on public security, log expenditure on social security, the share of stability-related mentions in next year's government work report, and the share of unrest events repressed. All regressions control for prefecture and year fixed effects and a polynomial function of centered log urban population. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

5.4 Discussion

In this section, we highlight that migrants' stronger settlement intentions reduce participation in labor unrest because they fear such actions may undermine their chances of permanent settlement.

[39] GWR texts have been used in the literature to measure Chinese officials' policy priorities (Jiang, 2018; Jiang et al., 2019; Jiang and Zhang, 2020).

[40] This measure is missing for prefecture-years with no reported unrest events.

Our causal mediation analysis shows that under moderate assumptions, changes in migrants' settlement intentions can explain 26–63 percent of the *hukou* reform's total effect on labor unrest.

It is worth considering the group and time horizon for which this mechanism plays out. Settlement intentions are only relevant for migrants who do want to settle permanently in their destinations but have not yet transferred their *hukou*. Therefore, the mechanism may be particularly significant in the short term. Because intentions can change quickly, this helps explain the immediate drop in labor unrest when the reform was enacted (cf. Figure 2).

Meanwhile, we notice that the *hukou* reform has a persistent effect of reducing labor unrest, suggesting that the reform may have induced more lasting changes in the dynamics of labor unrest. Our results show that such lasting changes are unrelated to migrant composition, migrants' well-being, or governments' social control. This suggests that the *hukou* reform may affect other deeper, more persistent determinants of labor unrest. One potential channel is the disruption of social networks that facilitate collective action. By strengthening some migrants' intentions to settle, the reform may weaken the social ties and coordination that sustain labor unrest. However, without micro-level information on the organization of labor unrest, we cannot directly test this hypothesis, and we leave it to future research.

6 Conclusions

As migration becomes a major challenge for many countries, public discourse often raises whether and how migrants should be allowed to stay so as to maintain social stability and cohesion. This makes it increasingly important to study and understand the social consequences of integration policy (e.g., regularization, permanent residency, and citizenship), a key part of the migration regime.

One perspective views integration as a "crown" awarded to migrants with desired characteristics, emphasizing the screening role of strict integration policies. However, the other perspective suggests that integration can be a "catalyst" by setting the right incentives for migrants to take desired actions; thus, integration policy can be inclusive for at least existing migrants. Our paper adds evidence to the catalyst view. In the context of China's restrictive internal migration regime, we find that offering migrants a pathway toward integration strengthens social stability, as reflected by a decrease in migrant labor unrest. We show quantitatively that 26–63 percent of the effect is because migrants who wish to settle have weak incentives to engage in activities that may jeopardize their chances of settlement. This logic readily extends to other informal activities, such as crime.

Our results speak to migrants' incentives when they see an attainable opportunity for integration. It is beyond the scope of our paper to be conclusive about how migrants behave after they establish permanent settlement and how local residents accept newly integrated migrants. To the extent that attachment to the host society sustains incentives for integration, we expect the stability effects to

persist. Nonetheless, understanding these issues merits further investigation in future research on migrant integration and its sociopolitical impacts.

Although we draw on the specific Chinese context, our results are still relevant for thinking about integration policies in broader contexts. It is well documented that for migrants facing high barriers to integration into host societies, the significant possibility of leaving adds costs to migrants' integration efforts along economic, cultural, and social dimensions (Galor and Stark, 1990; Dustmann, 2008, 1999; Adda et al., 2022). In this regard, restrictive integration policies can be counterproductive in achieving the goal of addressing clashes between migrants and host societies. Taken as a whole, our findings underscore the need to carefully consider migrants' endogenous integration actions when designing integration policies.

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A Additional Figures

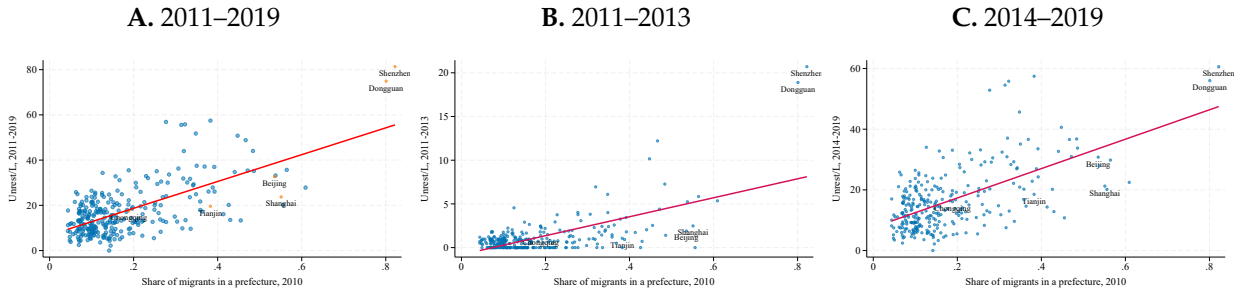


Figure A1. Migrant Share and Labor Unrest

Note: This figure depicts the relationship between the migrant share and the labor unrest rate. The migrant share is measured using the 2010 population census. It is defined as the share of the population whose *hukou* is registered outside the prefecture of residence. The labor unrest rate is measured using data from the China Labor Bulletin, defined as unrest events per million working-age population. Panel A covers 2011–2019, Panel B covers the pre-reform period 2011–2013, and Panel C covers the post-reform period 2014–2019.

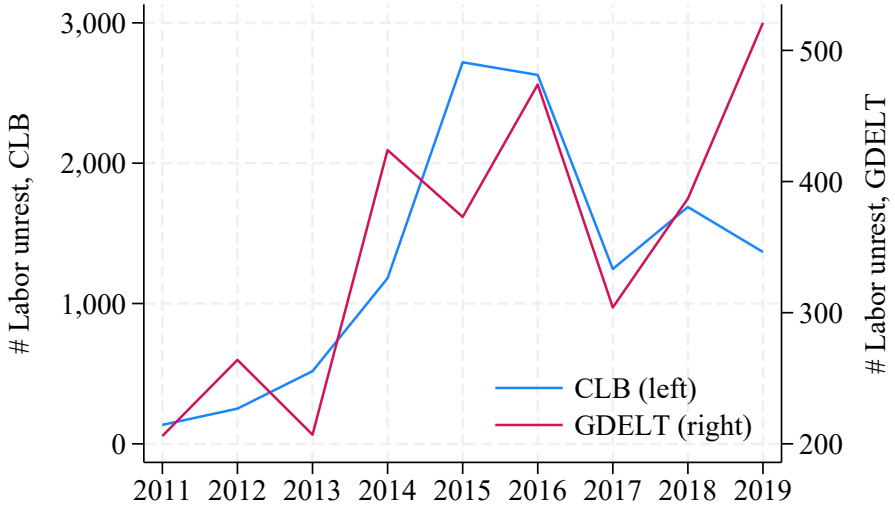


Figure A2. Labor Unrest in CLB and GDELT

Note: This figure shows national trends in labor unrest events recorded in CLB and GDELT. In GDELT, an event is defined as a labor unrest event if it is classified as "Protest" and involves labor.

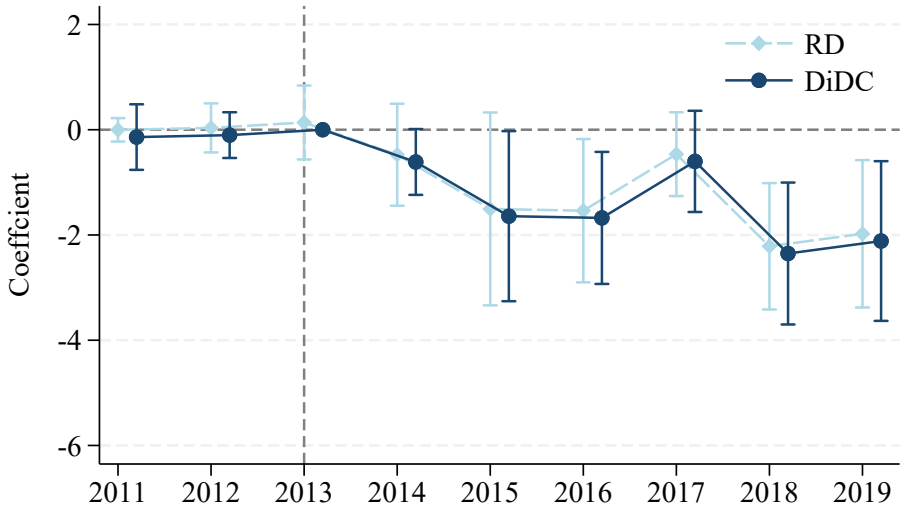


Figure A3. Effects of the *Hukou* Reform on Labor Unrest: RD Estimates

Note: To obtain RD estimates, for each period t, we estimate: $\frac{Unrest_{it}}{L_{i,2010}} = \alpha_t + \beta_t Reform_i + \zeta_{Reform,t}\tilde{p}_i + \nu_{it}$. The coefficients of interest are β_t's, which capture the differences in occurrences of labor unrest between reform and non-reform prefectures. In the figure, the solid points are point estimates of β_t's, and the spikes are the 95 percent confidence intervals.

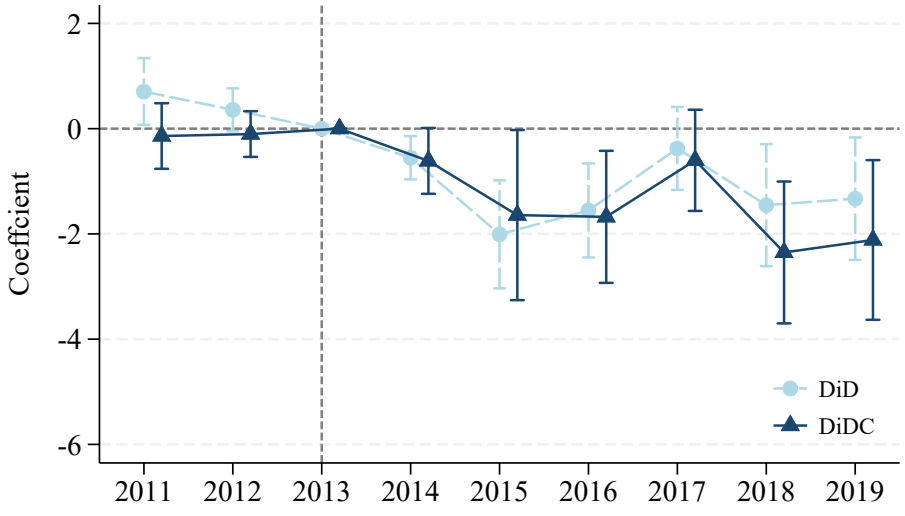


Figure A4. Effects of the *Hukou* Reform on Labor Unrest: DiD vs. DiDC Estimates

Note: This figure presents event-study estimates of DiD and DiDC designs. DiD is implemented by Equation 1. DiDC is implemented by Equation 2. In the figure, the solid points are point estimates, and the spikes are 95 percent confidence intervals.

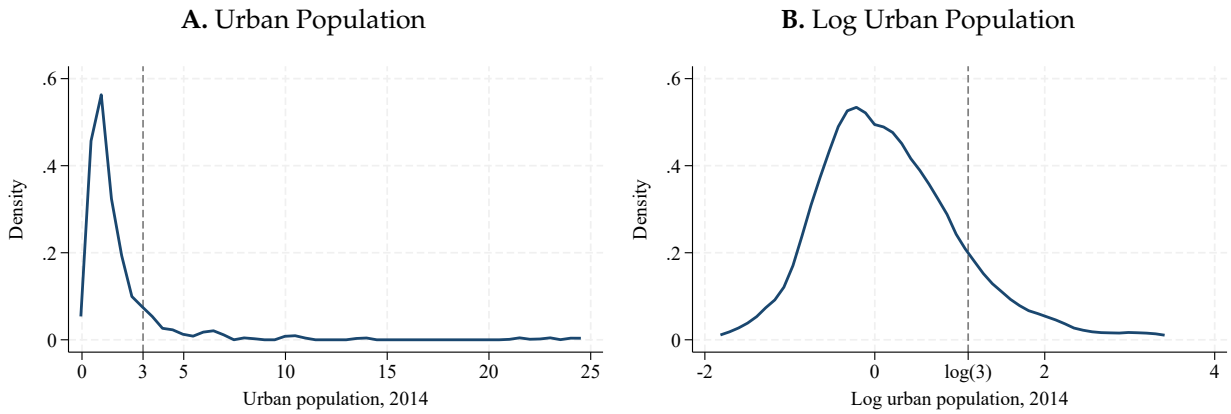


Figure A5. Distribution of Urban Population

Note: This figure depicts the distributions of urban population in level and in log. The vertical lines denote the reform cutoffs of 3 and log(3), respectively.

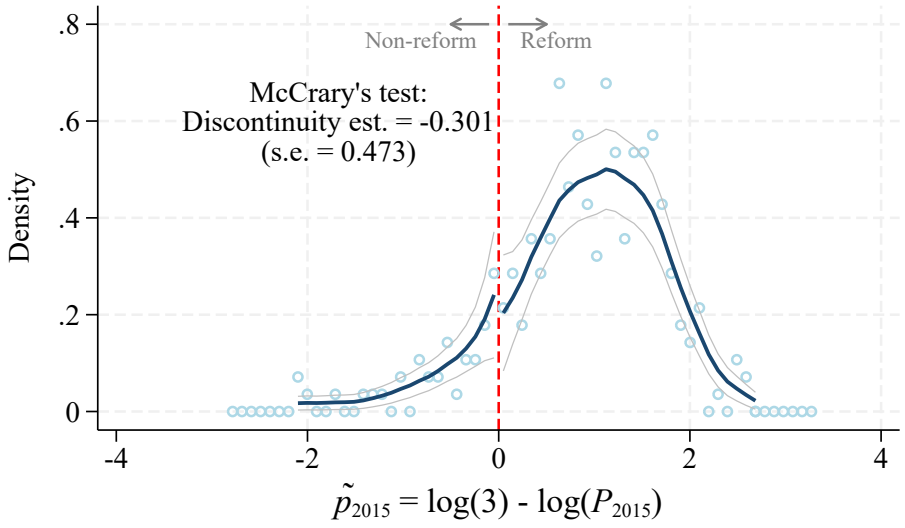


Figure A6. Density of Centered Log Urban Population in 2015

Note: This figure depicts the density of $\tilde{p}_{i,2015} = \log(3) - \log(P_{i,2015})$. We report McCrary's test for density discontinuity at $\tilde{p}_{i,2015} = 0$.

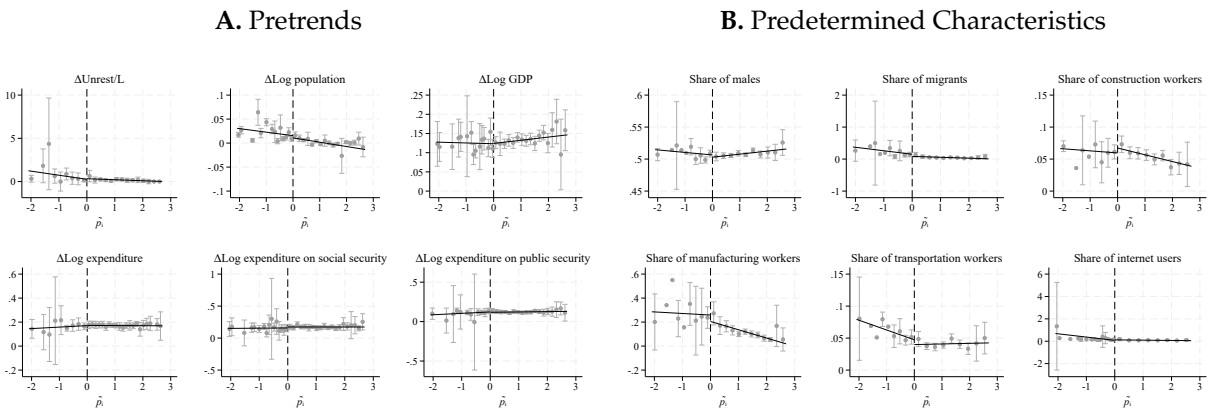


Figure A7. RD Plots of Pre-Reform Covariates

Notes: This figure visually presents balance tests of pre-reform covariates. Panel A examines pretrends in the labor unrest rate and variables conducive to labor unrest. Panel B examines a set of predetermined characteristics. RD estimates are reported in Table 1.

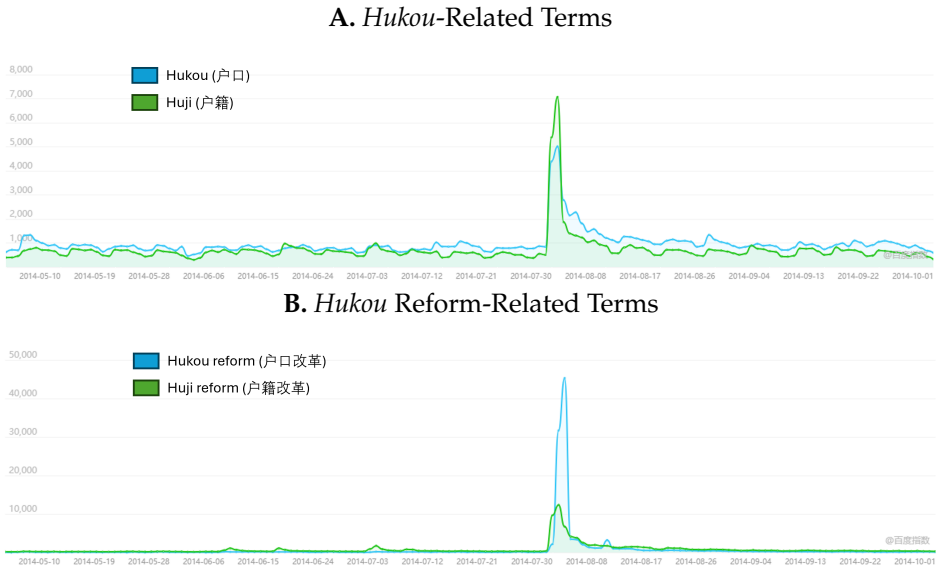


Figure A8. Searches for *Hukou*- and *Hukou*-Reform-Related Terms

Notes: This figure presents search volumes on Baidu for *hukou*-related terms. Panel A shows searches for "*hukou*" (户口) and its synonym, "*huji*" (户籍). Panel B shows searches for terms related to policies: "*hukou* reform" (户口改革) and "*huji* reform" (户籍改革).

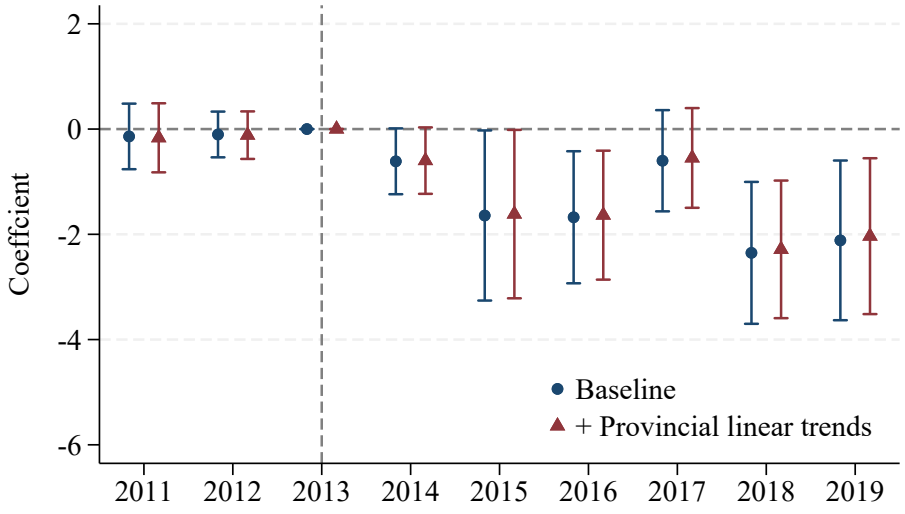


Figure A9. Dynamic Effects: Controlling for Provincial Linear Trends

Note: This figure presents the event-study estimates from Equation 3 (baseline) and the estimates after controlling for province-specific linear trends. The solid dots are point estimates, and the spikes are 95 percent confidence intervals constructed based on standard errors clustered at the prefecture level.

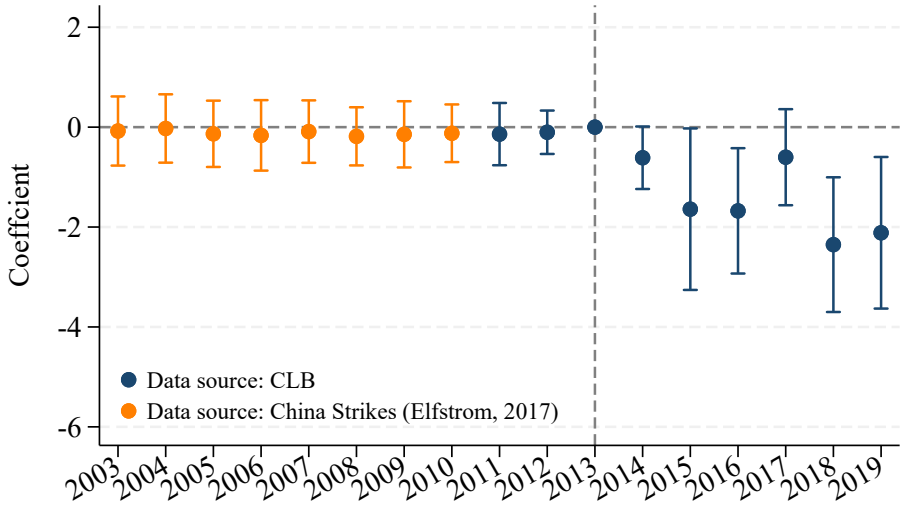


Figure A10. Dynamic Effects: Longer Pretrends

Note: This figure presents the event-study estimates from Equation 3 with extended pre-reform observations. We obtain data on labor unrest between 2003–2010 from the *China Strikes* database Elfstrom (2017). The solid dots are point estimates, and the spikes are 95 percent confidence intervals constructed based on standard errors clustered at the prefecture level.

B Additional Tables

Table B1. Robustness to Forms of the Running Variable

	(1) Unrest/L	(2) Unrest/L
Reform × Post	-1.419***	-1.237**
	(0.373)	(0.385)
Control mean	3.395	3.395
Prefecture FE	Yes	Yes
Year FE	Yes	Yes
Polynomials (log)	Yes	
Polynomials (level)		Yes
Observations	2,583	2,583

Note: This table examines the robustness of the main results to forms of the running variable. Column (1) is identical to Column (1) reported in Table 2, which uses $\log(3 - P_{i,2014})$ as the running variable. Column (2) instead uses $3 - P_{i,2014}$ as the running variable. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

Table B2. Officials' Promotion Prospects and Urban Population Growth

	Dependent: $\Delta \log(P)$, 2013–2014		
	(1) All	(2) $P_{2013} < 3$M	(3) $P_{2013} > 3$M
Promotion prospect	-0.026	-0.024	-0.203
	(0.082)	(0.091)	(0.279)
Observations	287	251	36

Note: This table reports the association between a prefectural party secretary's promotion prospect and the (observed) growth in urban population from 2013 to 2014. The promotion prospect index is estimated following Wang et al. (2020): the higher, the better the prospect (see Appendix E for estimation details). We report the association for the full sample (287 prefectures), for prefectures with less than 3 million urban residents in 2013, and for prefectures with more than 3 million urban residents in 2013. Robust standard errors are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

Table B3. The Effect of the *Hukou* Reform on Labor Unrest by Scale

	(1) Samll-scale unrest/L 1–100	(2) Middle-scale unrest/L 101–1,000	(3) Large-scale unrest/L 1,001–10,000
Reform × Post	-1.149***	-0.267**	-0.003
	(0.337)	(0.103)	(0.031)
Prefecture FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes
Observations	2,583	2,583	2,583
R^2	0.588	0.369	0.251

Note: This table presents the results for the effect of *hukou* reform on labor unrest by scale: small scale (1–100 participants), middle scale (101–1,000 participants), and large scale (1,001–10,000 participants). Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

Table B4. Examining the Role of Discontent with the *Hukou* System

	(1) Unrest/L	(2) (Benefit-unrelated Unrest)/L	(3) (Benefit-related Unrest)/L
Reform × Post	-1.419***	-1.342***	-0.077
	(0.373)	(0.353)	(0.051)
Control mean	3.395	3.138	0.257
Prefecture FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes
Observations	2,583	2,583	2,583

Note: This table examines whether the effect of the *hukou* reform on labor unrest is driven by changes in discontent with the *hukou* system itself. Column (1) reports the baseline estimate, which includes all types of unrest. Then, using information in CLB, we examine labor unrest events related to demands for benefits like social security and housing subsidies, which connect to the *hukou* system. Column (2) reports the effect on benefit-unrelated unrest events. Column (3) reports the effect on benefit-unrelated unrest events. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

Table B5. Examining the Role of NGOs

	(1) Unrest/L	(2) Unrest/L	(3) Unrest/L	(4) Unrest/L	(5) Unrest/L
Reform × Post	-1.419***	-1.348***	-1.515***	-1.431***	-1.464***
	(0.373)	(0.370)	(0.389)	(0.449)	(0.386)
Prefecture FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes	Yes	Yes
Log distance to HK × year FE		Yes			
Sample: distance to HK ≥ 500km			Yes		
Sample: distance to HK ≥ 1,000km				Yes	
Sample: Guangdong province dropped					Yes
Observations	2,583	2,583	2,304	1,755	2,403
R^2	0.601	0.607	0.586	0.602	0.586

Note: This table examines whether declining NGO activities can explain the effect of the *hukou* reform on labor unrest. Column (1) replicates the baseline estimate in Column (1) of Table 2. Column (2) controls for distance to Hong Kong, where most NGOs working on Chinese labor issues are based. Columns (3) and (4) exclude prefectures that are within 500km and 1000km of Hong Kong. Column (5) drops prefectures in Guangdong province from the sample. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

Table B6. The *Hukou* Reform, Outmigration, and Labor Unrest: 2011–2019

	DV: ΔUnrest/L, 2011–2019		
	(1) Baseline	(2) Mediation-OLS	(3) Mediation-IV
Reform [β or τ]	-1.419***	-1.316***	-0.963**
	(0.370)	(0.370)	(0.472)
ΔOutmigration [γ]		1.834***	8.061**
		(0.484)	(3.246)
% Total effect explained		0.073	0.322
Effective F stat.			16.126
Observations	287	287	287

Note: This table reports causal mediation analysis that quantifies the importance of the settlement intentions mechanism, measured by the outmigration rate, during 2011–2019. Column (1) reports the baseline estimate. Column (2) represents the conventional approach. Column (3) uses the IV-augmented approach. The coefficient on *Reform* in Column (1) is denoted by β, while the coefficients on *Reform* in Columns (2) and (3) are denoted by τ. The share of the total effect explained is calculated as $\frac{\beta-\tau}{\beta}$. The effective F statistic is calculated following Olea and Pflueger (2013). Robust standard errors are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

C Conceptual Model: Settlement and Participation in Labor Unrest

We present a simple conceptual model to clarify the linkage between settlement plans and unrest participation. Our modeling draws upon insights from the literature on temporary migration (for a review, see Dustmann and Görlach, 2016).

C.1 Model Setup

Consider a migrant in destination d. She has a two-period horizon: the current period (period 1) and the future (period 2). This migrant's lifetime utility is written as:

$$V = u_1 + \beta u_2, \tag{C1}$$

where u_t is the utility in period $t \in \{1, 2\}$, which we specify in detail below, and β is the discount factor.

Period 1 Utility. In period 1, the migrant receives basic income w. However, part of the migrant's income depends on participation in labor unrest. For example, migrants may face wage arrears, the resolution of which requires negotiations, protests, and even violent confrontations.[1] We let $e \geq 0$ denote the level of engagement in labor unrest, which yields a constant marginal return of $a > 0$. However, participation in unrest also entails costs, including psychological stress, time costs, and potential employer or government retaliation. We assume the cost to be $\frac{1}{2}\gamma e^2$. The cost function is convex in e, and its magnitude is governed by $\gamma > 0$. Collecting these terms gives the utility in period 1:

$$u_1 = w + ae - \frac{1}{2}\gamma e^2. \tag{C2}$$

Period 2 Utility. In period 2, the migrant chooses between returning to her origin o and remaining in destination d. We normalize the utility of returning to the origin to zero, denoted by $u_{2o} = 0$. In contrast, the utility of staying in the destination is:

$$u_{2d} = x - c - \frac{1}{2}\kappa e^2 = (\mu + \varepsilon) - c - \frac{1}{2}\kappa e^2. \tag{C3}$$

$x = \mu + \varepsilon$ is the present value of staying in destination d, where μ is a fixed component related to a destination's overall attractiveness to migrants, and ε is an idiosyncratic component capturing individual preferences. Within a destination, ε is distributed across migrants according to cdf $F(\cdot)$ (pdf $f(\cdot)$). However, if a migrant stays in the destination, she must bear two kinds of costs. The first, denoted by c, is the exogenous fixed cost of (continued) migration. Importantly, it includes the cost of limited access to local public services due to the inability to obtain local *hukou* registration. The *hukou* reform can reduce c. The second cost arises from unrest participation in period 1. A migrant with a history of unrest participation may face greater challenges in settlement. For instance, a

[1] In fact, wage arrears trigger 72 percent of labor unrest events recorded in our data during 2011–2019.

migrant blacklisted by local employers or punished by the local government may not be able to find a job (Leung, 2015). Taken together, a migrant's utility in period 2 depends on where she lives. Let $r \in \{o, d\}$ denote the location choice. Then,

$$u_2 = \mathbb{1}\{r = d\}u_{2d} + (1 - \mathbb{1}\{r = d\})u_{2o} = \mathbb{1}\{r = d\}\left(x - c - \frac{1}{2}\kappa e^2\right). \tag{C4}$$

Migrant's Problem. In period 1, anticipating future benefits and costs, the migrant chooses her intended location in period 2, $r \in \{o, d\}$, and her level of unrest participation, e, to maximize lifetime utility. The migrant's optimization problem is given by

$$\begin{aligned} \mathcal{V} = \max_{e,r} V &= u_1 + \beta u_2 \\ &= w + ae - \frac{1}{2}\gamma e^2 + \beta \mathbb{1}\{r = d\}\left(x - c - \frac{1}{2}\kappa e^2\right). \end{aligned} \tag{C5}$$

C.2 Results

To solve the model, we first derive the optimal level of unrest participation e, contingent on the migrant's location choice for period 2. Then, we compare the resulting lifetime utility to determine the optimal location choice.

If $r = d$, by first order conditions (FOCs), it is easy to see that the optimal level of unrest participation is $e_d = \frac{a}{\gamma + \beta\kappa}$. Thus, the lifetime utility is

$$\mathcal{V}_d = w + e_d + \beta x - \beta c - \frac{1}{2}(\gamma + \beta\kappa)e_d^2. \tag{C6}$$

If $r = o$, by FOCs, the optimal level of unrest participation is $e_o = \frac{a}{\gamma}$. Then, the lifetime utility is

$$\mathcal{V}_o = w + e_o - \frac{1}{2}\gamma e_o^2. \tag{C7}$$

Taken together, a migrant would plan on staying, namely, $r = d$, if

$$\mathcal{V}_d > \mathcal{V}_o \tag{C8}$$

$$x > c + \frac{a^2\kappa}{2\gamma(\gamma + \beta\kappa)} \tag{C9}$$

$$\varepsilon > c + \frac{a^2\kappa}{2\gamma(\gamma + \beta\kappa)} - \mu \tag{C10}$$

At the aggregate level, the share of migrants who choose to stay is

$$p \equiv \Pr(r = d) = 1 - F\left[c + \frac{a^2\kappa}{2\gamma(\gamma + \beta\kappa)} - \mu\right]. \tag{C11}$$

The aggregate level of unrest is

$$\mathcal{E} \equiv E(e) = pe_d + (1-p)e_o = \frac{a}{\gamma}\left(1 - \frac{p\beta\kappa}{\gamma + \beta\kappa}\right). \tag{C12}$$

Proposition C1. *By lowering the cost of staying in the destination (c), the hukou reform induces more migrants to stay and reduces aggregate unrest, i.e.,* $\frac{\partial p}{\partial c} < 0$ *and* $\frac{\partial \mathcal{E}}{\partial c} > 0$.

Proof. Take partial derivatives:

$$\frac{\partial p}{\partial c} = -f\left[c + \frac{a^2\kappa}{2\gamma(\gamma + \beta\kappa)} - \mu\right] < 0, \tag{C13}$$

$$\frac{\partial \mathcal{E}}{\partial c} = -\frac{a\beta\kappa}{\gamma(\gamma + \beta\kappa)}\frac{\partial p}{\partial c} > 0. \tag{C14}$$

Apparently, a decrease in c leads to an increase in p and a decrease in $\mathcal{E}$. ■

Proposition C2. *If pdf* f *is unimodal, and the share of migrants who are able to settle before the hukou reform is small, then the hukou reform leads to a larger increase in the share of staying migrants and a larger decrease in labor unrest in more attractive destinations (i.e., those with higher* μ*), i.e.,* $\frac{\partial^2 p}{\partial c \partial \mu} < 0$ *and* $\frac{\partial^2 \mathcal{E}}{\partial c \partial \mu} > 0$

Proof. Recall that a migrant chooses to stay if

$$\varepsilon > c + \frac{a^2\kappa}{2\gamma(\gamma + \beta\kappa)} - \mu \equiv \varepsilon(c, \mu). \tag{C15}$$

Since few migrants are able to settle before the reform, i.e., p is small, $\varepsilon(c, \mu)$ is higher than the mode of ε such that $f'(\varepsilon(c, \mu)) < 0$. Thus,

$$\frac{\partial^2 p}{\partial c \partial \mu} = f'(\varepsilon(c, \mu)) < 0 \tag{C16}$$

$$\frac{\partial^2 \mathcal{E}}{\partial c \partial \mu} = -\frac{a\beta\kappa}{\gamma(\gamma + \beta\kappa)}\frac{\partial^2 p}{\partial c \partial \mu} > 0. \tag{C17}$$

■

Proposition C1 guides much of our empirical investigation. We empirically examine Proposition C2 in Section 5.2.4.

D Verifying the Definition of Reform Status

To verify the population-based definition of reform status, we extensively collect official documents issued by local governments regarding the reform, carefully read them, and manually code up a prefecture's reform status based on the content for comparison. The documents are collected from government websites, media outlets, and a database of *hukou* reforms by Zhang and Lu (2019).

The convention of policy-making in China is that when there is a central initiative, each level of government issues implementation guidelines based on local conditions. Typically, the guidelines become more specific about provisions when they get to lower levels. In the case we study, after the central government launched the *hukou* reform initiative, the provincial governments would release broad guidelines regarding implementation for their prefectures, and prefectural governments would further decide their provisions on the basis of the national initiative and provincial guidelines.[2]

To illustrate the information contained in these guidelines, we provide excerpts from Hebei Province's guidelines, issued on November 20, 2014 (Hebei Government, 2014).

[...]

(II.2) **Fully lift restrictions on settlement in urban areas of prefectures with a population of less than 1 million.** In the urban areas of Chengde, Zhangjiakou, Qinhuangdao, Cangzhou, Hengshui, and Xingtai, where the population is less than 1 million, people who have a legal and stable residence, and their spouses, children, and parents of both spouses who live together, can apply for a local *hukou*.

(II.3) **Reasonably determine the conditions for settlement in urban areas of prefectures with a population of more than 1 million.** In urban areas of Shijiazhuang, Tangshan, Baoding, and Handan, where the population is more than 1 million, if one of the following conditions is met, the individual and his/her spouse, children, and parents of both spouses living together can apply for a local permanent residence permit:

(a) Persons with legal and fixed residences. Legal and fixed residences include self-purchased houses and legally self-built houses (the same below) that meet the living conditions and are actually lived in;

(b) Persons with legal and stable occupations and other legal and stable residences. Persons with legal and stable occupations include those employed by administrative organs, people's organizations, and institutions or those who have signed labor contracts with enterprises in accordance with the law and have participated in the basic pension insurance for urban employees for

[2] In principle, the county governments can further specify their own guidelines. However, the prefectural guidelines are already detailed enough, leaving little room for further discretion to be used. Therefore, in practice, we find very few guidelines at the county level.

a certain number of years; those who have obtained industrial and commercial business licenses in accordance with the law and paid taxes in accordance with the law for a certain number of years; and those who are engaged in freelance occupations and have obtained residence permits for a certain number of years. Other legal and stable residences refer to legal and stable residences other than legal and fixed residences. The specific time limit for participating in the basic pension insurance for urban employees, paying taxes in accordance with the law, and obtaining residence permits shall be determined by the governments of each prefecture-level city in light of local conditions, and in principle shall not exceed 1 year.

As the provincial capital, Shijiazhuang should optimize the personnel structure in the urban area. On the basis of the above conditions, reasonable regulations can be made on the scope of legal and stable occupations and legal and stable residences, and the time limit for participating in the basic pension insurance for urban employees, paying taxes in accordance with the law, and obtaining residence permits according to the actual situation.

(II.4) **Reasonably determine the settlement conditions in towns around the national capital.** In the urban area of Langfang City, Sanhe City and its subordinate towns, Dachang Hui Autonomous County, Xianghe County, Yongqing County, Gu'an County government seat town, and other towns around the capital, those who have a legal and stable job and a legal and stable residence and have participated in social insurance for a certain number of years in accordance with national regulations, and the person himself and his spouse, children, and parents of both spouses who live together, can apply for a local permanent residence. The scope of legal and stable jobs, legal and stable residences, and the number of years of participation in social insurance shall be specifically stipulated by the Langfang Municipal Government based on actual conditions. No requirements such as area and amount shall be set for legal and stable residences, and the number of years of participation in social insurance shall not exceed 5 years.

[...]

The guidelines provided a broad picture of how the *hukou* reform would unfold in the prefectures of Hebei. Per (II.2), one can know that Chengde, Zhangjiakou, Qinhuangdao, Cangzhou, Hengshui, and Xingtai are mandated to lift their restrictions on *hukou* transfers, and thus they are coded as reform prefectures. By contrast, the tone in (II.3) is more subtle for Shijiazhuang, Tangshan, Baoding, and Handan—they have some discretion in setting requirements for residence and jobs, but they need to abide by the provincial government's guidelines. Thus, it is necessary to examine the actual provisions made by these prefectural governments to know their reform status. For instance, after review, we find that Shijiazhuang provided lenient requirements, although the

provincial government specifically allowed it to maintain tight control because of its status as the provincial capital. Lastly, per (II.4), one can know that Langfang should maintain tight control since it neighbors Beijing.

Generalizing this reading of Hebei Province's reform guidelines, we manually code up each prefecture's actual reform status through the following two-step procedure.

1. A prefecture is coded as "reform" if the provincial guidelines unambiguously mandate it to "fully lift restrictions."
2. If the reform status is not determined at step 1, we obtain and read the prefecture's own guidelines to find out its reform status. We code a prefecture as "non-reform" if it has any restrictive provisions on granting *hukou* transfers as follows:
   (a) A point-based system to screen migrants;
   (b) A requirement of having had a long duration of stay in the prefecture or enrollment in the local social security system (> 3 years);
   (c) A requirement of select types of stable residences and/or jobs, e.g., only home ownership is eligible or a minimum duration of work experience is required.

   Otherwise, a prefecture is coded as "reform."

Table D1 shows the release time of each province's guidelines following the national initiative. Most of them were released in late 2014 or the first half of 2015.

Comparing the population-based (≤ 3 million or not) and manually coded reform status, the two definitions disagree for only 17 out of 287 prefectures, or 6 percent of the sampled prefectures. Table D2 tabulates these disagreements. Panel A shows that 9 large-sized prefectures (urban population > 3 million) relaxed *hukou* transfers, whereas Panel B shows that 8 small-sized prefectures (≤ 3 million) did not relax. We do not interpret these disagreements as local governments' deliberate defiance of the national reform initiative; instead, they may reflect discretion based on local conditions. Prefectures in Panel A may make lenient provisions to attract labor. In contrast, prefectures in Panel B may be the hot spots of population inflows. Langfang is adjacent to Beijing and so is home to many migrants who work in Beijing, thus, it is urged to maintain tight control. The remaining ones are all prefectures in coastal provinces attractive to new migrants. The Guangdong government even explicitly required Zhuhai and Zhongshan, two manufacturing hubs in the Pearl River Delta, to impose strict restrictions on granting local *hukou*. Our results are robust to excluding these 17 prefectures (see Section 4.3).

Table D1. Policy Time by Province

Province	Policy Time	Province	Policy Time
Xinjiang*	10/13/2014	Guizhou	05/04/2015
Heilongjiang	11/03/2014	Anhui	05/08/2015
Henan	11/04/2014	Hunan	05/11/2015
Jiangxi	11/14/2014	Yunnan	06/01/2015
Shandong	11/19/2014	Guangdong	06/24/2015
Hebei	11/20/2014	Liaoning	07/10/2015
Sichuan	11/22/2014	Chongqing	08/25/2015
Ningxia†	-/-/2015	Hubei	09/06/2015
Gansu	01/01/2015	Inner Mongolia	09/08/2015
Jiangsu	01/12/2015	Zhejiang	12/10/2015
Shanxi	01/14/2015	Hainan	12/24/2015
Qinghai	01/27/2015	Shanghai	04/15/2016
Jilin	01/29/2015	Tianjin	04/20/2016
Fujian	02/13/2015	Tibet*	05/16/2016
Guangxi	02/25/2015	Beijing	09/08/2016
Shaanxi	03/19/2015		

Note: * = not in the main sample. † = only the release year is known.

Table D2. Discrepancies between Population-Based and Manually-Coded Definitions

Panel A: Population-based = 0, manually-coded = 1			**Panel B:** Population-based = 1, manually-coded = 0		
Province	Prefecture	Population (million)	Province	Prefecture	Population (million)
Hebei	Shijiazhuang	4.678	Hebei	Langfang	0.983
Hebei	Tangshan	3.088	Jiangsu	Changzhou	2.767
Jilin	Changchun	4.181	Zhejiang	Wenzhou	2.008
Jiangsu	Huai'an	3.317	Fujian	Fuzhou	2.495
Zhejiang	Shaoxing	3.562	Guangdong	Zhuhai	2.514
Anhui	Hefei	4.490	Guangdong	Zhongshan	2.878
Shandong	Zibo	3.063	Hainan	Haikou	2.667
Shandong	Linyi	3.086	Hainan	Sanya	0.764
Guangdong	Shantou	5.720			

Note: This table tabulates prefectures with a disagreement between population-based and manually coded definitions of reform status.

E Auxiliary Data

Local Socioeconomic Variables. Our analysis uses various local socioeconomic variables, which provide detailed information on economic growth, demographics, fiscal expenditures, and local governance. We collect them from several sources: China City Statistical Yearbooks, population census tabulations, as well as data compiled by other researchers (Campante et al., 2023; Rogoff and Yang, 2024).

Trade Data. To construct trade shock measures, we use: (i) the prefecture-level export structure measured using the 2010 Chinese customs database, obtained from Campante et al. (2023), and (ii) global export volumes, also from Campante et al. (2023), which are originally from the UN Comtrade database.

Officials' Promotion Prospects. Following Wang et al. (2020), we estimate a local leader's *ex ante* promotion prospect in a year. We focus on the party secretary, who is the chief leader of a prefecture. The estimated promotion prospect is a flexible function of the age when he starts the term, his official rank in the bureaucratic system, and some individual characteristics, which can be used as a proxy for his career concerns. This hinges on the personnel rule that mandates retirement ages, which increase with bureaucratic rank.[3] Specifically, we estimate the following Probit model:

$$\Pr(Promotion_{it} \mid \cdot) = \Phi\left[\beta_0 StartAge_{it} + \beta_1 HighRank_{it} + \beta_2\left(StartAge_{it} \times HighRank_{it}\right) + \mathbf{X}'_{it}\delta\right]. \quad \text{(E1)}$$

i indexes prefectures and t indexes terms. The unit of analysis is prefecture-by-term. $Promotion_{it}$ is a dummy that equals one if prefecture i's leader is promoted after term t. An outcome after a term ends is considered a promotion if the prefectural party secretary is appointed to a position ranked higher than his previous rank. However, we exclude rank enhancement as promotion if the prefectural party secretary is placed in an honorary position in the Chinese People's Political Consultative Conference (CPPCC) or the People's Congress (PC) at the prefectural or provincial level, which is commonly regarded as semi-retirement in China since these positions carry no real power. $StartAge_{it}$ is the age when a party secretary starts the term t of prefecture i. Most prefectural party secretaries have a prefectural (*zhengting*) rank, but some have a higher rank: deputy provincial (*fubu*), provincial (*zhengbu*), or even deputy national (*fuguo*). $HighRank_{it}$ is an indicator of the above deputy provincial rank. $\mathbf{X}_{it}$ includes an official's characteristics, including indicators for graduate degree and central government experience.

We use biographical data on local officials compiled by Yao et al. (2022) and Jiang (2018). The data include 2,305 party secretary terms in 337 prefectures between 2000 and 2017. The average start age is 50.8 years old, and 18.3 percent of party secretaries have a rank above the deputy provincial level. Table E1 reports the estimation results. The first two columns show estimates by a linear probability model (LPM), and Columns (3) and (4) show estimates by a Probit model. The results

[3]The retirement age is 60 for both prefecture-level and deputy-province-level leaders and 65 for province-level leaders.

are consistent with Table 2 in Wang et al. (2020). We use the estimated model in Column (4) to generate the predicted probability of promotion and use that as an index of promotion prospects.

Table E1. Prediction of Promotion Prospects

	Dependent: Promotion			
	(1) LPM	(2) LPM	(3) Probit	(4) Probit
Start age	-0.026***	-0.025***	-0.093***	-0.089***
	(0.003)	(0.003)	(0.009)	(0.009)
Deputy province or above	-1.921***	-1.925***	-8.615***	-8.752***
	(0.197)	(0.200)	(1.221)	(1.245)
Start age × Deputy province or above	0.035***	0.035***	0.157***	0.159***
	(0.004)	(0.004)	(0.023)	(0.023)
Graduate degree		0.035**		0.152**
		(0.017)		(0.074)
Central govt. expenditure		0.057		0.228*
		(0.040)		(0.138)
Dependent mean	0.185	0.185	0.185	0.185
Covariates		Yes		Yes
R^2	0.073	0.076		
Pseudo R^2			0.076	0.079
Observations	2,244	2,244	2,244	2,244

Note: This table reports the determinants of promotion, following Wang et al. (2020). Officials' characteristics include indicators for graduate degree and central government experience. Robust standard errors are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

GDELT. The Global Database of Events, Language, and Tone (GDELT) is a commonly used dataset on global events (`www.gdeltproject.org`), especially on social unrest (Cantoni et al., 2024). GDELT has conducted automated scraping of the world's broadcast, print, and web news since 1979, and uses machine learning methods to code an event's types, participants, date, location, etc. Given GDELT's wide coverage, we use it to measure the degree of media attention or reporting of local events.

Migrant Survey. We use a nationally representative survey on migrants—China Migrants Dynamic Survey (CMDS hereafter)—conducted annually by China's National Health Commission since 2009. CMDS covers migrants living and working in more than 300 prefectures across 31 provinces. CMDS uses stratified probability-proportional-to-size sampling so that the sample size is proportional to the number of migrants in each region. The data provide rich information on migrants' demographics, families, labor market outcomes, and attitudes. With a sample size of 100–200 thousand for each year, CMDS includes substantially more migrants than any other survey data in China. We use eight waves of surveys between 2011 and 2018. We focus our attention on working-age individuals (25–54). To ensure consistent geographic coverage over time, we restrict the sample to prefectures covered in all eight waves, resulting in a final sample of 255 prefectures.

Population Census Microdata. To study how the *hukou* reform influences migrants' outmigration decisions, we use the microdata of the 2015 mini-census, which surveyed 1 percent of Chinese population. The survey was conducted in November 2015. It also asks individuals to report their residential locations in November 2010 and November 2014. Thus, we can construct the residential history at three time points: (the end of) 2010, 2014, and 2015. We also observe one's *hukou* registration location in 2015.

Cultural Distance. Following Guarnieri (2025), we measure cultural distance by linguistic distance. Linguistic data come from *The Language Atlas of China* (Lavely and Berman, 2012). We treat the finest category in *The Atlas* as a language. We construct our measure of linguistic distance in three steps.

First, we calculate the cladistic distance between two languages x and y (Fearon and Laitin, 2003):

$$d_{xy} = 1 - \left(\frac{\#\text{ of common nodes between } x \text{ and } y}{\frac{1}{2} \times (\#\text{ of nodes of } x + \#\text{ of nodes of } y)} \right)^{\frac{1}{2}}. \tag{E2}$$

This measure captures the extent to which two languages share common origins. For instance, consider two variants of Mandarin, *Jingshi* (x) and *Huaicheng* (y). Their linguistic family trees are:

x : Sino-Tibetan Phylum → Sinitic Stock → Mandarin supergroup → Beijing Mandarin → Jingshi,

y : Sino-Tibetan Phylum → Sinitic Stock → Mandarin supergroup → Beijing Mandarin → Huaicheng.

Because the two languages share four of their five nodes, their cladistic distance is:

$$d_{\text{Jingshi,Huaicheng}} = 1 - \sqrt{\frac{4}{0.5 \times (5+5)}} = 0.106. \tag{E3}$$

Languages belonging to entirely distinct linguistic families have a maximum distance of 1.

Second, we calculate the linguistic distance between two prefectures h and i:

$$LD_{hi} = \sum_x \sum_y w_{xh} w_{yi} d_{xy}, \tag{E4}$$

where w_{xh} is the share of population primarily speaking language x in prefecture h, and w_{yi} is defined analogously for language y and prefecture i. *The Atlas* reports the primary spoken language in each county; then, we can calculate the weights w using the 2010 census.

Finally, we calculate the average linguistic distance between migrants' origins and their destination prefecture i:

$$LD_i^M = \sum_h s_{h \to i} LD_{hi}, \tag{E5}$$

where $s_{h \to i}$ is the share of migrants in prefecture i who originate from prefecture h.

F Other Population-Correlated Policies

Our estimates of the effects of the *hukou* reform could be confounded by other concurrent policies that (i) have provisions varying with urban population in the same way as the *hukou* reform, and (ii) can affect labor unrest. To address this concern, we conduct a comprehensive review of urban population-related policies.

We collect policies using the *PKULaw* database (https://www.pkulaw.com), which provides extensive information on Chinese laws and regulations and has been used in research on policy-making in China (e.g., Tian, 2024; Wang and Yang, 2025). We use two keywords to search for urban population-related policies: "urban population" (城区人口 in Chinese) and "city size" (城市规模 in Chinese). The *PKULaw* database supports a fuzzy search feature: it would return policies that contain the specified keywords and their synonyms. We focus on policies issued by the central government, which have the potential to establish provisions that vary systematically with urban population.

Results of Searches Based on "Urban Population." We first use the keyword "urban population." Table F1 tabulates the number of policies issued by the central government. Overall, *PKULaw* identifies 126 relevant policies during 2011–2019. We examine these policies in detail. The *PKULaw* database classifies these policies into 9 domains, most of which are unlikely to have direct effects on labor unrest.

The "regional planning" domain accounts for the largest share. This domain includes: (i) central government approvals of development plans proposed by local governments, and (ii) development plans formulated by the central government for specific regions. In these policies, "urban population" or synonyms are used to describe regional demographic characteristics rather than to specify provisions that vary systematically with urban population. For example, the State Council's approval of the Fuzhou Prefecture's urban development plan in 2015 stated:[4]

> "By 2020, the permanent population in the central urban area will be controlled within 4.1 million people, and the urban construction land will be controlled within 378 square kilometers."

The 2014 *hukou* reform falls in the domain of "government work." The only other policy in this domain in 2014 concerned the development of regions near the Yangtze River.[5] The policy in this domain in 2017 concerned the allocation of responsibilities among central government departments.[6] Neither policy had provisions that varied with urban population.

[4] https://www.gov.cn/gongbao/content/2015/content_2912363.htm

[5] State Council's Notice on Relying on Golden Waterways Guidance to Promote the Development of the Yangtze River Economic Belt (https://www.gov.cn/zhengce/content/2014-09/25/content_9092.htm).

[6] State Council's Opinions on the Implementation of the Government Work Report: The Division of Labor Between Departments (https://www.gov.cn/zhengce/content/2017-03/28/content_5181530.htm).

The policy in the "labor union" domain is worth noting. It called on labor unions to promote employment but contained no provisions that varied with urban population.[7] Thus, this policy is unlikely to confound our estimates.

Among the policies identified during our study period, six contain provisions that vary with urban population, as listed below:

1. **Opinions of the State Council on Further Promoting the Reform of the Household Registration System in 2014.**[8] This is the *hukou* reform we study. Its details are discussed in Section 2.1.
2. **Opinions of the State Council on Promoting the Development of Prefabricated Construction in 2016.**[9] The policy designates cities with more than 3 million people as priority areas for promoting prefabricated construction, while encouraging the adoption in other cities.
3. **Notice of the State Council on Issuing the Ecological and Environmental Protection Plan for the 13th Five-Year Plan (2016).**[10] The policy sets the following target: "By 2020, public transportation in built-up areas of cities with a permanent population of more than 3 million will account for 60% of motorized travel."
4. **Notice of the State Council on Issuing the 13th Five-Year Plan for the Development of a Modern Comprehensive Transportation System in 2017.**[11] This document states that by 2020, high-speed rail should cover more than 80% of the cities with an urban population over 1 million, while railways, highways, and civil aviation airports should cover cities with an urban population over 0.2 million.
5. **Opinions of the State Council on Further Strengthening the Planning and Management of Urban Rail Transit Construction in 2018.**[12] This document requires that cities applying to build a metro system should generally have a public fiscal budget of more than 30 billion yuan, a regional GDP of over 300 billion yuan, and an urban population of over 3 million.
6. **Opinions of the State Council on Promoting the Improvement and Expansion of the Domestic Service Industry in 2019.**[13] The document states that all cities with an urban population over 1 million should have domestic service training enterprises by 2022.

Clearly, aside from the *hukou* reform, the remaining policies are unlikely to have first-order effects on labor unrest.

Results of Searches Based on "City Size." Table F2 tabulates the number of policies containing the keyword "city size" or related terms by year and policy domain. Similar to the results for

[7] State Council's Notice on Issuing the Employment Promotion Plan for the 13th Five-Year Plan (https://www.gov.cn/zhengce/content/2017-02/06/content_5165797.htm).
[8] https://www.gov.cn/zhengce/content/2014-07/30/content_8944.htm.
[9] https://www.gov.cn/zhengce/content/2016-09/30/content_5114118.htm.
[10] https://www.gov.cn/zhengce/content/2016-12/05/content_5143290.htm
[11] https://www.gov.cn/zhengce/content/2017-02/28/content_5171345.htm.
[12] https://www.gov.cn/zhengce/content/2018-07/13/content_5306202.htm.
[13] https://www.gov.cn/zhengce/content/2019-06/26/content_5403340.htm.

"urban population," most policies identified by this search fall in the "regional planning" domain. In this domain, "city size" or related terms are used to describe city characteristics rather than to specify provisions that vary with urban population. Among the remaining policies, we identify one policy with provisions varying by urban population:

1. **Opinions of the State Council on Accelerating the Promotion of Ecological Civilization Construction in 2015.**[14] The policy calls for proper urban planning that takes into account resource and environmental carrying capacity, strict controls on the size of mega-cities, and strengthening the carrying capacity of small and medium-sized cities. However, it does not specify population thresholds.

As the policy focuses on environmental issues, it is not likely to have a first-order impact on labor unrest.

Table F1. Results for the Search Using Keyword *"Urban Population"*

Year	Regional Planning	Government Work	Resources & Environment	Labor Union	Transport	Science & Education	Health	Others	Total
2011	15	0	1	0	0	0	0	1	17
2012	28	0	0	0	0	0	0	4	32
2013	7	0	1	0	0	0	1	3	12
2014	4	2	0	0	0	0	0	2	7
2015	7	0	1	0	0	0	0	1	10
2016	10	0	2	0	0	1	1	6	20
2017	16	1	1	1	1	1	0	1	22
2018	3	0	0	0	0	0	0	1	4
2019	2	0	0	0	0	0	0	0	2

Note: This table reports the number of policies containing the keyword "urban population" or related terms, by year and policy domain. Policy domains are classified by *PKULaw*.

Table F2. Results for the Search Using Keyword *"City Size"*

Year	Regional Planning	Resources Agriculture	Planning	Standardized Management	Others	Total
2011	7	0	0	0	0	7
2012	1	1	0	0	0	2
2013	2	0	0	0	0	2
2014	2	0	2	1	0	5
2015	7	0	2	0	0	9
2016	12	1	1	0	0	14
2017	15	0	1	0	0	16
2018	2	0	0	0	0	2
2019	0	0	0	0	0	0

Note: This table reports the number of policies containing the keyword "city size" or related terms, by year and policy domain. Policy domains are classified by *PKULaw*.

[14] https://www.gov.cn/gongbao/content/2015/content_2864050.htm.

G Supplementary Results for Main Findings

G.1 Robustness to Sampling of Prefectures

Our main sample includes 287 prefectures with available urban population data for 2014 (from the Urban Construction Statistical Yearbook). We also exclude prefectures in Tibet and Xinjiang. We show that our results are not driven by this sampling choice. Specifically, we present results using three alternative samples:

1. The main sample of 287 prefectures plus prefectures in Tibet and Xinjiang with available urban population data for 2014 (298 prefectures in total);
2. All prefectures with available urban population data for 2014 (303 prefectures in total);
3. All prefectures in China (337 prefectures in total).

For each sample, we report both the DiD and DiDC estimates. One issue arises when using the sample of all 337 prefectures: 34 prefectures do not have 2014 urban population data available. To include the 34 prefectures in the estimation, we define them as reform prefectures, i.e., $Reform_i = 1$, as they are presumably small. This is confirmed by the 2010 population census: all of them had an urban population below 3 million in 2010 (maximum = 1.597 million). For the DiDC estimation, we construct the 34 prefectures' running variable $\tilde{p}_i$ using the urban population reported by the 2010 population census. Figure G1 and Table G1 report the results. Across samples, the results consistently show that the *hukou* reform reduced labor unrest, and the effect sizes are similar. We also note that the DiDC design is more credible than the DiD design, as it controls for pretrends.

Table G1. Robustness to Sampling of Prefectures

	Main sample		Plus Tibet & Xinjiang		Urban population available		All prefectures	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Unrest/L	Unrest/L	Unrest/L	Unrest/L	Unrest/L	Unrest/L	Unrest/L	Unrest/L
Reform × Post	-1.567***	-1.419***	-1.551***	-1.214***	-1.578***	-1.124***	-1.661***	-1.201***
	(0.271)	(0.373)	(0.271)	(0.375)	(0.270)	(0.370)	(0.269)	(0.360)
Control mean	3.395	3.395	3.335	3.335	3.335	3.335	3.335	3.335
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Polynomials		Yes		Yes		Yes		Yes†
No. prefectures	287	287	298	298	303	303	337	337
Observations	2,583	2,583	2,682	2,682	2,727	2,727	3,033	3,033

Note: This table examines the robustness of our results to sampling of prefectures. We present both DiDC and DiD estimates. Columns (1)–(2) display results using the main sample of 287 prefectures. Columns (3)–(4) use the main sample of 287 prefectures plus prefectures in Tibet and Xinjiang with available urban population data for 2014 (298 prefectures in total). Columns (5)–(6) use all prefectures with available urban population data for 2014 (303 prefectures in total). Columns (7)–(8) use all prefectures in China (337 prefectures in total). For Columns (7)–(8), 34 prefectures without available urban population data for 2014 are defined as reform prefectures, as they are presumably small; their $\tilde{p}_i$ is calculated using urban population from the 2010 census. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

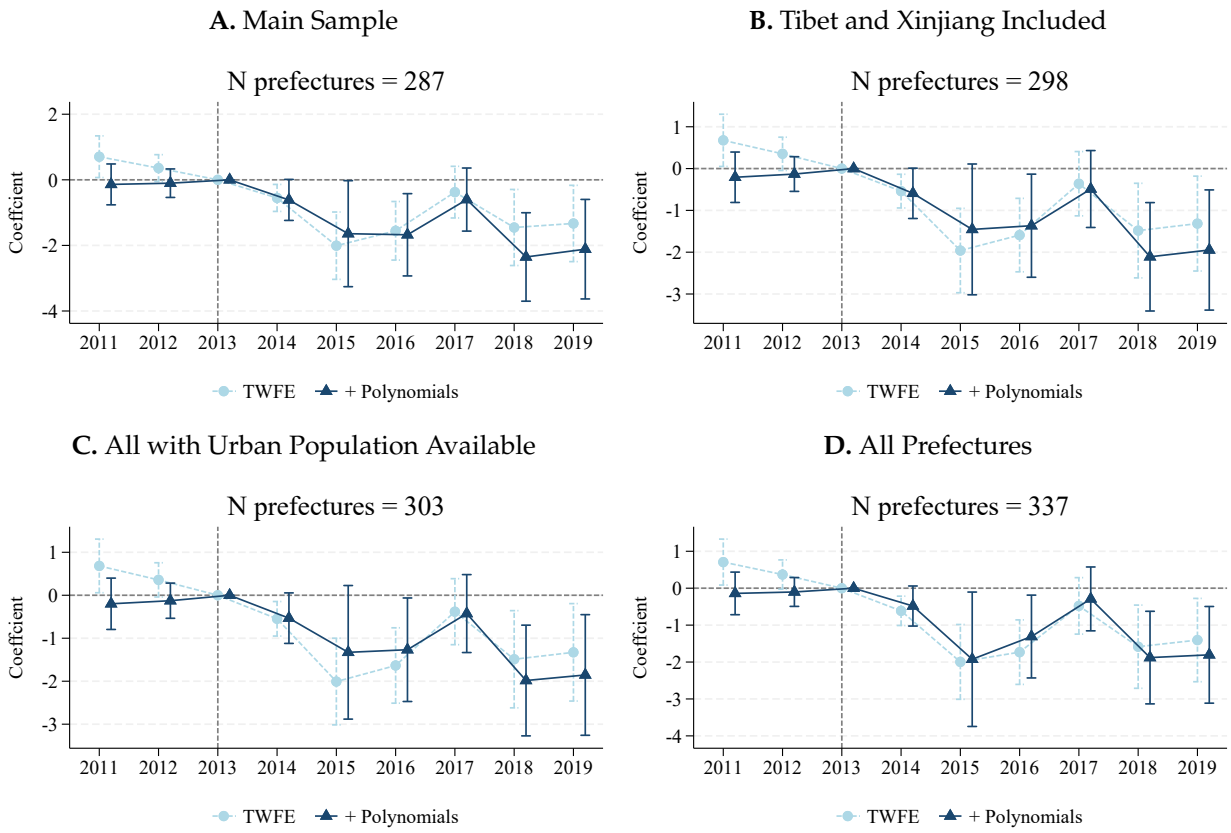


Figure G1. Robustness to Sampling of Prefectures: Event Study

Note: This figure examines the robustness of event-study estimates to sampling of prefectures. We present both DiDC and DiD estimates. Panel A displays results using the main sample of 287 prefectures. Panel B uses the main sample of 287 prefectures plus prefectures in Tibet and Xinjiang with available urban population data for 2014 (298 prefectures in total). Panel C uses all prefectures with available urban population data for 2014 (303 prefectures in total). Panel D uses all prefectures in China (337 prefectures in total). For Panel D, 34 prefectures without available urban population data for 2014 are defined as reform prefectures, as they are presumably small; their $\tilde{p}_i$ is calculated using urban population from the 2010 census. The solid dots are point estimates, and the spikes are 95 percent confidence intervals, constructed using standard errors clustered at the prefecture level.

G.2 Addressing Potential Violations of Local Parallel Trends

We follow Fenizia and Saggio (2024) and Rambachan and Roth (2023) to address potential concerns about violations of *local* parallel trends (for prefectures around the 3-million cutoff).

We first fit a linear trend implied by the pre-reform event-study estimates and extrapolate it to the post-reform periods, as shown in the first column of Figure G2. The pretrends are slightly upward. If these trends persist into the post-reform periods, we would have in fact *underestimated* the true negative effects of the *hukou* reform on labor unrest. Nonetheless, we can correct this downward bias by calculating the post-reform estimates' deviations from the extrapolated linear trend. The middle column of Figure G2 reports the detrended event-study results, confirming that the *hukou* reform has significantly reduced labor unrest rates.

Having shown robustness to linear differential trends, we use the methodology developed by Rambachan and Roth (2023) to assess the robustness of our results to *nonlinear* violations of local parallel trends. Specifically, let θ_t denote the differential trend between reform and non-reform prefectures in period t. For instance, $\theta_t = \theta \cdot t$ represents the linear trend, and the slope $\theta_t - \theta_{t-1} = \theta$ is constant. When allowing for nonlinear trends, the method imposes the following condition on changes in the slope of labor unrest trends in reform prefectures relative to non-reform prefectures between consecutive periods:

$$|(\theta_{t+1} - \theta_t) - (\theta_t - \theta_{t-1})| \leq M. \tag{G1}$$

M bounds changes in the slope, namely, the degree of non-linearity in the differential trend. $M = 0$ corresponds to a linear differential trend. A larger M thus allows a more nonlinear differential trend. For each given M, the method tests the null, net of the possible differential trend, whether the *hukou* reform still has a significant effect on labor unrest—defined as the average of post-reform event study coefficients. The last column of Figure G2 shows that our results withstand very nonlinear differential trends. We can reject a null effect up to when M is 0.03. Compared to the slope of the linear trend implied by pre-reform event study coefficients, 0.076, this means that the slope of the differential trend must change by more than $\pm\frac{0.03}{0.076} \approx \pm 39.4\%$ of the slope of the implied linear trend in each period. In other words, we would fail to reject the null only when a very wiggly differential trend is imposed.

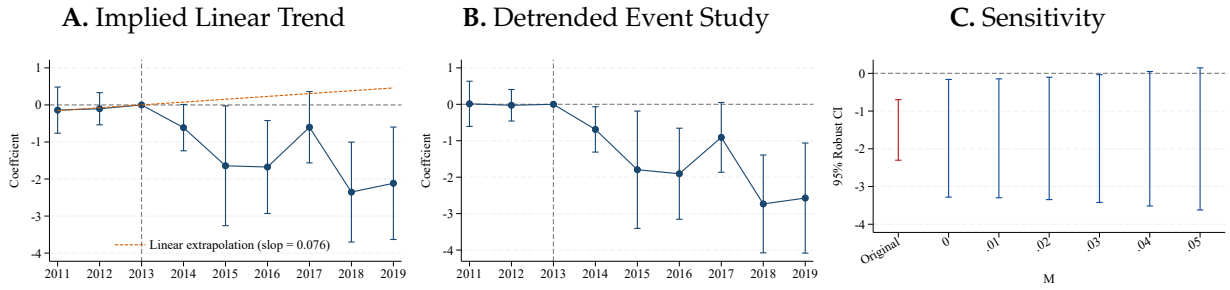


Figure G2. Addressing Potential Violations of Local Parallel Trends

Note: This figure examines the robustness to potential violations of local parallel trends. The first column displays the linear trend implied by pre-reform event study coefficients, and it is extrapolated into post-reform periods. The middle column shows the deviations of event-study coefficients from the linear trend. The last column reports the results for applying the methodology developed by Rambachan and Roth (2023), which tests whether the average of post-reform coefficients are statistically distinguishable from zero, net of different degrees of nonlinear differential trends between reform and non-reform prefectures (governed by M).

G.3 Robustness to Choices of Bandwidths, Kernels, and Polynomial Orders

Bandwidths. Figure G3 presents the estimated coefficient on $Reform_i \times Post_t$ in Equation 2, when we impose different bandwidth restrictions to our sample. We note three commonly used optimal bandwidths in the literature: Calonico et al. (2014) a.k.a. CCT, Imbens and Kalyanaraman (2012) a.k.a. IK, and cross validation a.k.a. CV. The estimates are overall stable across different bandwidth choices, though as expected, standard errors become large when small bandwidths are used.

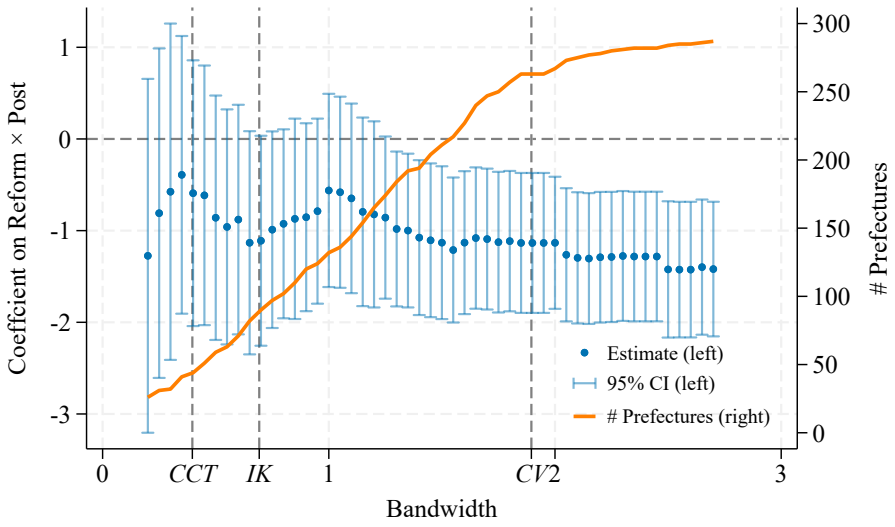


Figure G3. Robustness to Bandwidth Choices

Note: This figure presents the estimated coefficient on $Reform_i \times Post_t$ in Equation 2, when we impose different bandwidth restrictions to our sample. We note three commonly used optimal bandwidths in the literature: Calonico et al. (2014) a.k.a. CCT, Imbens and Kalyanaraman (2012) a.k.a. IK, and cross validation a.k.a. CV. Standard errors clustered at the prefecture level are used to construct the 95 percent confidence intervals.

Kernels. Our baseline results use the uniform kernel, which places equal weights on prefectures. We examine robustness to kernel choices. We consider two alternative kernels common in the literature: triangular and Epanechnikov kernels, which place high weights on prefectures closer to the cutoff $\tilde{p}_i = 0$. Using these two kernels, we then re-estimate Equation 2. Figure G4 shows that estimates of β in Equation 2 are similar to the baseline when using alternative kernels.

Polynomial Orders. Figure G5 presents event-study results for the effects of the *hukou* reform on labor unrest using quadratic and cubic polynomials of $\tilde{p}_i$. Compared with the baseline results using linear polynomials in Figure 2, these results are noisier, likely due to overfitting with the introduction of higher-order polynomials (Gelman and Imbens, 2019). Nonetheless, it is clear that after the *hukou* reform came into effect, labor unrest rates decreased.

G.4 Reporting of Local Events

General Coverage of Local Events. Since CLB data rely on online reports about labor unrest, we investigate if the *hukou* reform affects reporting of local events so that we observe a decrease in unrest rates in reform prefectures. We make use of the Global Database of Events, Language, and Tone Project (GDELT). GDELT records events from a comprehensive, global set of news feeds, and it also uses automated textual analysis to extract characteristics of recorded events, such as date, location, event type, parties involved, etc. Thus, we use the number of events recorded by GDELT (scaled by working-age population) as a measure of coverage of local events, the variation of which can reflect both media attention and information flows.

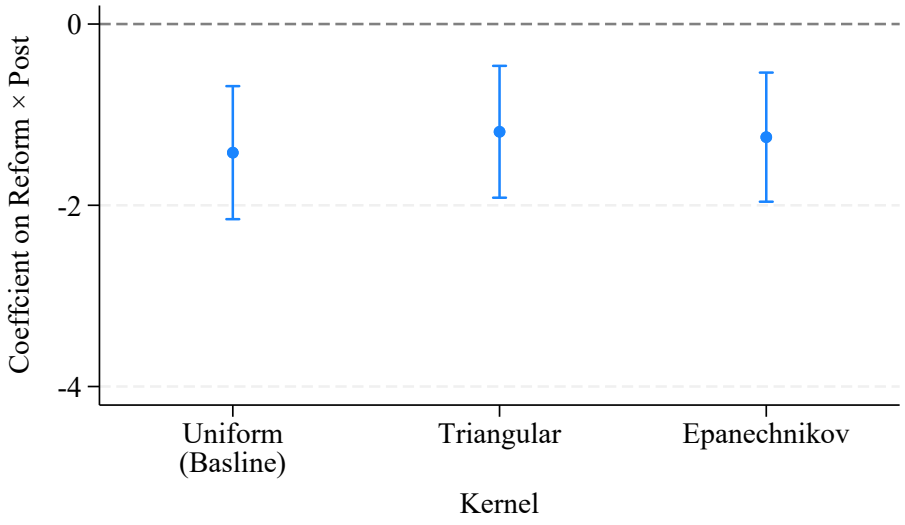
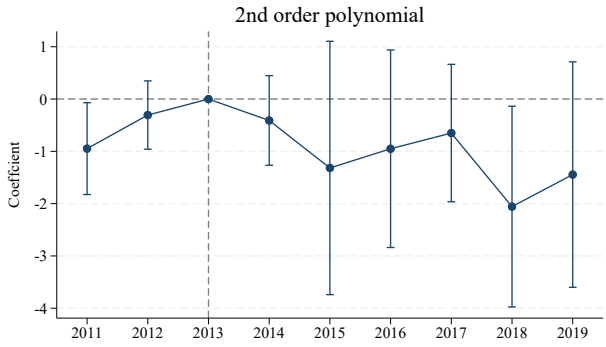


Figure G4. Robustness to Kernel Choices

Note: This table presents the robustness to kernel choices. We estimate Equation 2 using three kernels: the uniform kernel (baseline), the triangular kernel, and the Epanechnikov kernel. The solid dots denote the point estimates of β in Equation 2. The spikes are 95 percent confidence intervals. Standard errors are clustered at the prefecture level.

A. 2nd Order Polynomials



B. 3rd Order Polynomials

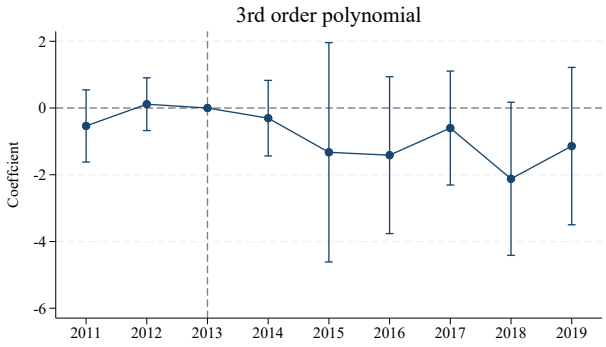


Figure G5. Robustness to Alternative Polynomial Orders

Note: This figure presents the event study results using polynomial orders. The solid dots are point estimates, and the spikes are 95 percent confidence intervals. Standard errors clustered at the prefecture level are used to construct confidence intervals.

Column (1) of Table G2 shows no significant differential changes in the number of reported events between reform and non-reform prefectures. If anything, coverage increased in reform prefectures. Column (2) examines the reporting of protest events.[15] Similarly, we find no statistically significant differences in the reporting of protests between reform and non-reform prefectures.

[15]The type of an event is classified based on the Conflict and Mediation Event Observations (CAMEO) code assigned by GDELT using machine learning techniques. Column (2) focuses on events with CAMEO code "14: Protest", which includes a range of protest activities including demonstrations, rallies, strikes, and violent protests.

Internet Censorship. Internet censorship was rising during the period we study (King et al., 2017). One may worry that the decrease in labor unrest in reform prefectures reflects changes in local reporting due to censorship. To address this concern, we measure the local censorship intensity by the share of deleted Weibo posts in a province using data from Qin et al. (2017),[16] and control for its interaction with year fixed effects. Column (3) shows that the effect of the reform remains robust after accounting for internet censorship.

In conclusion, the effect of the *hukou* reform on labor unrest is unlikely to be driven by differential reporting of local events.

Table G2. The *Hukou* Reform and Reporting of Local Events

	(1) All events/L GDELT	(2) Protests/L GDELT	(3) Unrest/L CLB
Reform × Post	758.064	9.086	-1.207***
	(642.532)	(6.801)	(0.362)
Control mean	722.131	6.573	3.416
Prefecture FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes
Censorship level × Year FE			Yes
Observations	4,420	4,420	2,331

Note: The dependent variables in Columns (1) and (2) are the number of local events and the number of protests recorded in GDELT (scaled by working-age population). The dependent variable in Column (3) is the number of labor unrest events in CLB data scaled by working-age population. "Censorship level" is the share of deleted Weibo posts in a province in 2013, based on data from Qin et al. (2017). Standard errors are clustered at the prefecture level and reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

G.5 The Effects of the *Hukou* Reform on Population

In this section, we investigate the effects of the *hukou* reform on total population, urban population, and native urban population (urban residents with local *hukou*). We study the effects on both levels and growth rates.

Figure G6 reports event-study estimates using the DiDC strategy. Results for the two research designs are similar. Regarding total population, Figure G6A shows that reform prefectures exhibit a downward linear trend in log total population, meaning that reform prefectures experienced declines at a constant percent rate relative to non-reform prefectures. Indeed, Figure G6D shows that the growth rate does not vary significantly over time. These results indicate that the *hukou* reform has a null effect, or possibly a negative effect, on a prefecture's total population. For urban population and native urban population, we find no discernible effects.

[16] This measure is available at the provincial level for 2013.

In sum, these results suggest that the *hukou* reform has no discernible effect on either total or urban population. If anything, it appears to have a negative effect on total population.

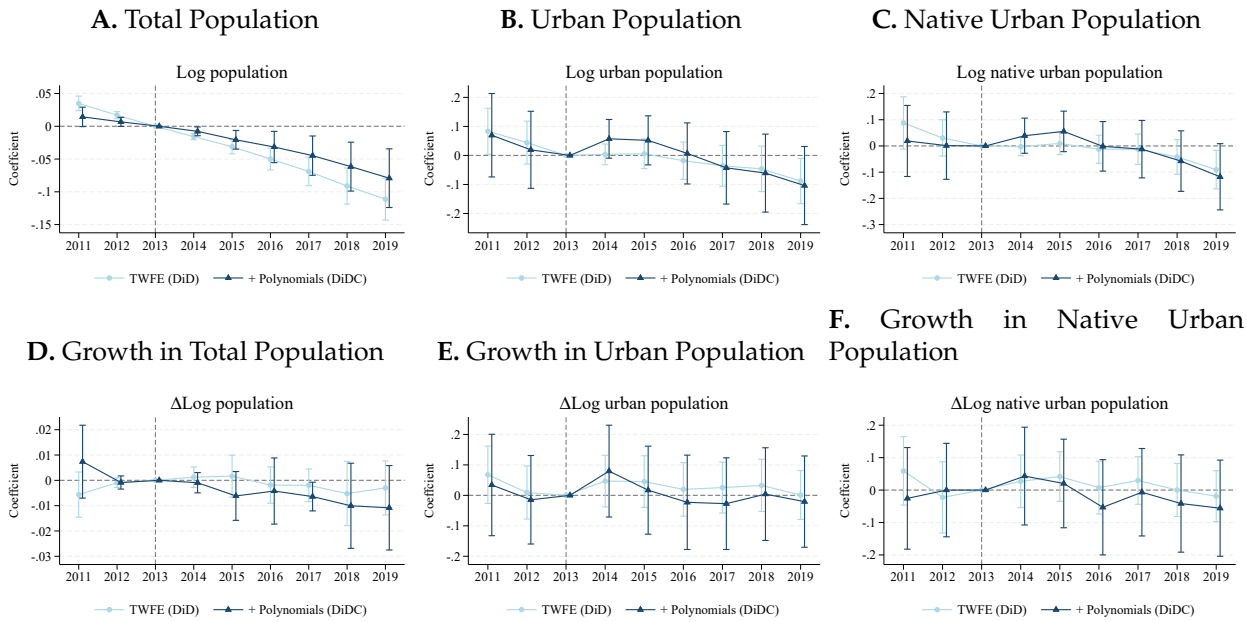


Figure G6. Dynamic Effects of the *Hukou* Reform on Population

Note: This figure reports the dynamic effects of the *hukou* reform on total population, urban population, and native urban population (urban residents with local *hukou*). We look at both their levels (in log points) and growth rates. We estimate event study models modified respectively from Equations 1 and 2. The solid dots are point estimates, and the spikes are 95 percent confidence intervals. Standard errors clustered at the prefecture level are used for constructing confidence intervals.

G.6 Addressing Time-Varying Prefecture Sizes

For our main results reported in Section 4, we scale the number of labor unrest events using the working-age population, measured in the 2010 population census. We show that our results hold even if we use time-varying population for scaling. There are no annual data on the working population. Instead, we use time-varying total population and urban population, sourced from Rogoff and Yang (2024) and the Urban Construction Statistical Yearbooks, respectively. Figure G7 reports the results, confirming our findings that the *hukou* reform leads to a significant decrease in labor unrest rates.

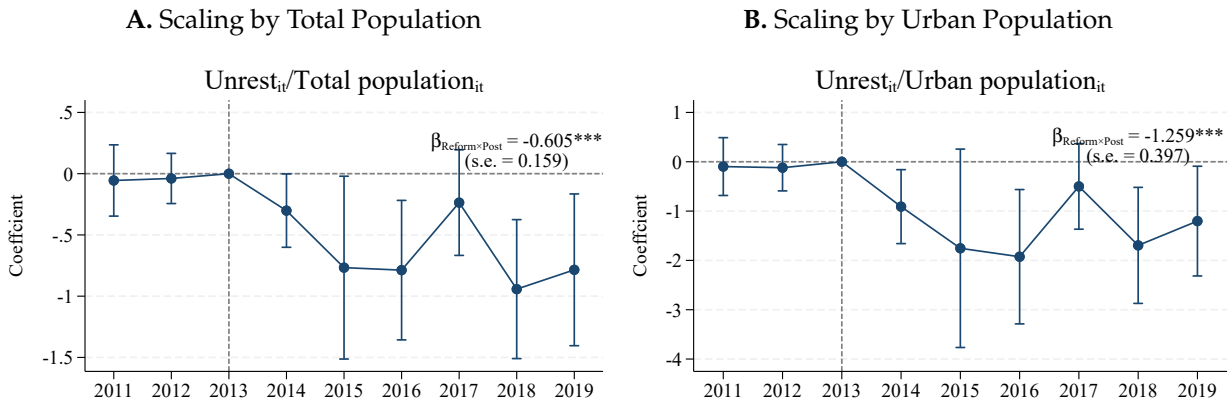


Figure G7. Scaling Unrest Events Using Time-Varying Prefecture Size

Note: This figure reports the results when the number of unrest events is scaled by time-varying population size. Figure G7A uses total population, and Figure G7B uses urban population. We visualize estimates from a dynamic specification: $Y_{it} = \sum_{s \neq 2013} \beta_s \left(Reform_i \times D_s\right) + f\left(\tilde{p}_{i,2014}; \zeta_{Reform,t}\right) + \lambda_i + \mu_t + \varepsilon_{it}$. The solid points are point estimates of β_s's, and the spikes are 95 percent confidence intervals. We also report the estimate from a static specification: $Y_{it} = \beta\left(Reform_i \times Post_t\right) + f\left(\tilde{p}_{i,2014}; \zeta_{Reform,t}\right) + \lambda_i + \mu_t + \varepsilon_{it}$. All standard errors are clustered at the prefecture level.

* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

H Additional Robustness Checks

H.1 Alternative Standard Errors

Table H1 shows that our results are robust to different standard errors. To ease comparison, Column (1) presents results with standard errors clustered at the prefecture level. In Column (2), standard errors are clustered at the province level. As there are only 29 provinces, to deal with potentially invalid asymptotics due to the small number of clusters, we follow Cameron et al. (2008) and implement a wild bootstrap-*t* procedure to compute the *p*-value. In Columns (3) and (4), we use Conley standard errors to address the spatial correlations in error terms (Conley, 1999): we assume that the error terms are serially correlated and spatially correlated across observations within a certain distance (300km or 500km). All the results confirm the statistical significance of our results.

Table H1. Robustness: Alternative Standard Errors

	(1) Unrest/L	(2) Unrest/L	(3) Unrest/L	(4) Unrest/L
Reform × Post	-1.419***	-1.419***	-1.419***	-1.419***
	(0.373)	(0.399)	(0.332)	(0.335)
SE type	Clustered by prefecture	Clustered by province	Conley, 300km	Conley, 500km
Wild-bootstrap *p*-value		0.002		
Observations	2,583	2,583	2,583	2,583

Note: This table examines the robustness to alternative standard errors. Column (1) presents results with standard errors clustered at the prefecture level. Column (2) uses standard errors clustered at the provincial level, and the *p*-value is computed by a wild bootstrap-*t* procedure (Cameron et al., 2008). Columns (3) and (4) use Conley standard errors (Conley, 1999).
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

H.2 Alternative Specifications and Estimators

We show that our results are robust to using alternative specifications and estimators. First, we use alternative forms of unrest rate $\frac{Unrest_{it}}{L_{i,2010}}$. We take the log of the unrest rate (plus one) or use the inverse hyperbolic sine (IHS) transformation. Second, due to the count data nature, we use the pseudo Poisson maximum likelihood estimation Silva and Tenreyro (2006). Third, we estimate a spatial autoregressive (SAR) model to take into account potential spatial spillovers. Table H2 and Figure H1 display the results. All results confirm that the *hukou* reform significantly reduces labor unrest rates.

H.3 Addressing Potential Outliers

To address concerns that our results may be driven by special regional factors, we exclude one province at a time and re-estimate Equation 2. Figure H2 shows the estimated coefficients on $Reform_i \times Post_t$. Compared with the baseline estimate using the full sample, excluding any province

Table H2. Robustness: Alternative Specifications and Estimators

	Alt. Unrest Measures		PPML	SAR
	(1)	(2)	(3)	(4)
	Log(Unrest/L)	IHS(Unrest/L)	Unrest/L	Unrest/L
Reform × Post	-0.305***	-0.383***	-0.545*	-1.456***
	(0.086)	(0.112)	(0.308)	(0.363)
Control mean	1.209	1.549	3.395	3.395
Method	OLS	OLS	PPML	SAR
Prefecture FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes	Yes
Observations	2,583	2,583	2,583	2,583

Note: This table examines the robustness to alternative specifications and estimators. Column (1) takes the log of the unrest rate (plus one). Column (2) takes the inverse hyperbolic sine (IHS) transformation. Column (3) implements pseudo Poisson maximum likelihood (PPML) estimation. Column (4) estimates a spatial autoregressive (SAR) model. Standard errors are clustered at the prefecture level and reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

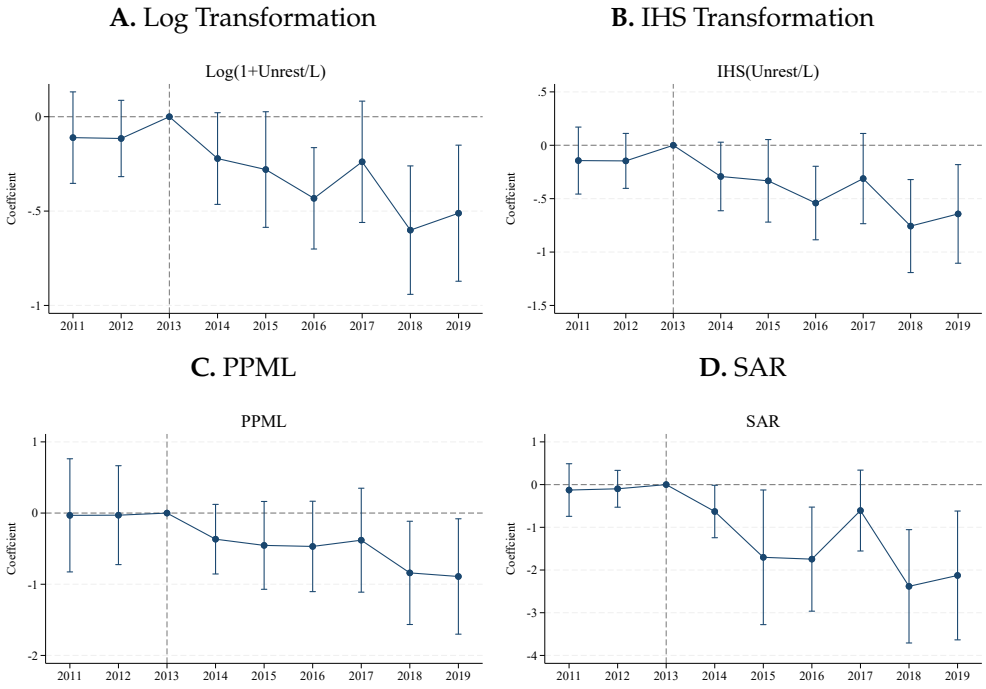


Figure H1. Robustness: Alternative Specifications and Estimators, Event-Study Results

Note: This figure presents the event-study results using alternative specifications and estimators. The solid dots are point estimates, and the spikes are 95 percent confidence intervals. Standard errors clustered at the prefecture level are used to construct confidence intervals.

has little influence on the estimate. We consistently find a negative effect of the *hukou* reform on labor unrest rates.

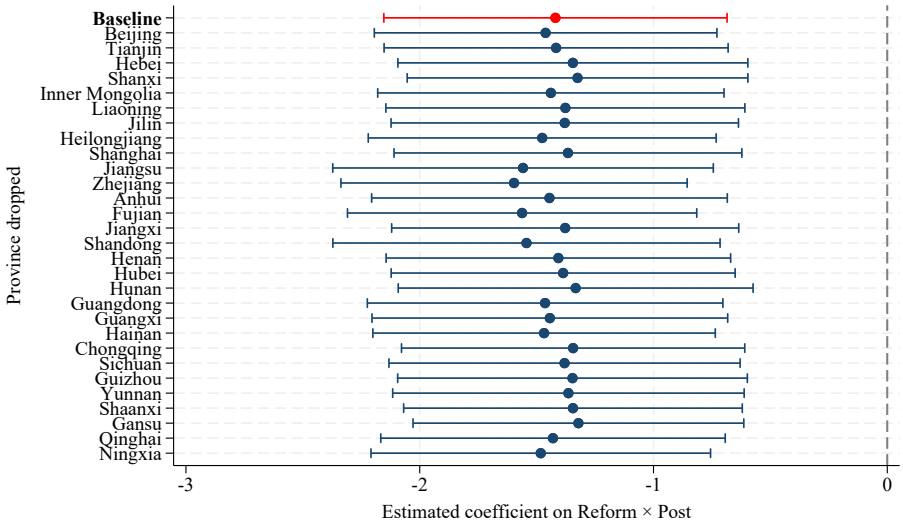


Figure H2. Robustness: Dropping One Province Each Time

Note: This figure reports the estimated coefficient on $Reform_i \times Post_t$ from Equation 2, using the full sample less one province. For comparison, we also present the baseline estimated coefficient using the full sample (in red). The solid points are point estimates, and the spikes are 95 percent confidence intervals. Standard errors are clustered at the prefecture level.

In Table H3, we show our results are not driven by other potential outlier observations. In Column (1), in the spirit of a donut-hole RD exercise that addresses potential sorting around the cutoff, we remove prefectures that have an urban population size very close to the reform cutoff, specifically, $\left|\tilde{p}_i\right| < 0.2$. In Column (2), we exclude 17 prefectures whose population-based reform status does not agree with the manually coded reform status. In Column (3), we follow Hansen (2022, pp. 84–86) to calculate an index of each prefecture's influence on estimation. The index is calculated as follows. We estimate a cross-sectional RD regression that is numerically equivalent to the panel regression, Equation 2:

$$Y_i \equiv \frac{\sum_{t=2014}^{2019} \frac{Unrest_{it}}{L_{i,2010}}}{6} - \frac{\sum_{t=2011}^{2013} \frac{Unrest_{it}}{L_{i,2010}}}{3} = Reform_i + f\left(\tilde{p}_i; \zeta_{Reform}\right) + \varepsilon_i. \tag{H1}$$

Then, the influence index is calculated as $d_i = \hat{Y}_i - \tilde{Y}_i$, where $\hat{Y}_i$ is the predicted value based on a regression using the full sample, while $\tilde{Y}_i$ is the predicted value based on a regression that leaves out the prefecture i. A high $|d_i|$ implies that prefecture i is an influential observation for the overall fit. Thus, we drop prefectures with a high d_i: for reform prefectures we drop the top 25 percent, and for non-reform prefectures, we drop the top 10 percent. Our results survive these exercises and in fact become more pronounced.

H.4 Balancing Baseline Characteristics

Given the balance shown in Table 1, our main specification does not include covariates. Table H4 shows that our results are robust to controlling for covariates, using several different strategies.

Table H3. Robustness: Addressing Potential Outliers

	(1) Near the cutoff	(2) w/ a diff. btw. population-based & manually-coded reforms	(3) w/ high influence
Reform × Post	-1.624***	-1.834***	-1.631***
	(-4.62)	(-4.36)	(-7.29)
Control mean	3.750	3.769	4.053
Prefecture FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes
Observations	2,349	2,430	2,007

Note: This table checks the robustness of our results to potential outliers. Column (1) removes prefectures that have an urban population very close to the reform cutoff, specifically, $|\tilde{p}_i| < 0.2$. Column (2) excludes 17 prefectures whose population-based reform status does not agree with the manually coded reform status. Column (3) follows Hansen (2022) to exclude prefectures that have a strong influence on the overall fit of the data. Standard errors are clustered at the prefecture level and reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

For comparison, Column (1) re-estimates Equation 2 using the sample of prefectures for which all data for baseline covariates are available, which is slightly smaller than the full sample. Column (2) directly controls for interactions between baseline covariates and year indicators in the regression. Column (3) weights observations using the propensity score predicted by the baseline covariates. Column (4) implements the coarsened exact matching (CEM) proposed by Iacus et al. (2012), which weights observations such that the *distributions* of baseline covariates are balanced between reform and non-reform prefectures. All approaches consistently show a negative effect of the *hukou* reform on labor unrest.

Table H4. Robustness: Covariates Balancing

	(1) Unrest/L	(2) Unrest/L	(3) Unrest/L	(4) Unrest/L
Reform × Post	-1.439***	-1.248***	-1.347*	-1.685***
	(0.376)	(0.341)	(0.763)	(0.430)
Balancing approach	-	Controls added	P-score	CEM
Control mean	3.395	3.395	3.100	3.482
Prefecture FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes	Yes
Observations	2,511	2,511	1,683	2,376

Note: This table presents multiple approaches to control for baseline covariates, including the shares of males, urban residents, migrants, construction workers, manufacturing workers, transportation workers, and internet users. Column (1) presents the baseline result for comparison. Column (2) directly includes baseline covariates interacted with year indicators in the regression. Column (3) weights observations to balance the propensity score predicted by covariates. Column (4) implements the coarsened exact matching (CEM, Iacus et al., 2012) to balance distributions of covariates. Standard errors are clustered at the prefecture level and reported in parentheses.

Note: * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

I The Effect of the *Hukou* Reform on Outmigration

I.1 Cox Proportional Hazards Model

As a robustness check, we estimate a Cox proportional hazards model that is commonly used in survival analysis:

$$h\left(Outmigration_{jkt} \mid t, \mathbf{W}\right) = h_0(t) \exp\left[\rho_1 \left(Reform_k \times Post15_t\right) + \rho_2 Reform_k + \rho_3 Post15_t\right]. \quad \text{(I1)}$$

$h\left(Outmigration_{jkt} \mid t, \mathbf{W}\right)$ is the hazard rate of outmigration at time t, conditional on a vector of explanatory variables $\mathbf{W}$ specified on the right-hand side of Equation I1. $h_0(t)$ is a common function of the time-at-risk. Following the semiparametric approach devised by Cox (1972), we leave the baseline hazard function $h_0(t)$ unrestricted and estimate the coefficients by partial maximum likelihood. Standard errors are clustered at the prefecture level.

Table I1 presents the results for the Cox model, which are consistent with those in Table 3. The *hukou* reform reduced the hazard ratio of outmigration—according to Column (3)—by 53.2 percent. The effect is much larger for rural migrants than for urban migrants (Column (4) vs. Column (5)).

Table I1. Effect of the *Hukou* Reform on Outmigration: Cox Hazards Model

	Outmigration from 2010 residing prefecture				
	(1)	(2)	(3)	(4)	(5)
Reform × Post	-1.000***	-0.764***	-0.760***	-0.879***	-0.465*
	(0.129)	(0.232)	(0.223)	(0.245)	(0.254)
Sample	All migrants	All migrants	All migrants	Rural migrants	Urban migrants
Mfx. on hazard rate	-0.632	-0.534	-0.532	-0.585	-0.372
Polynomials		Yes	Yes	Yes	Yes
Stratified hazard function			Yes	Yes	Yes
Observations	100,875	100,875	100,875	58,647	42,228

Note: This table reports the effect of the *hukou* reform on the outmigration rate. When applicable, the baseline hazard function $h_0(t)$ is stratified by birth cohort, gender, educational attainment, and employment status. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

I.2 Addressing Potential Measurement Error in the Migrant Definition

In our sample, an individual is defined as a migrant if their 2010 prefecture of residence differs from their 2015 *hukou* registration prefecture. The definition could introduce potential measurement error if an individual changed *hukou* registration between 2010 and 2015. As a robustness check, we exclude potentially misclassified migrants from the migrant sample and re-estimate Equation 5. Individuals are considered potentially misclassified migrants if they satisfy the following criteria.

1. *In 2010, living in prefectures that later had reforms.* Misclassified migrants are not likely to be natives of prefectures that did not have reforms in 2014. This is because their home prefectures are attractive places in China, so local residents are unlikely to relinquish their *hukou* there.

2. *In 2015, registering hukou in prefectures with lenient hukou transfer policies before the 2014 reform.* Misclassified migrants must have transferred their *hukou*, and they are more likely to transfer to prefectures that already had lenient transfer policies. We identify such lenient prefectures using data from Fan (2019). Fan (2019) evaluates a transfer policy on a 0–3 scale, with 0 being the most restrictive and 3 being the most lenient. He distinguishes between the transfer policies of a prefecture's core district and its peripheral counties. We define a prefecture as lenient if either its core district or its peripheral counties received a score above 2 in 2010 in Fan (2019)'s data.

Table I2 reports results after excluding potentially misclassified migrants. Compared to Table 3, the estimates in fact show larger negative effects of the *hukou* reform on outmigration rates, albeit with larger standard errors due to a 30 percent reduction in sample size. The effects are again driven by rural migrants, indicating that the heterogeneous effects between rural and urban migrants are not due to differential measurement error between the two groups. Measurement error within rural migrants is likely small because urban-to-rural migration is rare and even prohibited.

Table I2. The *Hukou* Reform and Outmigration: Addressing Potential Measurement Error in the Migrant Definition

	Outmigration from 2010 residing prefecture				
	(1)	(2)	(3)	(4)	(5)
Reform × Post	-0.072*	-0.072*	-0.074*	-0.106*	-0.025
	(0.039)	(0.039)	(0.041)	(0.056)	(0.025)
Sample	All migrants	All migrants	All migrants	Rural migrants	Urban migrants
Control mean	0.133	0.133	0.133	0.165	0.089
Prefecture FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Polynomials	Yes	Yes	Yes	Yes	Yes
Origin FE		Yes	Yes	Yes	Yes
Individual covariates × Year FE			Yes	Yes	Yes
Observations	71,214	71,214	71,214	41,362	29,852
R^2	0.075	0.118	0.127	0.181	0.081

Note: This table reports the effect of the *hukou* reform on the outmigration rate, addressing potential measurement error in the migrant definition. We exclude potentially misclassified migrants who (i) in 2010, lived in prefectures that later had reforms, and (ii) in 2015, registered *hukou* in prefectures with lenient *hukou* transfer policies before the 2014 reform. Individual covariates include birth cohorts, gender, educational attainment, and employment status. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

J Causal Mediation Analysis

J.1 Conventional Approach

Suppose that the unit of analysis is denoted by i. This does not lose generality. In a panel, one can define $i \equiv (j, t)$, where j and t index units and periods, respectively. For brevity, we let i index individuals in subsequent discussion.

Let Y_i denote the outcome variable. T_i is the treatment of interest. M_i is the mechanism variable. We are interested in the causal effect that T_i has on Y_i, and how much of this effect is achieved by T_i intervening in M_i. To clarify the parameters of interest, consider a potential outcome framework with linear heterogeneity of causal effects:

$$Y_i(T_i, M_i) = \tau_i T_i + \gamma_i M_i + u_i, \tag{J1}$$

$$M_i(T_i) = \pi_i T_i + v_i. \tag{J2}$$

Therefore,

$$Y_i(T_i) \equiv Y_i\left(T_i, M_i(T_i)\right) = \left(\tau_i + \gamma_i \pi_i\right) T_i + u_i + \gamma_i v_i \equiv \beta_i T_i + \eta_i, \tag{J3}$$

where $\beta_i = \tau_i + \gamma_i \pi_i$ and $\eta_i = u_i + \gamma_i v_i$.

Definition J1 (Causal Parameters of Interest)**.**

1. The *total effect* is $\beta_i = \tau_i + \gamma_i \pi_i$, thus, the average total effect (ATotE) is $E(\beta_i) = E(\tau_i) + E(\gamma_i \pi_i)$.
2. The *indirect effect* that is due to mechanism variable M_i is $\gamma_i \pi_i$, thus, the average indirect effect (AIE) is $E(\gamma_i \pi_i)$.
3. The *direct effect* that is not due to mechanism variable M_i is τ_i, thus, the average direct effect (ADE) is $E(\tau_i)$.

Note that $ATotE = ADE + AIE$. We may want to estimate AIE and compare it to $ATotE$. For instance, $\frac{AIE}{ATotE}$ tells us the share of the total effect mediated by a particular mechanism.

We further assume that T_i is exogenous. This rules out challenges in identifying the causal effects of T_i, allowing us to focus on identifying mediation effects.

Assumption J1 (Treatment Exogeneity)**.** $\{Y_i(t', m), M_i(t)\} \perp\!\!\!\perp T_i$, for all t, t', and m.

The conventional approach for estimating the mediation effect, popularized by Baron and Kenny (1986), is to estimate the following linear simultaneous equations model (LSEM):

$$Y_i = \alpha_1 + \beta T_i + e_{i1}, \tag{J4}$$

$$Y_i = \alpha_2 + \tau T_i + \gamma M_i + e_{i2}, \tag{J5}$$

$$M_i = \alpha_3 + \pi T_i + e_{i3}. \tag{J6}$$

Let $\hat{w}$ denote the OLS estimand of coefficient w, i.e., the probability limit of the OLS estimator. Since we assume that T_i is exogenous, $\hat{\beta}$ from Equation J4 captures the average total effect (ATotE). In Equation J5, intuitively, as M_i is controlled for, $\hat{\tau}$ should capture the average direct effect (ADE), i.e., the effect *not* mediated by M_i. Therefore, the change in the coefficient on T_i after including M_i is interpreted as an estimand for AIE, that is,

$$\widehat{AIE} \equiv \hat{\beta} - \hat{\tau} = \hat{\gamma}\hat{\pi}. \tag{J7}$$

The second equality follows from the properties of least squares. We see that $\widehat{AIE}$ can be written as a product of (i) $\hat{\gamma}$, the estimated effect of M_i on Y_i, and (ii) $\hat{\pi}$, the estimated effect of T_i on M_i.

Despite the simplicity, strong assumptions are required for $\widehat{AIE}$ to identify AIE (e.g., among others, Imai et al., 2011). To see this, let us examine $\widehat{AIE}$ term by term. First consider $\hat{\beta}$.

$$\hat{\beta} = \frac{Cov(T_i, Y_i)}{Var(T_i)} = \frac{Cov(T_i, \beta_i T_i + \eta_i)}{Var(T_i)} = E(\beta_i). \tag{J8}$$

The second equality is by plugging in the potential outcome Equation J3. The third equality is by Assumption J1. Thus, $\hat{\beta}$ identifies the ATotE.

Next, consider $\hat{\tau}$. By the property of least squares,

$$\hat{\tau} = \frac{Cov(T_i, Y_i - \hat{\gamma} M_i)}{Var(T_i)} = \hat{\beta} - \hat{\gamma}\hat{\pi}, \tag{J9}$$

where $\hat{\pi} = E(\pi_i)$ identifies the population average effect of T_i on M_i under Assumption J1.

Now, we derive $\hat{\gamma}$. Let $L(R \mid T)$ denote the linear projection of R on T and a constant 1, and $\widetilde{R}$ is the linear projection residual, namely, $\widetilde{R} = R - L(R \mid T)$. By Assumption J1 and Equation J2, $L(M_i \mid T_i) = E(M_i \mid T_i)$, i.e., the linear projection recovers the conditional mean.[17] $\hat{\gamma}$ has the following expression:

$$\hat{\gamma} = \frac{E(\widetilde{M}_i Y_i)}{E(\widetilde{M}_i^2)} \tag{J10}$$

$$= \frac{E\left[\widetilde{M}_i(\tau_i T_i + \gamma_i M_i + u_i)\right]}{E(\widetilde{M}_i^2)} \tag{J11}$$

$$= \frac{E(\widetilde{M}_i M_i \gamma_i)}{E(\widetilde{M}_i^2)} + \frac{E(\widetilde{M}_i T_i \tau_i)}{E(\widetilde{M}_i^2)} + \frac{E(\widetilde{M}_i u_i)}{E(\widetilde{M}_i^2)}. \tag{J12}$$

[17] This follows from the fact that the conditional mean $E(M_i \mid T_i) = E(\pi_i \mid T_i)T_i + E(v_i \mid T_i) = E(\pi_i)T_i + E(v_i)$ is linear.

The first term is a weighted average of γ_i, albeit different from the population average $E(\gamma_i)$. The second term picks up the causal effect of T_i, representing a contamination bias as in Goldsmith-Pinkham et al. (2022). The third term is the endogeneity bias. Therefore, $\hat{\gamma}$ is biased for $E(\gamma_i)$. In fact, it does not even identify a causal effect of M_i on Y_i. This bias is expected, since no exogenous variation in M_i conditional on T_i warrants identification.

Taken together,

$$\widehat{AIE} = \hat{\beta} - \hat{\tau} = \left[\frac{E(\widetilde{M}_i M_i \gamma_i)}{E(\widetilde{M}_i^2)} + \frac{E(\widetilde{M}_i T_i \tau_i)}{E(\widetilde{M}_i^2)} + \frac{E(\widetilde{M}_i u_i)}{E(\widetilde{M}_i^2)} \right] E(\pi_i). \tag{J13}$$

Note that

$$AIE \equiv E(\gamma_i \pi_i) = E(\gamma_i)E(\pi_i) + Cov(\gamma_i, \pi_i). \tag{J14}$$

Therefore, the bias of using $\widehat{AIE}$ to estimate AIE is:

$$\text{Bias} = \widehat{AIE} - AIE = \underbrace{[\hat{\gamma} - E(\gamma_i)]\, E(\pi_i)}_{\text{bias 1}} + \underbrace{Cov(\gamma_i, \pi_i)}_{\text{bias 2}}. \tag{J15}$$

There are two sources of bias.

- Bias 1 arises because $\hat{\gamma}$ does not identify $E(\gamma_i)$. This is due to the endogeneity of M_i.
- Bias 2 is a mechanical consequence of LSEM. The LSEM overlooks heterogeneity in γ_i and uses an *average* slope of Y_i for M_i, $E(\gamma_i)$, to evaluate M_i's marginal effect to Y_i, π_i. This produces bias if π_i is correlated with γ_i: consider a positive correlation; then the average slope systemically understates contributions of high π_i's and overstates contributions of low π_i's. This is often called an *ecological bias* (Glynn, 2012).

Researchers often assume homogeneity of γ_i, which eliminates bias 2. But bias 1 still exists. Imai et al. (2011) show that a sufficient condition to eliminate this bias is a "sequential ignorability" assumption, which essentially assumes some exogenous variation in M_i.

Assumption J2 (Sequential Ignorability). $Y_i(t', m) \perp\!\!\!\perp M_i(t) \mid T_i = t, \forall t, t', m.$

That said, M_i is exogenous conditional on T_i, indicating that there are no other post-treatment confounders. With this assumption, we can derive the following results.

$$\begin{aligned} E(\widetilde{M}_i T_i \tau_i) &= E[E(\widetilde{M}_i T_i \tau_i \mid T_i)] && \text{(J16)} \\ &= E[E(\widetilde{M}_i T_i \mid T_i) E(\tau_i \mid T_i)] && \text{(J17)} \\ &= E(\widetilde{M}_i T_i) E(\tau_i) && \text{(J18)} \\ &= 0. && \text{(J19)} \end{aligned}$$

The first equality is by the law of iterated expectations (LIE). The second equality is by Assumption J2. The third equality is by Assumption J1. The fourth equality is by the orthogonality of linear projection residuals. Using similar tricks yields

$$E(\widetilde{M}_i u_i) = E[E(\widetilde{M}_i u_i \mid T_i)] \tag{J20}$$
$$= E[E(\widetilde{M}_i \mid T_i)E(u_i \mid T_i)] \tag{J21}$$
$$= E(\widetilde{M}_i)E(u_i) \tag{J22}$$
$$= 0. \tag{J23}$$

Taking stock, under the homogeneity assumption $\gamma_i = \gamma_0$,

$$\hat{\gamma} = \frac{E(\widetilde{M}_i M_i \gamma_i)}{E(\widetilde{M}_i^2)} + \cancel{\frac{E(\widetilde{M}_i T_i \tau_i)}{E(\widetilde{M}_i^2)}} + \cancel{\frac{E(\widetilde{M}_i u_i)}{E(\widetilde{M}_i^2)}} = \frac{E(\widetilde{M}_i M_i \gamma_0)}{E(\widetilde{M}_i^2)} = \gamma_0, \tag{J24}$$

where the last equality follows from $E(\widetilde{M}_i M_i) = E(\widetilde{M}_i^2)$. Since $E(\gamma_i) = \gamma_0 = \hat{\gamma} = E(\gamma_i)$ under the homogeneity assumption, we have

$$\widehat{AIE} = AIE, \tag{J25}$$

i.e., $\widehat{AIE}$ identifies AIE.[18]

J.2 IV-Augmented Approach

J.2.1 General Case

Assumption J2 may not be plausible, as it imposes strong exogeneity to M_i. Without this assumption, the key problem is that $\hat{\gamma}$ does not identify a causal effect of M_i on Y_i, not to mention $E(\gamma_i)$. We propose using an IV for M_i, denoted by Z_i, to identify a causal effect of M_i on Y_i. We note potential issues in this approach. First, IV validity requires additional assumptions. Second, even if IV is valid, it generally does not identify the population average effect $E(\gamma_i)$ unless γ_i is constant or other assumptions are imposed. Nonetheless, we view the IV-augmented approach as an improvement over the conventional approach because it credibly identifies a causal effect of M_i on Y_i.

We extend the potential outcome framework to incorporate IV Z_i.

$$Y_i(T_i, M_i) = \tau_i T_i + \gamma_i M_i + u_i, \tag{J26}$$
$$M_i(T_i, Z_i) = \pi_i T_i + \theta_i Z_i + v_i. \tag{J27}$$

[18] This result does not necessarily require the homogeneity assumption. With Assumption J2, one can further show that $E(\widetilde{M}_i M_i \gamma_i) = E(\widetilde{M}_i M_i \gamma_i \mid T_i) = E(\widetilde{M}_i M_i)E(\gamma_i) = E(\widetilde{M}_i^2)E(\gamma_i)$. Thus, $\hat{\gamma} = E(\gamma_i)$, hence, bias 1 = 0. Assumption J2 also implies that $\gamma_i \perp\!\!\!\perp \pi_i \mid T_i$. Thus, bias 2 $= E[E(\gamma_i \pi_i \mid T_i)] - E(\gamma_i)E(\pi_i) = E[E(\gamma_i \mid T_i)E(\pi_i \mid T_i)] - E(\gamma_i)E(\pi_i) = 0$, where the first equality is by the definition of covariance and the LIE, the second equality is by $\gamma_i \perp\!\!\!\perp \pi_i \mid T_i$, and the last equality is by Assumption J1. Taken together, $\widehat{AIE} = AIE$, even if the homogeneity assumption is not made.

Therefore, we have the following reduced-form model:

$$Y_i(T_i, Z_i) \equiv Y_i\left(T_i, M_i(T_i, Z_i)\right) \tag{J28}$$
$$= \left(\pi_i + \gamma_i \pi_i\right) T_i + \gamma_i \theta_i Z_i + \left(u_i + \gamma_i v_i\right) \tag{J29}$$
$$= \beta_i T_i + \rho_i Z_i + \eta_i, \tag{J30}$$

where $\beta_i = \tau_i + \gamma_i \pi_i$, $\rho_i = \gamma_i \theta_i$, and $\eta_i = u_i + \gamma_i v_i$.

We impose the following assumptions.

Assumption J3 (Treatment Exogeneity). $\{Y_i(t', m), M_i(t, z), Z_i\} \perp\!\!\!\perp T_i$, for all t, t', m, and z.

Assumption J4 (IV Validity).

1. (Independence) $\{Y_i(t', m), M_i(t, z)\} \perp\!\!\!\perp Z_i$ for all t, t', m, and z.
2. (Exclusion) $Y_i(t, m) \,|_z = Y_i(t, m) \,|_{z'}$ for all z and z'.
3. (Relevance) $E(\theta_i) \neq 0$.
4. (Monotonicity) Either $\Pr(\theta_i \geq 0) = 1$ or $\Pr(\theta_i \leq 0) = 1$.

Proposition J1. *Under Assumptions J3 and J4, two-stage least squares (2SLS) estimation of the LSEM, with M_i instrumented by Z_i, yields*

$$\widehat{AIE} = \hat{\beta} - \hat{\tau} = \hat{\gamma}\hat{\pi} = \underbrace{E(\gamma_i \pi_i)}_{AIE} + \underbrace{\left[\hat{\gamma} - E(\gamma_i)\right] E(\pi_i)}_{bias\ 1} - \underbrace{Cov(\gamma_i, \pi_i)}_{bias\ 2}, \tag{J31}$$

where $\hat{\beta} = E(\beta_i)$ and $\hat{\gamma} = E\left[\frac{\theta_i}{E(\theta_i)}\gamma_i\right]$. $\hat{\beta} - \hat{\tau}$ identifies AIE if (i) γ_i is constant, or (ii) $\{\theta_i, \pi_i\} \perp\!\!\!\perp \gamma_i$.

Proof. By Assumption J3, it is straightforward to show that

$$\hat{\beta} = \frac{Cov(T_i, Y_i)}{Var(T_i)} = E(\beta_i), \tag{J32}$$
$$\hat{\pi} = \frac{Cov(T_i, M_i)}{Var(T_i)} = E(\pi_i). \tag{J33}$$

By the property of least squares, $\hat{\tau} = \frac{Cov(T_i, Y_i - \hat{\gamma} M_i)}{Var(T_i)} = \hat{\beta} - \hat{\gamma}\hat{\pi}$. We now derive the 2SLS estimand $\hat{\gamma}$. It can be written as:

$$\hat{\gamma} = \frac{E(\widetilde{Z}_i Y_i)}{E(\widetilde{Z}_i M_i)}, \tag{J34}$$

where $\widetilde{Z}_i = Z_i - L(Z_i \mid T_i)$ is the linear projection residual. By Assumption J3, $E(Z_i \mid T_i) = E(Z_i)$ is linear, thus, $L(Z_i \mid T_i) = E(Z_i \mid T_i)$. We can show the following result.

$$
\begin{aligned}
E(\widetilde{Z}_i Y_i) &= E[\widetilde{Z}_i(\beta_i T_i + \rho_i Z_i + \eta_i)] && \text{(J35)}\\
&= E(\widetilde{Z}_i T_i)E(\beta_i) + E(\widetilde{Z}_i Z_i)E(\rho_i) + E(\widetilde{Z}_i)E(\eta_i), && \text{(J36)}\\
&= E(\widetilde{Z}_i Z_i)E(\rho_i). && \text{(J37)}
\end{aligned}
$$

The second equality is by Assumptions J3 and J4(1). The third equality uses the fact that $\widetilde{Z}_i = Z_i - E(Z_i \mid T_i) = Z_i - E(Z_i)$. With similar tricks, the following result follows:

$$
\begin{aligned}
E(\widetilde{Z}_i M_i) &= E[\widetilde{Z}_i(\pi_i T_i + \theta_i Z_i + u_i)] && \text{(J38)}\\
&= E(\widetilde{Z}_i T_i)E(\pi_i) + E(\widetilde{Z}_i Z_i)E(\theta_i) + E(\widetilde{Z}_i)E(u_i), && \text{(J39)}\\
&= E(\widetilde{Z}_i Z_i)E(\theta_i). && \text{(J40)}
\end{aligned}
$$

Taken together,

$$
\hat{\gamma} = \frac{E(\rho_i)}{E(\theta_i)} = \frac{E(\theta_i \gamma_i)}{E(\theta_i)}. \tag{J41}
$$

An alternative expression for $\hat{\gamma}$ is

$$
\hat{\gamma} = \frac{E(\widetilde{Z}_i M_i \gamma_i)}{E(\widetilde{Z}_i M_i)}. \tag{J42}
$$

Therefore,

$$
\begin{aligned}
\widehat{AIE} &= \hat{\beta} - \hat{\tau} && \text{(J43)}\\
&= \hat{\gamma} E(\pi_i) && \text{(J44)}\\
&= \hat{\gamma} E(\pi_i) + E(\gamma_i \pi_i) - Cov(\gamma_i, \pi_i) + E(\gamma_i)E(\pi_i) && \text{(J45)}\\
&= \underbrace{E(\gamma_i \pi_i)}_{\text{AIE}} + \underbrace{[\hat{\gamma} - E(\gamma_i)]\, E(\pi_i)}_{\text{bias 1}} - \underbrace{Cov(\gamma_i, \pi_i)}_{\text{bias 2}}, && \text{(J46)}
\end{aligned}
$$

where $\hat{\gamma} = \frac{E(\widetilde{Z}_i M_i \gamma_i)}{E(\widetilde{Z}_i M_i)} = \frac{E(\theta_i \gamma_i)}{E(\theta_i)}$. It is obvious that bias 1 = bias 2 = 0 if (i) γ_i is constant, or (ii) $\{\theta_i, \pi_i\} \perp\!\!\!\perp \gamma_i$, making $\widehat{AIE} = AIE$. ■

Proposition J1 implies that even with instrumentation, $\widehat{AIE}$, the change in the coefficient on T_i after controlling for instrumented M_i, may still be biased for the AIE. Specifically, the bias comes from two sources.

1. **Bias 1 ("IV bias").** The first bias comes from the discrepancy between the IV-identified average effect and the population average effect, i.e., $\hat{\gamma}$ versus $E(\gamma_i)$. Intuitively, this bias occurs because the IV identifies the effect of M_i on Y_i within a subpopulation that may differ from the population in which the effect of T_i on Y_i is identified. Therefore, without assuming

constant γ_i, the IV-identified effect of M_i on Y_i cannot be perfectly generalized to another population.

2. **Bias 2 ("ecological bias").** The second bias is due to the correlation between γ_i and π_i. This is the classical ecological bias, which occurs when the characteristics of a population are attributed to an individual (Glynn, 2012). The intuition is as follows. Ideally, one would like to use γ_i, *individual-specific* effect of M_i on Y_i, as the loading on π_i to evaluate the contribution of the treatment effect on M_i to the total treatment effect. However, the linear regression can only use $\hat{\gamma}$, an *average* impact of M_i on Y_i, as the loading on π_i. As a result, the linear regression overlooks co-movement of γ_i and π_i, thus, it may systematically mis-measure the importance of the treatment-induced change in M_i to the outcome at the individual level and also at the aggregate level.[19] Clearly, by assuming constant γ_i, such bias is zero.

In sum, assuming γ_i to be homogeneous eliminates bias 1 and bias 2. Then, under this assumption, using a valid IV Z_i for M_i allows $\widehat{AIE}$ to identify AIE.

J.2.2 Extension: Regression Discontinuity

As we implement an RDD in this paper, we extend the above results to RDDs. If one takes a local randomization view of RDDs, i.e., treatment is randomly assigned around the cutoff, the results above apply directly within a very small bandwidth around the cutoff. No polynomial functions are necessary, provided rich data are available even after bandwidth restrictions. In the following, however, we focus on parametric RDDs, which model the conditional means of potential outcomes and thus require polynomial controls.

Let r_i denote the running variable. $T_i = \mathbb{1}\{r_i \geq 0\}$. $\mathbf{R}_i = \left(r_i, r_i^2, \cdots, r_i^p\right)$ is a set of power functions of r_i, up to order p. Also define $\mathbf{R}_i^{(0)} = (1, \mathbf{R}_i)$. $\mathbf{X}_i = [\mathbf{R}_i T_i, \mathbf{R}_i(1 - T_i)]'$ is the polynomial function to be included in RD regressions. Also define $\mathbf{X}_i^{(0)} = \left[\mathbf{R}_i^{(0)} T_i, \mathbf{R}_i^{(0)}(1 - T_i)\right]'$. With the introduction of the running variable, the LSEM to estimate now becomes:

$$Y_i = \alpha_1 + \beta T_i + \mathbf{X}_i' \zeta_1 + e_{i1}, \tag{J47}$$

$$Y_i = \alpha_2 + \tau T_i + \gamma M_i + \mathbf{X}_i' \zeta_2 + e_{i2}, \tag{J48}$$

$$M_i = \alpha_3 + \pi T_i + \mathbf{X}_i' \zeta_3 + e_{i3}. \tag{J49}$$

In RDDs, parameters of interest are causal effects at the cutoff $r_i = 0$. They are defined as follows.

Definition J2 (Parameters of Interest in RDDs)**.**

[19] For instance, suppose $Cov(\gamma_i, \pi_i) > 0$ and $\gamma_i, \pi_i > 0$. Then, within the low-γ_i group, the contribution of the treatment-induced change in M_i (i.e., π_i) is systematically overstated, while within the high-γ_i group, the contribution is systematically understated.

1. The total effect is $\beta_i = \tau_i + \gamma_i\pi_i$. ATotE is defined as the average total effect at the cutoff $r_i = 0$, $E(\beta_i \mid r_i = 0) = E(\tau_i \mid r_i = 0) + E(\gamma_i\pi_i \mid r_i = 0)$.
2. The indirect effect that is due to mechanism variable M_i is $\gamma_i\pi_i$. AIE is defined as the average indirect effect at the cutoff $r_i = 0$, $E(\gamma_i\pi_i \mid r_i = 0)$.
3. The direct effect that is not due to mechanism variable M_i is τ_i. ADE is defined as the average direct effect at the cutoff $r_i = 0$, $E(\tau_i \mid r_i = 0)$.

Assumption J5 (Linearity of Conditional Means)**.**

1. $E[Y_i(1, Z_i) \mid r_i]$ and $E[Y_i(0, Z_i) \mid r_i]$ are linear in $\mathbf{R}_i^{(0)}$.
2. $E[M_i(1, Z_i) \mid r_i]$ and $E[M_i(0, Z_i) \mid r_i]$ are linear in $\mathbf{R}_i^{(0)}$.

Assumption J6 (IV Validity)**.**

1. (Independence) $\{Y_i(t', m), M_i(t, z)\} \perp\!\!\!\perp Z_i \mid r_i$ for all t, t', m, and z.
2. (Exclusion) $Y_i(t, m) \mid_z = Y_i(t, m) \mid_{z'}$ for all z and z'.
3. (Relevance) $E(\widetilde{Z}_i M_i) \neq 0$, where $\widetilde{Z}_i = Z_i - L\left[Z_i \mid \mathbf{X}_i^{(0)}\right]$ is the linear projection residual.
4. (Monotonicity) Either $\Pr(\theta_i \geq 0) = 1$ or $\Pr(\theta_i \leq 0) = 1$.

Assumption J7 (IV Linearity)**.** $E(Z_i \mid r_i)$ is linear in $\mathbf{X}_i^{(0)}$.

Assumption J5 specifies conditional means of potential outcomes. It implies the continuity of conditional means. The assumption also allows us to abstract away from estimation complications in parametric RDDs, e.g., bandwidth selection. Assumption J6 warrants the validity of IV. Importantly, independence only needs to hold conditional on the running variable r_i. Assumption J7 assumes linearity of IV, as in Ishimaru (2024).

Proposition J2. *Under Assumptions J5, J6, and J7, 2SLS estimation of the LSEM, with M_i instrumented by Z_i, yields*

$$\begin{aligned}\widehat{AIE} &= \hat{\beta} - \hat{\tau} = \hat{\gamma}\hat{\pi} \\ &= \underbrace{E(\gamma_i\pi_i \mid r_i = 0)}_{AIE} + \underbrace{\left[\hat{\gamma} - E(\gamma_i \mid r_i = 0)\right] E(\pi_i \mid r_i = 0)}_{bias\ 1} - \underbrace{Cov(\gamma_i, \pi_i \mid r_i = 0)}_{bias\ 2},\end{aligned} \tag{J50}$$

where $\hat{\beta} = E(\beta_i \mid r_i = 0)$ and $\hat{\gamma} = \frac{E(\widetilde{Z}_i M_i \gamma_i)}{E(\widetilde{Z}_i M_i)} = \frac{E[Var(Z_i|r_i)E(\theta_i\gamma_i|r_i)]}{E[Var(Z_i|r_i)E(\theta_i|r_i)]}$. $\widehat{AIE}$ identifies AIE if (i) γ_i is constant, or (ii) $\{\theta_i, \pi_i\} \perp\!\!\!\perp \gamma_i \mid r_i$ and $\gamma_i \perp\!\!\!\perp r_i$.

Proof. By Assumption J5, linear regression identifies conditional means of potential outcomes. Thus, a linear regression of Y_i on 1, T_i and $\mathbf{X}_i$ yields

$$\hat{\beta} = \lim_{r \downarrow 0} E[Y_i(1, Z_i) \mid r_i = r] - \lim_{r \uparrow 0} E[Y_i(0, Z_i) \mid r_i = r] \tag{J51}$$

$$= \lim_{r \downarrow 0} E[\beta_i + \rho_i Z_i + \eta_i \mid r_i = r] - \lim_{r \uparrow 0} E[\rho_i Z_i + \eta_i \mid r_i = r] \tag{J52}$$

$$= E(\beta_i \mid r_i = 0), \tag{J53}$$

where the first equality is by linearity from Assumption J5, the second equality plugs in potential outcomes, and the last equality uses continuity implied by J5. Similarly, one can show that $\hat{\pi} = E(\pi \mid r_i = 0)$.

By the property of least squares, $\hat{\tau} = \hat{\beta} - \hat{\gamma}\hat{\pi}$. Now derive $\hat{\gamma} = \frac{E(\widetilde{Z}_i Y_i)}{E(\widetilde{Z}_i M_i)}$, where $\widetilde{Z}_i = Z_i - L\left[Z_i \mid \mathbf{X}_i^{(0)}\right]$. $\hat{\gamma}$ is written as:

$$\hat{\gamma} = \frac{E[\widetilde{Z}_i(\beta_i T_i + \rho_i Z_i + \eta_i)]}{E[\widetilde{Z}_i(\pi_i T_i + \theta_i Z_i + v_i)]}. \tag{J54}$$

Analyze term by term.

$$E(\widetilde{Z}_i T_i \beta_i) = E[E(\widetilde{Z}_i T_i \beta_i \mid r_i)] = E[E(\tilde{Z}_i \mid r_i) T_i E(\beta_i \mid r_i)] = 0. \tag{J55}$$

The first equality is by the LIE. The second equality is by Assumption J6(1) and the fact that T_i is completely determined by r_i in a RDD. The last equality is due to Assumption J7 that implies $L\left[Z_i \mid \mathbf{X}_i^{(0)}\right] = E\left[Z_i \mid \mathbf{X}_i^{(0)}\right]$. Similarly,

$$E(\widetilde{Z}_i T_i \pi_i) = 0, \tag{J56}$$

$$E(\widetilde{Z}_i \eta_i) = 0, \tag{J57}$$

$$E(\widetilde{Z}_i v_i) = 0. \tag{J58}$$

Therefore,

$$\hat{\gamma} = \frac{E(\widetilde{Z}_i Z_i \theta_i \gamma_i)}{E(\widetilde{Z}_i Z_i \theta_i)} = \frac{E[E(\widetilde{Z}_i Z_i \mid r_i) E(\theta_i \gamma_i \mid r_i)]}{E[E(\widetilde{Z}_i Z_i \mid r_i) E(\theta_i \mid r_i)]} = \frac{E[Var(Z_i \mid r_i) E(\theta_i \gamma_i \mid r_i)]}{E[Var(Z_i \mid r_i) E(\theta_i \mid r_i)]}. \tag{J59}$$

An alternative expression for $\hat{\gamma}$ is

$$\hat{\gamma} = \frac{E(\widetilde{Z}_i M_i \gamma_i)}{E(\widetilde{Z}_i M_i)}. \tag{J60}$$

Taken together,

$$\hat{\beta} - \hat{\tau} = \hat{\gamma} E(\pi_i \mid r_i = 0) \tag{J61}$$

$$= \hat{\gamma} E(\pi_i \mid r_i = 0) + E(\gamma_i \pi_i \mid r_i = 0) - Cov(\gamma_i, \pi_i \mid r_i = 0) + E(\gamma_i \mid r_i = 0) E(\pi_i \mid r_i = 0) \tag{J62}$$

$$= \underbrace{E(\gamma_i \pi_i \mid r_i = 0)}_{\text{AIE}} + \underbrace{\left[\hat{\gamma} - E(\gamma_i \mid r_i = 0)\right] E(\pi_i \mid r_i = 0)}_{\text{bias 1}} - \underbrace{Cov(\gamma_i, \pi_i \mid r_i = 0)}_{\text{bias 2}}, \tag{J63}$$

where $\hat{\gamma} = \frac{E(\widetilde{Z}_i M_i \gamma_i)}{E(\widetilde{Z}_i M_i)} = \frac{E[Var(Z_i|r_i)E(\theta_i \gamma_i|r_i)]}{E[Var(Z_i|r_i)E(\theta_i|r_i)]}$. It is obvious that bias 1 = bias 2 = 0 if (i) γ_i is constant, or (ii) $\{\theta_i, \pi_i\} \perp\!\!\!\perp \gamma_i \mid r_i$ and $\gamma_i \perp\!\!\!\perp r_i$. ■

In conclusion, under RD settings, $\widehat{AIE}$ identifies AIE when assuming that γ_i is constant. In Section 5, we use this result to establish our baseline causal mediation results.

J.3 Bounding Exercise

Discussions above show that $\widehat{AIE} = \hat{\beta} - \hat{\tau}$ identifies AIE if γ_i is homogeneous across i. However, this assumption may be overly strong in many applications, resulting in overestimation or underestimation of the importance of a mechanism. In this section, we discuss under what conditions $\hat{\beta} - \hat{\tau}$ remains informative about AIE, even if γ_i is heterogeneous.

J.3.1 General Case

Consider our baseline setup. Note that the bias is expressed as

$$Bias = \widehat{AIE} - AIE \tag{J64}$$

$$= \hat{\gamma}\hat{\pi} - E(\gamma_i \pi_i) \tag{J65}$$

$$= \left[\hat{\gamma} - E(\gamma_i)\right] \hat{\pi} - Cov(\gamma_i, \pi_i) \tag{J66}$$

$$= Cov(\phi_i, \gamma_i)\hat{\pi} - Cov(\gamma_i, \pi_i) \tag{J67}$$

$$= \rho_{\phi\gamma}\sigma_\phi\sigma_\gamma\hat{\pi} - \rho_{\gamma\pi}\sigma_\gamma\sigma_\pi, \tag{J68}$$

where $\phi_i = \frac{\widetilde{Z}_i M_i}{E(\widetilde{Z}_i M_i)}$, $\rho_{\phi\gamma} = Corr(\phi_i, \gamma_i)$, $\rho_{\gamma\pi} = Corr(\gamma_i, \pi_i)$, $\sigma_\phi = SD(\phi_i)$, $\sigma_\gamma = SD(\gamma_i)$, and $\sigma_\pi = SD(\pi_i)$. If there is knowledge of $Cov(\phi_i, \gamma_i)$ and $Cov(\gamma_i, \pi_i)$, together with $\hat{\gamma}\hat{\pi}$, we can de-bias or bound AIE. We propose one approach below.

Assumption J8 (Sign and Distributional Restrictions)**.**

1. All γ_i's have the same sign. All π_i's have the same sign.
2. Both γ_i and π_i are uniformly distributed over intervals with one endpoint at 0.

We view the sign restriction as a minimum requirement for causal mediation. Without it, $AIE = E(\gamma_i \pi_i)$ could be zero at the aggregate level, even if the indirect effect $\gamma_i \pi_i$ is in fact nonzero

for most individuals. Researchers also often implicitly make the sign restriction when investigating mechanisms qualitatively. For instance, after estimating an average treatment effect on an outcome, a researcher may argue that the effect operates through a particular mechanism by showing that the treatment also has a significant average effect on the proposed mediator. Such an argument typically relies on a prior belief about the *sign* of the mediator's effect on the outcome, even though no formal causal mediation analysis is performed.

Proposition J3. *Under Assumption J8, for every given $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$, AIE is identified by the estimand*

$$\widehat{\widehat{AIE}} = \hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}, \rho_{\gamma\pi}\right)\hat{\pi}. \tag{J69}$$

where $\hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}, \rho_{\gamma\pi}\right) = \left[\frac{\sqrt{3}\,\mathrm{sgn}(\gamma_i)\hat{\gamma}}{\sqrt{3}\,\mathrm{sgn}(\gamma_i)+\rho_{\phi\gamma}\hat{\sigma}_\phi} + \frac{\rho_{\gamma\pi}\hat{\sigma}_\phi\,\mathrm{sgn}(\pi_i)}{\sqrt{3}}\right]$, $\hat{\sigma}_\phi = \mathrm{plim}\sqrt{\frac{1}{n}\sum_{i=1}^n\left(\frac{\widetilde{Z}_i M_i}{\frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i M_i} - 1\right)^2}$, *and* $\mathrm{sgn}(\gamma_i)$ *and* $\mathrm{sgn}(\pi_i)$ *respectively give signs of* γ_i *and* π_i. $\hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}, \rho_{\gamma\pi}\right)$ *is monotonic in* $\rho_{\phi\gamma}$ *and* $\rho_{\gamma\pi}$.

Proof. By the expression of *Bias*,

$$\hat{\gamma}\hat{\pi} - E(\gamma_i)\hat{\pi} = \rho_{\phi\gamma}\sigma_\phi\sigma_\gamma\hat{\pi}. \tag{J70}$$

Under Assumption J8, by properties of uniform distributions, $E(\gamma_i) = \sqrt{3}\,\mathrm{sgn}(\gamma_i)\sigma_\gamma$. Thus, $\sigma_\gamma = \frac{\hat{\gamma}}{\sqrt{3}\,\mathrm{sgn}(\gamma_i)+\rho_{\phi\gamma}\sigma_\phi}$. Since π_i satisfies a uniform distribution and $\hat{\pi} = E(\pi_i)$, $\sigma_\pi = \frac{\hat{\pi}\,\mathrm{sgn}(\pi_i)}{\sqrt{3}}$. Taken together,

$$AIE = \left[\frac{\sqrt{3}\,\mathrm{sgn}(\gamma_i)\hat{\gamma}}{\sqrt{3}\,\mathrm{sgn}(\gamma_i) + \rho_{\phi\gamma}\sigma_\phi} + \frac{\rho_{\gamma\pi}\sigma_\phi\,\mathrm{sgn}(\pi_i)}{\sqrt{3}}\right]\hat{\pi}. \tag{J71}$$

In this expression, σ_ϕ is unknown. However, consider the estimator $\sqrt{\frac{1}{n}\sum_{i=1}^n\left(\frac{\widetilde{Z}_i M_i}{\frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i M_i} - 1\right)^2}$, the probability limit of which is denoted by $\hat{\sigma}_\phi$. By the weak law of large numbers, $\hat{\sigma}_\phi = \sigma_\phi$. As such, AIE is identified by

$$\widehat{\widehat{AIE}} = \left[\frac{\sqrt{3}\,\mathrm{sgn}(\gamma_i)\hat{\gamma}}{\sqrt{3}\,\mathrm{sgn}(\gamma_i) + \rho_{\phi\gamma}\hat{\sigma}_\phi} + \frac{\rho_{\gamma\pi}\hat{\sigma}_\phi\,\mathrm{sgn}(\pi_i)}{\sqrt{3}}\right]\hat{\pi} \equiv \hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}, \rho_{\gamma\pi}\right)\hat{\pi}. \tag{J72}$$

for every given $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$. The monotonicity of $\hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}, \rho_{\gamma\pi}\right)$ in $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$ is obvious. ■

If we assume $\gamma_i \perp\!\!\!\perp \pi_i$, then by Proposition J1, the only bias in $\widehat{AIE}$ is the IV bias. We can correct the bias for every given $\rho_{\phi\gamma}$. This is summarized by the following corollary.

Corollary J1. *If $\gamma_i \perp\!\!\!\perp \pi_i$, for every given $\rho_{\phi\gamma}$, AIE is identified by*

$$\widehat{\widehat{AIE}} = \hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}\right)\hat{\pi}. \tag{J73}$$

where $\hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}\right) = \frac{\sqrt{3}\,\mathrm{sgn}(\gamma_i)\hat{\gamma}}{\sqrt{3}\,\mathrm{sgn}(\gamma_i)+\rho_{\phi\gamma}\hat{\sigma}_\phi}$, $\hat{\sigma}_\phi = \mathrm{plim}\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{\widetilde{Z}_i M_i}{\frac{1}{n}\sum_{i=1}^{n}\widetilde{Z}_i M_i}-1\right)^2}$, *and* $\mathrm{sgn}(\gamma_i)$ *and* $\mathrm{sgn}(\pi_i)$ *respectively give signs of* γ_i *and* π_i. $\hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}\right)$ *is monotonic in* $\rho_{\phi\gamma}$.

Proof. When $\gamma_i \perp\!\!\!\perp \pi_i$, $\rho_{\gamma\pi} = 0$. Applying Proposition J3 completes the proof. ∎

Corollary J2 (Bounding Exercise)**.**

(I) Under Assumption J8, a researcher can

1. *vary $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$ arbitrarily to obtain the lowest $\widehat{\widehat{AIE}}$ as the lower bound of AIE; or*
2. *pick acceptable $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$, and calculate $\widehat{\widehat{AIE}}$ to estimate AIE.*

(II) If further assuming $\gamma_i \perp\!\!\!\perp \pi_i$, a researcher can

1. *vary $\rho_{\phi\gamma}$ arbitrarily to obtain the lowest $\widehat{\widehat{AIE}}$ as the lower bound of AIE; or*
2. *pick an acceptable $\rho_{\phi\gamma}$ and calculate $\widehat{\widehat{AIE}}$ to estimate AIE.*

Proof. (I) follows from Proposition J3, and (II) follows from Corollary J2. ∎

J.3.2 Extension: Regression Discontinuity

The following proposition extends to RDDs.

Proposition J4. *Suppose $\{\gamma_i, \pi_i\} \perp\!\!\!\perp r_i$. Under Assumption J8, for every given $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$, AIE is identified by the estimand*

$$\widehat{\widehat{AIE}} = \hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}, \rho_{\gamma\pi}\right)\hat{\pi}. \tag{J74}$$

where $\hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}, \rho_{\gamma\pi}\right) = \left[\frac{\sqrt{3}\,\mathrm{sgn}(\gamma_i)\hat{\gamma}}{\sqrt{3}\,\mathrm{sgn}(\gamma_i)+\rho_{\phi\gamma}\hat{\sigma}_\phi} + \frac{\rho_{\gamma\pi}\hat{\sigma}_\phi\,\mathrm{sgn}(\pi_i)}{\sqrt{3}}\right]$, $\hat{\sigma}_\phi = \mathrm{plim}\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{\widetilde{Z}_i M_i}{\frac{1}{n}\sum_{i=1}^{n}\widetilde{Z}_i M_i}-1\right)^2}$, *and* $\mathrm{sgn}(\gamma_i)$ *and* $\mathrm{sgn}(\pi_i)$ *respectively give signs of* γ_i *and* π_i. $\hat{\gamma}\left(\hat{\gamma}, \rho_{\phi\gamma}, \rho_{\gamma\pi}\right)$ *is monotonic in* $\rho_{\phi\gamma}$ *and* $\rho_{\gamma\pi}$.

Proof. By assuming $\{\gamma_i, \pi_i\} \perp\!\!\!\perp r_i$, the proof is the same as in basic setups. ∎

Assuming $\{\gamma_i, \pi_i\} \perp\!\!\!\perp r_i$, Corollaries J1 and J2 can also extend to RDDs. In Section 5.2, for simplicity, we present the bounding exercise using Corollary J2(II), assuming $\gamma \perp\!\!\!\perp \pi_i$. We leave the full bounding exercise using Corollary J2(I) to Appendix K.5.

K Additional Empirical Results for Causal Mediation Analysis

K.1 Balance Tests for Trade Shocks

We discuss whether the trade shock in origins measured in Equation 16 can be viewed as a quasi-exogenous *pull factor* of outmigration from the destination. Recall that the trade shock in origin h is constructed as a shift-share variable:

$$TradeShock_{ht} = \sum_k \underbrace{\frac{X^{CN}_{hk,2010}}{L_{h,2010}}}_{\text{share}} \times \underbrace{\frac{\Delta X^{ROW}_{kt}}{X^{CN}_{k,2010}}}_{\text{shift}} \equiv \sum_k s_{hk} g_{kt}, \tag{K1}$$

where $s_{hk} = \frac{X^{CN}_{hk,2010}}{L_{h,2010}}$ and $g_{kt} = \frac{\Delta X^{ROW}_{kt}}{X^{CN}_{k,2010}}$. The "shift" component, g_{kt}, is the change in global demand for product k in year t, *relative* to the industry size of product k in China in 2010. It is apportioned to prefectures by the "share" term, s_{hk}, which reflects the initial specialization of prefecture h.

Borusyak et al. (2022) show that the quasi-exogeneity of $TradeShock_{ht}$ can follow from the quasi-exogeneity of the shift, g_{kt}. Intuitively, this condition would be violated if industries with high g_{kt} systematically concentrate in prefectures with differential dynamics that may induce emigrants to leave their destinations. To test the quasi-exogeneity of g_{kt}, we conduct balance tests following Borusyak et al. (2022). Specifically, we run the following industry-level regression, using data from 2011–2015:

$$q_k = \beta_0 + \beta_1 g_{kt} + \mu_t + \varepsilon_k. \tag{K2}$$

$q_k = \sum_h \frac{s_{hk}}{\sum_i s_{hk}} v_h$. v_h is a covariate of prefecture h, thus, q_k captures the s_{hk}-weighted average covariate of prefectures that have industry k. μ_t is the year fixed effect. Therefore, Equation K2 stacks all years to estimate the associations between trade shocks and predetermined covariates. The regression is weighted by $\sum_h s_{hk}$. Standard errors are clustered at the 2-digit HS section level to account for correlated disturbances of industries within the same sector. The coefficient of interest is β_1. We expect β_1 to be close to zero if g_{kt} is quasi-exogenous.

Table K1 reports the balance tests. In Panel A, we consider trends in variables capturing local dynamics at origins that may affect emigrants' decisions to return. In Panel B, we examine baseline characteristics. We find that all estimates of β_1 are statistically insignificant, supporting that $TradeShock_{ht}$ can be a quasi-exogenous pull factor for outmigration from the destination.

Table K1. Balance Tests for Industry-Level Trade Shocks

Dependent	Coef.	SE
Panel A: Pretrends		
ΔLog population$_{t-1}$	0.094	(0.223)
ΔLog GDP$_{t-1}$	0.710	(0.593)
ΔLog total expenditure$_{t-1}$	0.526	(0.426)
ΔLog expenditure on social security$_{t-1}$	-1.718	(1.403)
ΔLog expenditure on public security$_{t-1}$	1.117	(0.800)
Panel B: Predetermined characteristics		
Share of males	-0.144	(0.123)
Share of migrants	-2.081	(2.072)
Share of construction workers	0.100	(0.228)
Share of manufacturing workers	-2.130	(1.992)
Share of transportation workers	0.129	(0.165)
Share of internet users	-0.135	(0.820)

Note: This table presents balance tests for industry-level trade shocks, following Borusyak et al. (2022). Each row represents a regression of the dependent variable, transformed to the industry level, on the industry-level shock (Equation K2). The sample includes 4,374 six-digit HS industries and 5 years (2011–2015). For readability, all estimated coefficients are multiplied by 1,000,000. Standard errors are clustered at the 2-digit HS section level (N = 93).
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

K.2 Effects of Origin Trade Shocks on Outmigration from the Destination

To better understand the source of variation, we estimate the following regression:

$$\Pr\left(Outmigration_{jodt} \mid \mathbf{W}_{jodt}\right) = \beta \cdot TradeShock_{ot} + \lambda_o + \mu_t, \quad t \in \{2014, 2015\}. \tag{K3}$$

$Outmigration_{jodt}$ is a dummy variable that equals one if migrant j, who was from origin o and lived in destination d in 2010, had left d by year t. $TradeShock_{ot}$ is the trade shock to origin o in year t, as defined in Equation 16. Since $Outmigration_{jod,2014}$ captures all migration decisions from the end of 2010 to the end of 2014, we let $TradeShock_{o,2014}$ be the average trade shock between 2011–2014, i.e., $\frac{1}{4}\sum_{t=2011}^{2014} TradeShock_{ot}$. λ_o and μ_t are origin and year fixed effects, respectively.

Table K2 presents the results. We use the sample of rural migrants.[20] Column (1) is a minimal specification, showing that a \$1,000 increase in exports per worker at origins leads to a 1.69-percentage-point increase in the probability of outmigration. This is consistent with the idea that better home economic conditions "pull" migrants back. In Column (2), we further control for province-by-year fixed effects, which yields a larger estimated effect of 3.1 percentage points. Column (3) shows that, interestingly, the destination trade shock has a weak effect on outmigration. However, it has an expected negative sign, suggesting that unfavorable destination conditions "push" migrants out.

Table K2. The Effects of Origin Trade Shocks on Outmigration from the Destination

	Outmigration from the 2010 residing prefecture		
	(1)	(2)	(3)
Origin trade shock$_{ot}$	0.0169***	0.0310***	0.0346***
	(0.0040)	(0.0048)	(0.0050)
Destination trade shock$_{dt}$			-0.0002
			(0.0014)
Sample	Rural migrants	Rural migrants	Rural migrants
Origin prefecture FE	Yes	Yes	Yes
Destination prefecture FE			Yes
Year FE	Yes		
Origin province × year FE		Yes	Yes
Destination province × year FE			Yes
Observations	58,647	58,647	58,647

Note: This table reports the effects of origin trade shocks on outmigration from the destination. Standard errors clustered by origin and destination are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

[20] Using urban migrants or all migrants yields similar patterns. Results are available upon request.

K.3 IV Validity: Balance Tests for Origin Trade Shocks

Because the instrument Z_i is a shift-share variable, we test its validity using the same approach as in Appendix K.1. To operationalize this, rewrite Z_i:

$$Z_i = \sum_h s_{h \to i} \times TradeShock_{h,2015} \tag{K4}$$

$$= \sum_k \left(\sum_h s_{h \to i} s_{hk} \right) g_{k,2015} \tag{K5}$$

$$\equiv \sum_k \tilde{s}_{ik} g_{k,2015}. \tag{K6}$$

$\tilde{s}_{ik} \equiv \sum_h s_{h \to i} s_{hk}$ captures origin industrial specialization of an average migrant in prefecture i.

The excludability of Z_i follows from the quasi-exogeneity of $g_{k,2015}$. In other words, one needs to assume that prefectures with differential odds of labor unrest do not systematically receive migrants from origins specializing in industries with high global demand fluctuations. We can conduct balance tests using the following regression model:

$$\tilde{q}_k = \beta_0 + \beta_1 g_{k,2015} + \nu_k. \tag{K7}$$

$\tilde{q}_k = \sum_i \frac{\tilde{s}_{ik}}{\sum_i \tilde{s}_{ik}} v_i$. v_i is a covariate of destination prefecture i; thus, $\tilde{q}_k$ captures the $\tilde{s}_{ik}$-weighted average covariate of destination prefectures that receive migrants from origin prefectures producing product k. The regression is weighted by $\sum_i \tilde{s}_{ik}$. Standard errors are clustered at the 2-digit HS section level to account for correlated disturbances of industries within the same sector. We expect β_1 to be close to zero if $g_{k,2015}$ is quasi-exogenous. Table K3 reports the results for balance tests. All estimated β_1's are statistically insignificant, supporting that Z_i is a valid instrument that predicts outmigration without picking up unobserved determinants of labor unrest in destination i.

Table K3. Balance Tests for Industry-Level Trade Shocks at Origins

Dependent	Coef.	SE
Panel A: Pretrends		
ΔLog population$_{t-1}$	0.114	(0.085)
ΔLog GDP$_{t-1}$	0.139	(0.113)
ΔLog total expenditure$_{t-1}$	0.312	(0.239)
ΔLog expenditure on social security$_{t-1}$	0.022	(0.114)
ΔLog expenditure on public security$_{t-1}$	0.124	(0.108)
Panel B: Predetermined characteristics		
Share of males	-0.019	(0.021)
Share of migrants	0.106	(0.112)
Share of construction workers	0.028	(0.055)
Share of manufacturing workers	-0.019	(0.081)
Share of transportation workers	0.050	(0.052)
Share of internet users	0.146	(0.157)

Note: This table presents balance tests for industry-level trade shocks at origins, following Borusyak et al. (2022). Each row represents a regression of the predetermined variable, transformed to the industry level, on the industry-level trade shock (Equation K7). Standard errors are clustered at the 2-digit HS section level. For readability, all estimated coefficients are multiplied by 1,000,000.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

K.4 Robustness Checks

Controlling for Covariates. In Table K4 and Figure K1, we conduct robustness checks by controlling for destination's trade shocks and population growth in 2015. If anything, it slightly accentuates the importance of the outmigration channel.

Table K4. The *Hukou* Reform, Outmigration, and Labor Unrest: Controlling for Covariates

	Baseline			+ Destination trade shocks			+ Population growth		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Total effect	Mediation-OLS	Mediation-IV	Total effect	Mediation-OLS	Mediation-IV	Total effect	Mediation-OLS	Mediation-IV
Reform [β or τ]	-1.047**	-0.951*	-0.387	-0.781	-0.662	-0.083	-0.987*	-0.905*	-0.305
	(0.518)	(0.535)	(0.740)	(0.520)	(0.538)	(0.731)	(0.523)	(0.539)	(0.768)
ΔOutmigration [γ]		1.689**	11.651***		1.843***	10.843***		1.489**	12.344***
		(0.680)	(4.077)		(0.660)	(3.878)		(0.691)	(4.635)
% Total effect explained		0.091	0.630		0.152	0.894		0.083	0.691
Effective F stat.			16.126			16.900			13.270
Observations	287	287	287	287	287	287	282	282	282

Note: This table reports causal mediation analysis, controlling for destination trade shocks and population growth. The coefficients on *Reform* in "Total effect" columns are denoted by β, while the coefficients on *Reform* in "Mediation-OLS" and "Mediation-IV" columns are denoted by τ. The share of the total effect explained is calculated as $\frac{\beta-\tau}{\beta}$. The effective F statistic is calculated following Olea and Pflueger (2013). Robust standard errors are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

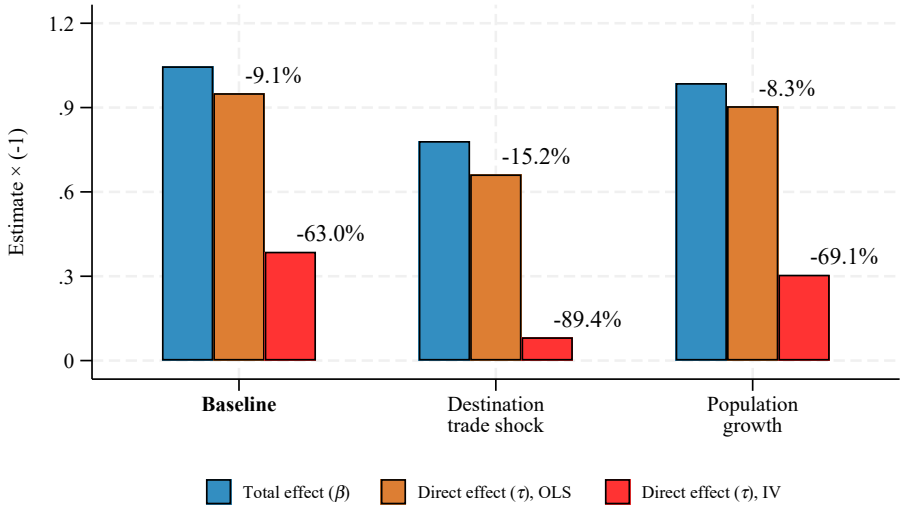


Figure K1. Mediation Visualized: Controlling for Covariates

Note: This figure examines the robustness of causal mediation results when controlling for covariates. The plots are created based upon estimates in Table K4. The "Total effect" bars correspond to the estimated total effects (β) in the "Total effect" columns; the "Direct effect, OLS" bars correspond to the estimated direct effects (τ) in the "Mediation-OLS" columns; and the "Direct effect, IV" bars correspond to the estimated direct effects (τ) in the "Mediation-IV" columns. All estimates are multiplied by -1 to improve visualization. The percent changes are calculated as $\frac{\beta-\tau}{\beta}$.

Alternative IVs. Table K5 and Figure K2 report the results using alternative constructions of the IV to address spillovers and omitted variable bias:

1. Excluding origins within 100km of a destination prefecture;
2. Excluding origins that are regional centers (provincial capitals and four provincial-level municipalities);

3. Recentered IV, following Borusyak and Hull (2023):

$$Z_i^r = \sum_h s_{h \to i} \times TradeShock_{h,2015}^r. \tag{K8}$$

In this expression, $TradeShock_{h,2015}^r = \sum_k s_{hk} g_{k,2015} - \sum_k s_{hk} \bar{g}_{2015}$, where s_{hk} and $g_{k,2015}$ are defined as in Equation K1, and $\bar{g}_{2015} = \frac{1}{K} \sum_{k=1}^{K} g_{k,2015} \approx E(g_{k,2015})$ is the mean industry-level shock (assuming $g_{k,2015}$'s are i.i.d. distributed).

Table K5. The *Hukou* Reform, Outmigration, and Labor Unrest: Alternative IVs

	Baseline			Origins ≥ 100km			Non-center origins			Recentered IV		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Total effect	Mediation-OLS	Mediation-IV	Total effect	Mediation-OLS	Mediation-IV	Total effect	Mediation-OLS	Mediation-IV	Total effect	Mediation-OLS	Mediation-IV
Reform [β or τ]	-1.047**	-0.951*	-0.387	-1.047**	-0.951*	-0.115	-1.047**	-0.951*	-0.108	-1.047**	-0.951*	-0.281
	(0.518)	(0.535)	(0.740)	(0.518)	(0.535)	(0.875)	(0.518)	(0.535)	(0.918)	(0.518)	(0.535)	(0.786)
ΔOutmigration [γ]		1.689**	11.651***		1.689**	16.916***		1.689**	16.577**		1.689**	13.528***
		(0.680)	(4.077)		(0.680)	(5.264)		(0.680)	(7.921)		(0.680)	(4.615)
% Total effect explained		0.091	0.630		0.091	0.890		0.091	0.896		0.091	0.732
Effective F stat.			16.126			16.213			6.975			17.268
Observations	287	287	287	287	287	286	287	287	287	287	287	287

Note: This table reports causal mediation analysis that quantifies the importance of the settlement intentions mechanism, as captured by the outmigration rate. Columns (1)–(3) present the baseline results for ease of comparison. Columns (4)–(6) use an IV constructed using only origins that are 100km away from a destination prefecture. Columns (7)–(9) use an IV constructed using origins that are not regional centers (provincial capitals and four provincial-level municipalities). Columns (10)–(12) use the re-centered IV construction proposed by Borusyak and Hull (2023) to address potential omitted variables bias. The coefficients on *Reform* in "Total effect" columns are denoted by β, while the coefficients on *Reform* in "Mediation-OLS" and "Mediation-IV" columns are denoted by τ. The share of the total effect explained is calculated as $\frac{\beta-\tau}{\beta}$. The effective F statistic is calculated following Olea and Pflueger (2013). Robust standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

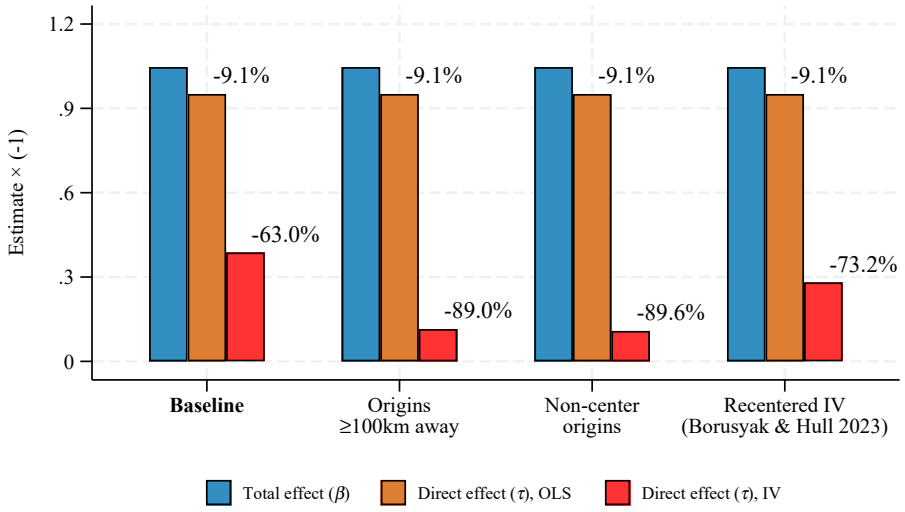


Figure K2. Mediation Visualized: Alternative IVs

Note: This figure examines the robustness of causal mediation results when using alternative IV constructions. The plots are created based upon estimates in Table K5. The "Total effect" bars correspond to the estimated total effects (β) in the "Total effect" columns; the "Direct effect, OLS" bars correspond to the estimated direct effects (τ) in the "Mediation-OLS" columns; and the "Direct effect, IV" bars correspond to the estimated direct effects (τ) in the "Mediation-IV" columns. All estimates are multiplied by −1 to improve visualization. The percent changes are calculated as $\frac{\beta-\tau}{\beta}$.

Sensitivity to Violations of the Exclusion Restriction. Uncontrolled confounders can lead to violations of the exclusion restriction, but it is challenging to find proxies for all possible confounding factors. To examine the sensitivity of our 2SLS estimate of γ in Equation 14 to potential violations of the exclusion restriction, we use the methodology developed by Conley et al. (2012). This approach

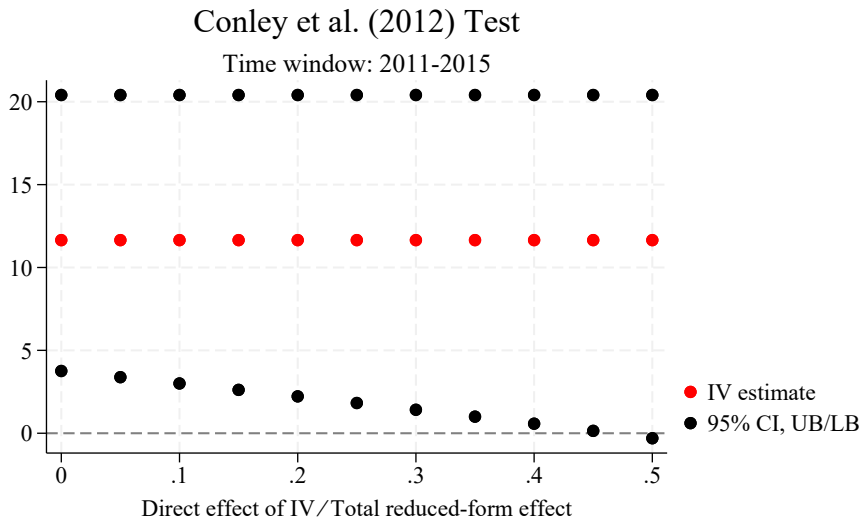


Figure K3. Conley et al. (2012) Test

Note: This figure reports the test proposed by Conley et al. (2012). This test allows the IV to have a direct effect on the outcome, and on that basis, it examines whether the IV estimate remains statistically significant. In this figure, we plot the confidence interval of the IV estimate against the hypothesized direct effect of the IV (relative to the total reduced effect of the IV).

allows the IV, Z_i, to directly enter the second stage of the model (Equation 14) with a coefficient of ψ, which measures the extent to which the exclusion restriction is violated and is set by the researcher.

We test whether instrumented outmigration still has a significant effect on labor unrest for different values of ψ. Since we find that outmigration has a positive effect, violations of the exclusion restriction are problematic only when ψ is positive. We look at the largest positive value of ψ such that the resultant 2SLS estimate $\hat{\gamma}$ is still significant at the 5% level. This value is denoted by $\bar{\psi}$, and it is scaled by Z_i's reduced-form effect on labor unrest, ψ_{RF}. The ratio $\bar{\psi}/\psi_{RF}$ represents the maximum hypothetical violation of the exclusion restriction that can be allowed such that the 2SLS estimate $\hat{\gamma}$ remains statistically significant. A higher $\bar{\psi}/\psi_{RF}$ indicates stronger resilience of the IV to violations of the exclusion restriction.

Figure K3 reports the results, where we plot the 95% confidence interval of the 2SLS estimate $\hat{\gamma}$ against $\bar{\psi}/\psi_{RF}$. It shows that $\bar{\psi}/\psi_{RF} = 0.5$. These results indicate that unobserved confounders have to explain at least 50 percent of Z_i's reduced effect, if we were to accept a null effect of outmigration on labor unrest. We believe this fraction is unrealistically high, which provides further support for the validity of our IV.[21]

[21] Conley et al. (2012) do not provide a rule-of-thumb cutoff for $\bar{\psi}/\psi_{RF}$. However, using Conley et al. (2012)'s approach, researchers have demonstrated the robustness of their 2SLS estimates given the following $\bar{\psi}/\psi_{RF}$ ratios: 0.3 in Fatás and Mihov (2013) and 0.46 in Bentzen et al. (2017).

K.5 Bounding Exercise Under the General Framework

Recall from Appendix J.2, $\widehat{AIE}$ can be written as:

$$\widehat{AIE} = \underbrace{E(\gamma_i \pi_i)}_{\text{AIE}} + \underbrace{[\hat{\gamma} - E(\gamma_i)]\, E(\pi_i)}_{\text{IV bias}} - \underbrace{Cov(\gamma_i, \pi_i)}_{\text{ecological bias}} \tag{K9}$$

For simplicity, the results reported in Section 5.2.3 assume $\gamma_i \perp\!\!\!\perp \pi_i$, which rules out ecological bias and leaves only IV bias. The bounding exercise reported in Figure 7 invokes Corollary J2(II) to bound *AIE* when γ_i is allowed to be heterogeneous. In this section, we further relax the assumption $\gamma_i \perp\!\!\!\perp \pi_i$ to allow *both* IV bias and ecological bias, and we use Corollary J2(I) to bound *AIE*.

We expect that $\gamma_i \geq 0$, i.e., a high outmigration rate (low settlement intention) universally increases unrest, and $\pi_i \leq 0$, i.e., the *hukou* reform reduces outmigration from the destination (or raises settlement intentions). Then, we consider $\rho_{\phi\gamma} \in [0, 1]$ and $\rho_{\gamma\pi} \in [0, 1]$, so that $\widehat{\widehat{AIE}}$ corrects the *overestimation* of AIE by $\widehat{AIE}$. $\rho_{\phi\gamma}$ governs the size of IV bias, while $\rho_{\gamma\pi}$ governs the size of ecological bias. Specifically, for each pair of $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$, we look at the share of total effect explained by the proposed mechanism: *ShareExplained* $= \widehat{\widehat{AIE}}/\hat{\beta}$.

Figure K4 plots *ShareExplained* as a function of $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$. Three findings emerge. First, when $\rho_{\phi\gamma} = \rho_{\gamma\pi} = 0$, namely, when γ_i is constant, we obtain the highest *ShareExplained* of 63 percent. Second, overall, *ShareExplained* is more sensitive to IV bias: *ceteris paribus*, *ShareExplained* drops more quickly when $\rho_{\phi\gamma}$ increases than when $\rho_{\gamma\pi}$ increases. When $\rho_{\phi\gamma} = 0$ and $\rho_{\gamma\pi} = 1$, i.e., when IV bias is ruled out but ecological bias is maximized, Figure K4 shows that *ShareExplained* falls into region III and thus is at least 55 percent. In contrast, Figure 7 shows that when $\rho_{\phi\gamma} = 1$ and $\rho_{\gamma\pi} = 0$, i.e., when IV bias is maximized but ecological bias is ruled out, *ShareExplained* is 16 percent. These patterns motivate us to focus on IV bias. Finally, if we set both IV bias and ecological bias at moderate levels by setting $\rho_{\gamma\pi} = \rho_{\phi\gamma} = 0.5$, then *ShareExplained* falls at the border between regions IX and X, meaning that settlement intentions can explain about 25 percent of the *hukou* reform's effect on labor unrest, which is close to the reasonable lower bound shown in 7 (26 percent under $\rho_{\gamma\pi} = 0.5$ and $= \rho_{\phi\gamma} = 0$).

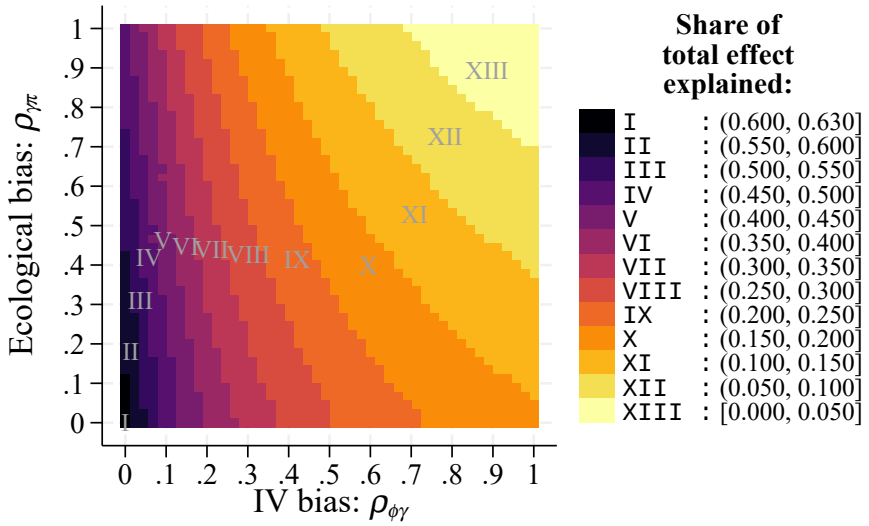


Figure K4. Bounding Exercise for Causal Mediation Under General Framework

Note: This figure presents the share of total effect explained by the settlement intentions mechanism, calculated as $\widehat{\widehat{AIE}}/\hat{\beta}$, for each given pair of $\rho_{\phi\gamma}$ and $\rho_{\gamma\pi}$. $\widehat{\widehat{AIE}}$ is calculated according to Proposition J3.

L Replication of An et al. (2024)

In this section, we replicate the main results in An et al. (2024): the effects of the *hukou* reform on workforce participation, wages, and access to social security (ASS), which we also investigate in Table 6. They also use the CMDS data, and their findings are: (i) the reform has no effect on migrants' workforce participation; (ii) the reform significantly lowers wages; and (iii) the reform reduces ASS. However, we do not find (ii) and (iii). Below, we show how the discrepancies arise.

There are two major differences between their empirical implementation and ours.

1. *Reform Status Definition.* To measure reform status, we use the total urban population that includes *both* natives and migrants from the Urban Construction Yearbook (UCSY), which is consistent with the official definition of urban population National Bureau of Statistics (2008), whereas An et al. (2024) use only the native urban population.
2. *Identification Strategy.* An et al. (2024) implement a DiD design similar to Equation 1, which in essence compares more populous versus less populous areas.[22] In contrast, our strategy, as given by Equation 2, combines DiD and RD designs (DiDC).

Table L1 shows how results change under these decisions. To avoid other sources of differences, e.g., sample construction, we produce these results using the sample in An et al. (2024)'s replication package. Panel A adopts An et al. (2024)'s definition of reform status, whereas Panel B uses our definition. Columns (1)–(3) report DiD estimates, and Columns (4)–(6) report the DiDC estimates. Clearly, different definitions of reform status do change the reform and non-reform groups. Using total urban population, we define 10 fewer prefectures as reform ones. Nonetheless, different definitions of reform status do not change the results markedly. The main change comes from the identification strategy. DiD estimates replicate the findings in An et al. (2024). However, when we implement the DiDC design to compare prefectures around the 3-million cutoff, we no longer find discernible effects of the *hukou* reform on wages and ASS. This pattern also appears when we use our own estimation sample (see Table L2).

The DiDC strategy arguably produces cleaner, more reliable results than a DiD design because it flexibly controls for heterogeneity due to urban size (cf. Table 1). Moreover, we find it difficult to reconcile the *hukou* reform's negative effects on wages and ASS. An et al. (2024) argue that the reform increased labor inflow and thus reduced wages and put pressure on the local social security system. However, as we show in Appendix G.5, even when using a DiD strategy, we find that the reform in fact did not have strong effects on population growth; if anything, reform prefectures had relative population declines (or disadvantages in population growth). Taken together, we argue that the *hukou* reform does not affect wages and ASS much.

[22] Specifically, they use prefectures with more than 5 million people as the control group, and jointly estimate the effects of the *hukou* reform for different treatment groups defined by urban population size (3–5 million, 1-3 million, 0.5–1 million, and < 0.5 million). Their estimated treatment effects for the 3–5 million group are small and often not significant. Therefore, to ease comparison here, when replicating their analysis, we use prefectures with more than 3 million people as the control group and the remaining prefectures as the treatment group.

Table L1. Replicating Main Results of An et al. (2024)

	(1) Working	(2) Log wage	(3) ASS	(4) Working	(5) Log wage	(6) ASS
Panel A: An et al.'s definition of treatment						
Reform (An et al.) × Post	0.006	-0.077***	-0.041**	-0.001	0.018	0.008
	(0.006)	(0.018)	(0.018)	(0.009)	(0.021)	(0.020)
Control mean	0.872	8.131	0.526	0.872	8.131	0.526
Sample period	2011–17	2011–17	2011, 13, 16	2011–17	2011–17	2011, 13, 16
No. prefectures (No. reform prefectures)	267 (241)	267 (241)	266 (239)	267 (241)	267 (241)	266 (239)
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Polynomials				Yes	Yes	Yes
Observations	788,219	642,700	144,145	788,219	642,700	144,145
Panel B: Our definition of treatment						
Reform × Post	0.007	-0.087***	-0.038**	0.004	-0.004	0.010
	(0.006)	(0.016)	(0.018)	(0.009)	(0.021)	(0.027)
Control mean	0.872	8.131	0.526	0.872	8.131	0.526
Sample period	2011–17	2011–17	2011, 13, 16	2011–17	2011–17	2011, 13, 16
No. prefectures (No. reform prefectures)	267 (230)	267 (230)	266 (228)	267 (230)	267 (230)	266 (228)
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Polynomials				Yes	Yes	Yes
Observations	788,219	642,700	144,145	788,219	642,700	144,145

Note: This table replicates the key results for migrants' workforce participation, wages, and access to social security (ASS) in An et al. (2024). We use the CMDS sample in An et al. (2024)'s replication package. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

Table L2. Replicating Main Results of An et al. (2024): Own Sample

	(1) Working	(2) Log wage	(3) ASS	(4) Working	(5) Log wage	(6) ASS
Reform × Post	-0.005	-0.066***	-0.047**	-0.003	0.003	0.006
	(0.005)	(0.013)	(0.018)	(0.007)	(0.019)	(0.029)
Control mean	0.883	8.153	0.522	0.883	8.153	0.522
Sample period	2011–18	2011–18	2011, 13, 16	2011–18	2011–18	2011, 13, 16
No. prefectures	255	255	255	255	255	255
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Polynomials				Yes	Yes	Yes
Observations	990,912	810,696	162,239	990,912	810,696	162,239

Note: This table replicates the key results for migrants' workforce participation, wages, and access to social security (ASS) in An et al. (2024). We use the the same sample as in Table 6. Standard errors clustered at the prefecture level are reported in parentheses.
* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

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